



M7XC

Compact Satellite Modem

Installation and Operation Manual (v.4.5)



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NOTE: All appendices may not be present in manual. Some Appendixes may be shipped with the option.

Preface

This manual is intended for use by personnel with reasonable knowledge of satellite communications and earth station terminology. This manual provides Installation, Operating, and Maintenance procedures for the M7XC Compact Satellite Modem and available options.

Standard Designations

The M7XC designation is used throughout this manual where references apply to the M7XC Compact Satellite Modem. The designation for the M7XC may also be referred to as 'modem' or 'unit' interchangeably.

Recommended Standard (RS) designations (i.e., RS-232) have been superseded by the new designation of the Electronic Industries Association (EIA) designations (i.e., EIA-232). The RS and EIA designation may be used interchangeably within the manual and protocol documentation and have the same electrical and mechanical meaning.

Electrical Safety and Compliance Notice

The M7XC has been safety tested and shown to comply with standard EN 60950 Safety of Information Technology Equipment (Including Electrical Business Machines).

Never operate the unit with the cover removed. Never remove a cover with power applied. As a safety measure the power cord must be disconnected from the unit when preparing to remove the cover.

Battery

The M7XC optionally contains a Lithium Battery. Danger of explosion exists if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Dispose of used batteries in accordance with local and national regulations.

EMC (Electromagnetic Compatibility) and Safety Compliance



This equipment has been tested in accordance with FCC and ESTI standards.

- FCC: Part 15, Subpart B, Class A
- ETSI EN 301 489-1 V1.9.2 (2011-09) (Emissions and Immunity)
- EN 55022 (2010) (Emissions)
- EN 55024 (2010) (Immunity)
- EN 60950-1:2006 (2nd Edition) and IEC 60950-1:2005 (Safety)

To maintain compliance with these standards the following the precautions must be observed.

- The unit must be operated with the cover attached with all cover screws in place.
- All connections are designed to have integral shielding on the cable and connector assembly.

Telecommunications Terminal Equipment Directive (TTED)

In accordance with the TTED 91/263/EEC, the M7XC should not be connected directly to the Public Telecommunications Network.

RoHS Compliance

The M7XC satisfies the requirements specified in the European Union Directive on the Restriction of Hazardous Substances, Directive 2002/95/EC (EU RoHS).

Environmental

The M7XC is designed to operate in an enclosed environment. Operation while exposed to precipitation, condensation, relative humidity above 95%, excessive dust, or temperature outside the operating range of -40° C to +60° C temperature range may cause damage to the unit and will void the warranty. Maximum storage temperature is -55° C to +70° C.

Do not operate the modem in an unsafe environment near explosive or flammable gases or liquids.

If the M7XC is to operate in a mobile environment, the user must take precautions to provide a stable environment consistent with the equipment specifications.

Grounding

Insure good grounding practices. Where a ground lug is provided, the modem should be connected to a good earth ground with low impedance cable in installations.

About This Manual

This manual is composed of several separable documents. The main body of the manual is separated into several “Chapters” and “Sections”. The Chapters within the main manual are the Modem Description, Installation, Operation, and Maintenance. The Appendices include the Specifications, Remote Control Protocol and information on Options and user configurations.

A Section is considered a sub-section of a Chapter. For example, Section 4.1.2 is a numbered section within Chapter 4. Page numbers include the Chapter (i.e., Page 3-14) and all Tables and Figures will show the Chapter in the caption (i.e., Table 2-2).

This manual is available in a printed form and as an electronic “Portable Document Format” or .PDF file. The electronic format is produced as a universal Adobe Acrobat readable file, and can be requested directly from Datum Systems, Inc., or via download from the web at www.datumsystems.com. The electronic format on the web is always the latest revision.

Revision History

Release Number	Release Date	Comments
Revision 0.01	04/01/2021	Initial Public Release
Revision 0.02	04/01/2023	Updated to reflect current capabilities
Revision 0.04	09/22/2024	Update: New GUI screens and capabilities

Pen and Ink Changes Made to this Manual

This image shows a full page of blank white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for writing or drawing. There are no margins, text, or other markings on the page.

1 Introduction

1.1 Modem Description

The M7XC, represented in Figure 1-1 - Compact Satellite Modem, is a major extension of our innovative design concepts that have been proven and refined in over twenty years of production. The M7XC is capable of independent transmission and reception of carriers compliant to most satellite communications standards. The M7XC uses the latest Digital Signal Processing (DSP) technology and other proprietary techniques to provide unsurpassed performance. The M7XC is a highly modular design made possible by a higher level of integration.



Figure 1-1 - Compact Satellite Modem

The dimensions of the black enclosed M7XC with fan and heatsink are 5" (L) x 3" (W) x 1.3" (H) and weighs < 2 lbs.

The dimensions of the silver enclosed M7XC without fan and heatsink are 5" (L) x 3" (W) x 1.0" (H) and weighs < 2 lbs.

The M7XC is end-to-end compatible with all Datum Systems M7 series modems.

The M7XC is designed to be integrated into a compact satellite terminal for use in a variety of satellite systems. The satellite terminals may be small fly-away systems, Manpack, Airborne/UAV, or other mobile satellite terminal applications.

The M7XC design can also be integrated with other M7XC units into a master hub station.

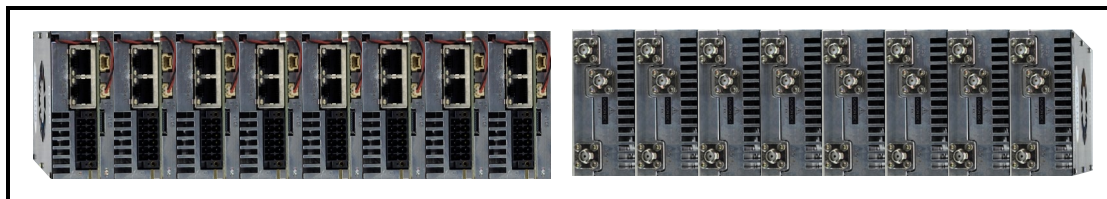


Figure 1-2 Compact Hub Modems

The highly integrated design allows system designers the flexibility to integrate units into many different innovative hardware system designs. The M7XC is an integral part of a satellite earth station's equipment operating between the Data Terminal Equipment and the station RF equipment.

The M7XC is designed for operation in different network types. Operating modes may be dedicated or shared. In a dedicated system two modems are set for continuous operation with each other. In shared systems, the modem carriers are typically controlled by a Network Management System (NMS).

1.2 Network Types

The M7XC is extremely flexible regarding the network types that can be configured. Depending on the network type, different baseband interface options are available that will maximize the capability for the network. The main network types are:

- Point-to-Point
- Point-to-Multipoint
- Mesh
- Hybrid

In a Point-to-Point (P2P) network, the M7XC provides the industry leading Eb/No performance and flexible baseband interface options. This performance improvement and the Ethernet/IP interface in the M7XC provide the network operator with the latest technology needed to extend cellular backhaul connectivity and Internet access to remote locations. With Ethernet bridging capabilities internal to the M7XC, there is a significant reduction in the number of devices that need to be managed and maintained. A typical P2P network is shown in Figure 1-3 Point to Point Network.

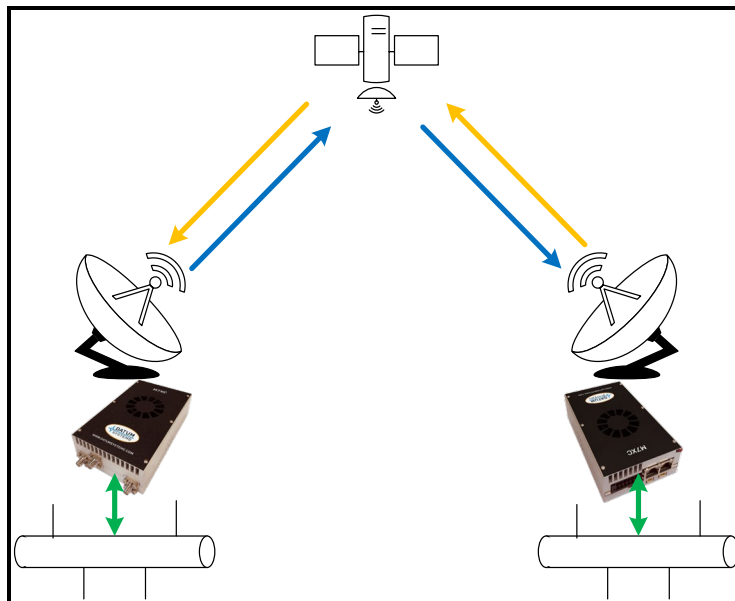


Figure 1-3 Point to Point Network

Point-to-Multipoint (P2MP) networks are used to connect a central location to multiple remote locations. The central Hub location will transmit a shared carrier that is received by all the remote sites. The Hub location will also receive the individual carriers from each of the remote sites on individual demodulators. Within the shared outbound carrier will be all the information destined for each of the remote sites. At each of the remote sites, the M7XC with an Ethernet/IP interface will filter only the information that is to be received by that site. There is considerable statistical bandwidth savings by using a shared outbound carrier in an IP based network because of the typical traffic patterns consistent with IP networks. A typical P2MP network is shown in Figure 1-4 Point to Multi-Point Network.

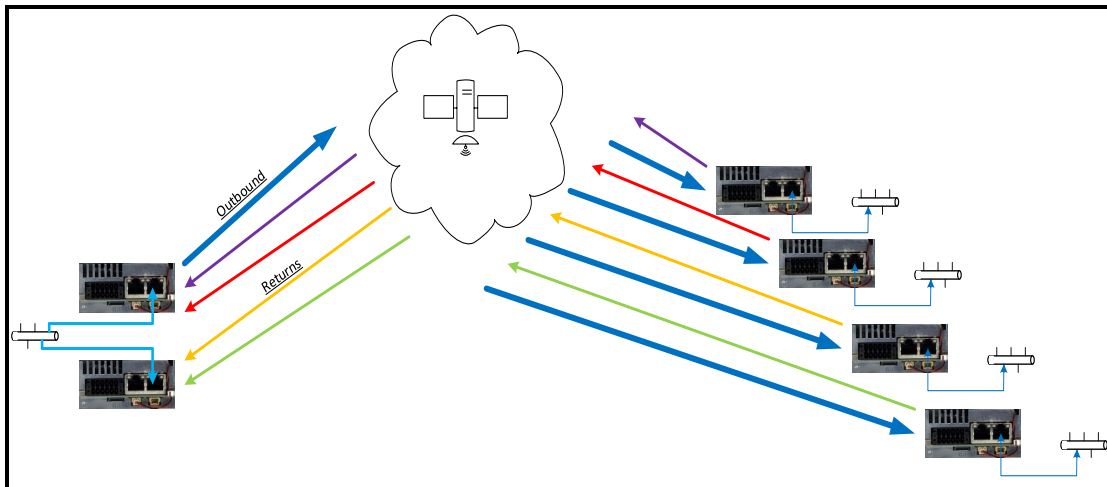


Figure 1-4 Point to Multi-Point Network

A Mesh network is used when all sites in a network need to be connected to all other sites within that same network with a single satellite hop link. This could be done by installing multiple P2P networks between each network site, but as the number of network nodes increases, there is a dramatic increase in the number of carriers and an increase in the number of modems required at each site. At some point, this becomes unmanageable, and cost prohibitive. The alternative is to use a Mesh network topology instead of multiple P2P networks. At each site in a Mesh network there is a single carrier that contains all the information destined for all the nodes within the network. In an IP based network there is considerable statistical satellite bandwidth savings by using the shared outbound carrier from each site because of the typical traffic patterns consistent with IP networks. A typical Mesh network is shown in Figure 1-5 Mesh Network.

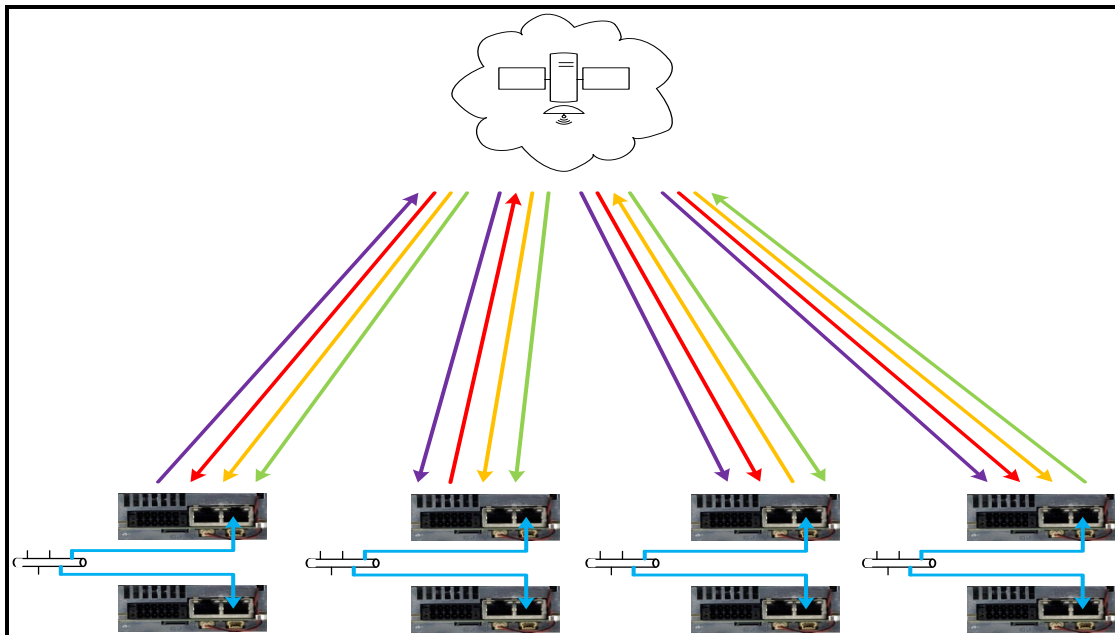


Figure 1-5 Mesh Network

A hybrid network is a combination of any of the networks shown above.

1.3 M7XC Overview

The M7XC is similar in operation to the previous Datum Systems M7 modems but specifically targeted for small form factor, low weight, low power (SWaP) applications. For this reason, there is no integrated front panel. Monitor and control is via SNMP, and Web Browser connections. The list below shows some of the differences between the M7XC and the previous Datum Systems M7 modem products.

1.3.1 M7XC Highlights

- Powered by +10 to +36 VDC input with a max of 30 Watts.
- Transmit and Receive L-Band 950 to 2250 MHz.
- >560 Mbps (Waveform Dependent).
- QPSK, 8PSK/APSK, 16APSK, 32APSK, 64APSK, 128APSK and 256APSK.
- Contains separate Modulator and Demodulator cards.
- Capable of being a TX/RX Modem, a TX only Modulator, or a RX only Demodulator.
- Independent TX/RX LDPC-Gen2, DVBS2/DVBS2X, and Vit/RS/TPC.
- Multi-Demods up to 16 LDPC-Gen2 or two DVBS2X Independent RX Channels.
- Segmented feature up to Sixteen TX LDPC-Gen2 channels.
- GigE IP Control interface provides remote control via a HTTP/S web browser user interface and SNMP protocol.
- Complete monitor and control functions as well as firmware/software updates from the web browser.
- IP Control management interface connection is separate from the IP Data traffic interface connection.
- RS-232 serial remote-control interface binary packet control protocol is also provided.
- Baseband traffic Data is connected via the GigE Layer 2 Switched Bridged Ethernet Protocol interface.
- Available with Sharp Carrier Technology that provides eight (8) filter roll-off options from 0.40 Alpha to 0.02 Alpha.
- Available with second-generation advanced carrier canceller for Point-to-Point (P2P) and Point-to-Multipoint (P2MP) networks.
- Compliant with Mil-STD 165B Standard for TX Spurious and Phase Noise
- AES 128/256 (FIPS-140-2 Compliant), 32k P/N Generated Rolling Keys
- High Stability 10/50 MHz Reference (In/Out), supports BUC/LNB Reference
- LNB Power (+13 and +18 VDC Selectable)
- -40° C to + 60° C Temp Range
- MIL-STD-810G – Vibration

1.4 Modem Assemblies

The M7XC consists of four (4) main functional elements arranged on two (2) electronic printed circuit board assemblies shown in the next Figure.

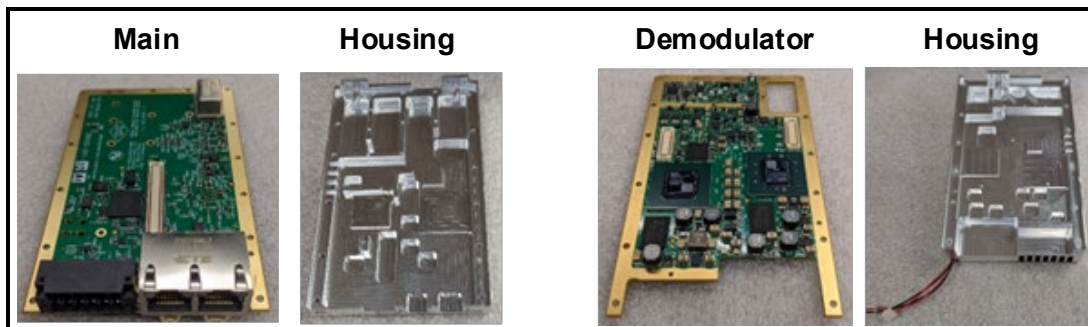


Figure 1-6 Major Assemblies

The modem assembly consists of the following major assemblies:

- A. M7XC digital PSK main board assembly with L-Band carrier generation, monitor/control microprocessor, GigE control and data interfaces.
- B. M7XC digital PSK demodulator board assembly that accepts carriers in the L-Band frequency range. The Digital Signal Processor Acquisition subsystem is part of the demodulator board assembly.
- C. M7XC 'clamshell' metal housing.

A functional block diagram of the M7XC is shown in Figure 1-7 Functional Block Diagram.

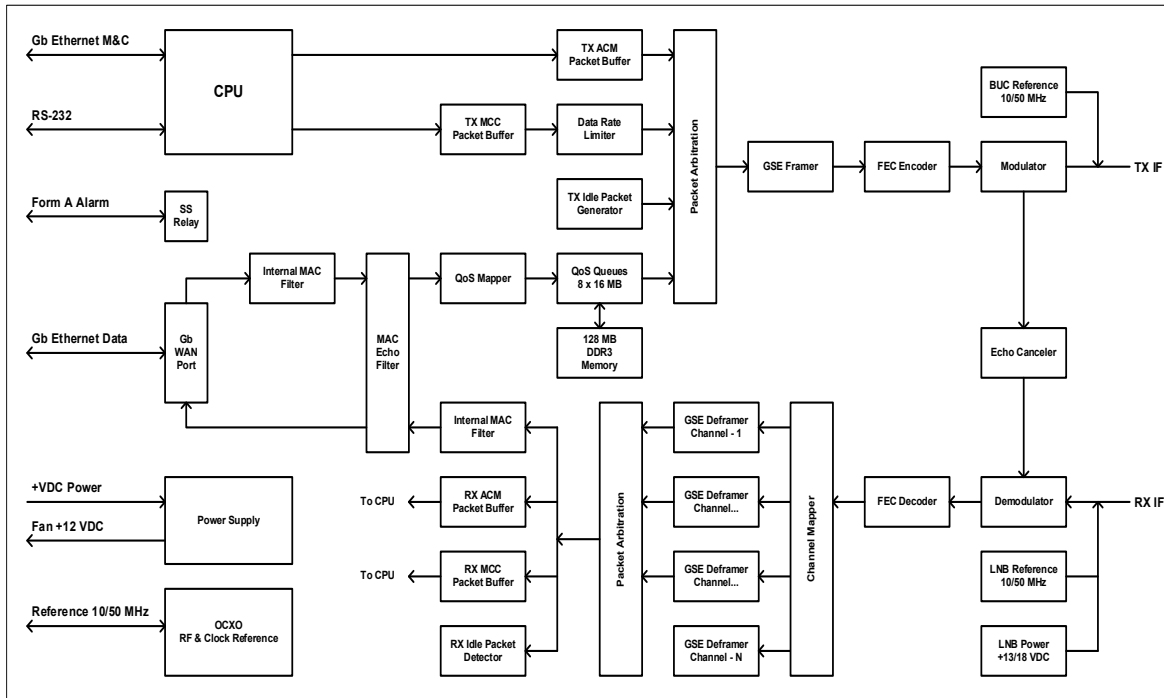


Figure 1-7 Functional Block Diagram

1.5 Capabilities

1.5.1 Inter-Facility Interface (IF)

The M7XC transmit and receive L-Band IF frequency range is 950 to 2250MHz.

The M7XC transmit and receive IF also provides for either 10 or 50 MHz Frequency References for the BUC and/or LNB. The modem is ideal for ground stations and mobile assets that are designed with block Up and Down conversions.

1.5.2 Reference In/Out (Ref)

The M7XC can source or receive optional external references to frequency lock multiple modems, BUCs, LNBs or any 3rd party equipment requiring or sourcing a reference. This interface may be configured as an output to provide a reference to other units. The Input or Output Reference Frequency is configurable to 10 MHz or 50 MHz.

1.5.3 GigE Baseband Interfaces (IP)

The M7XC connects to the Local Area Network (LAN) via the local GigE Ethernet 10/100/1000BASE-T interface. The modem also offers a 1000BASE-T Management interface. The GigE Ethernet interface ports are standard copper RJ-45 connector. The Data and IP Control Ethernet interfaces are physically separate. Connections to the interfaces should be made with Category 5 or higher ethernet cables.

1.5.4 Segmented Mode Operation

Refer to Appendix H – Segmented Mode Operation for details.

1.5.5 Sharp Carrier Technology

The M7XC provides multiple settings for the modulation carrier roll-off that will permit interoperability with industry standards and increase the number of carriers that can be placed in a satellite transponder. This Datum Systems capability is referred to as “Sharp Carrier Technology”.

There are ten (10) filter roll-off options available to the modulator and demodulator control functions. These selectable filter “Alpha” values are “0.40”, “0.35”, “0.30”, “0.25”, “0.20”, “0.15”, “0.10”, “0.05” and “0.02”. Figure 1-8 Spectrum Analyzer Plot 0.02 Roll Off vs. 0.35 ROF shows a comparison of two (2) spectrum plots one which is 0.35 Alpha (blue) and the other 0.02 Alpha (red).

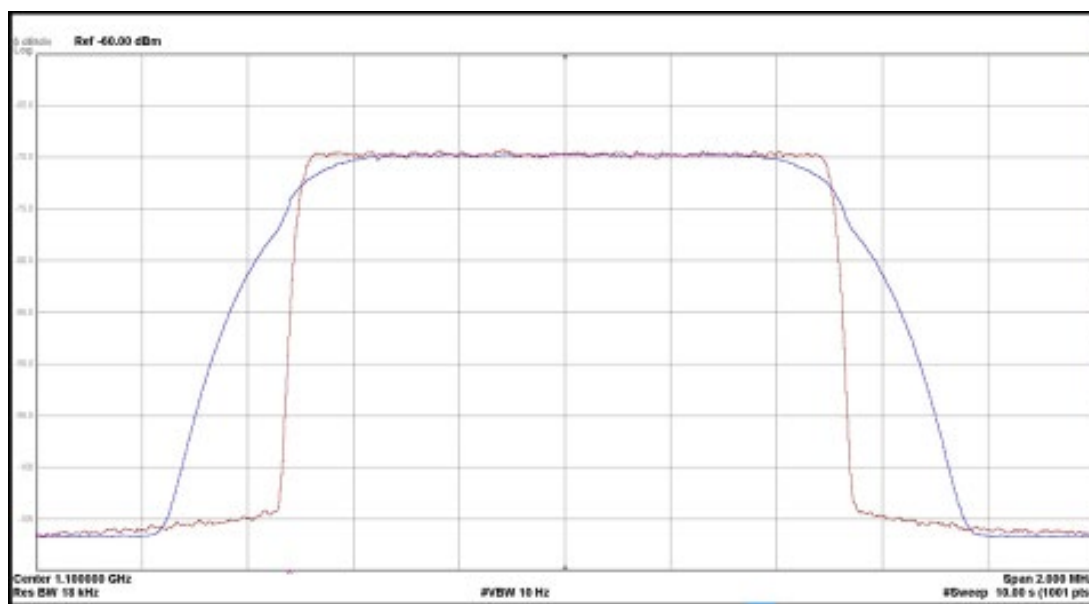


Figure 1-8 Spectrum Analyzer Plot 0.02 Roll Off vs. 0.35 ROF

A significant benefit of this technology is the reduction in carrier spacing required on the satellite. With a filter roll-off of 0.4, the carrier spacing would be 1.4 times the carrier symbol rate. When the carriers are changed to a 0.05 roll-off, this carrier spacing can be reduced by more than 30% to 1.05 times the carrier symbol rate.

A spectrum analyzer plot demonstrating this capability is shown in Figure 1-9 and Figure 1-10. Notice that we are able to fit twelve (12) carriers using 0.05 ROF compared to ten (10) carriers at 0.35 ROF with LESS overall satellite capacity.

10x 1Msps Carriers @ 0.35 ROF = 13.5MHz VS. 12x 1Msps Carriers @ 1.05 = 12.6MHz!! See spectrum plots below.

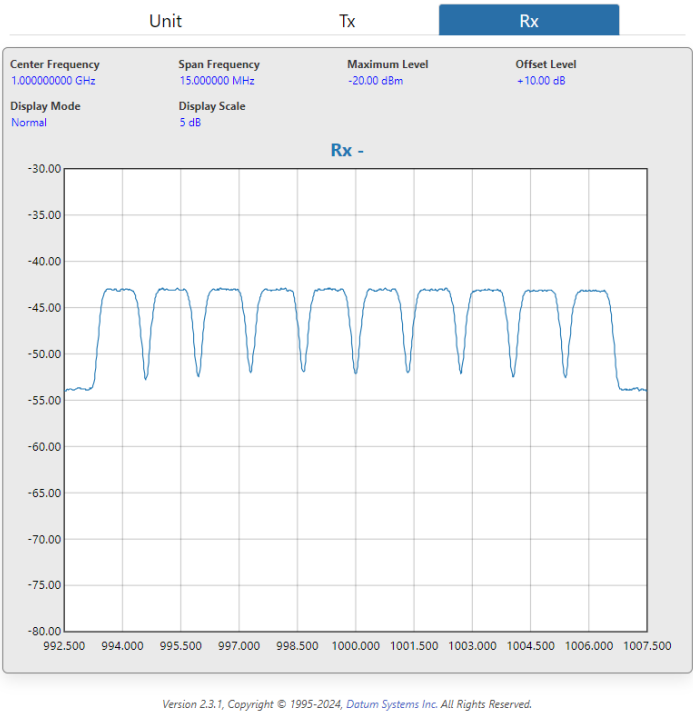


Figure 1-9 (10 Carriers 1Msps Each With 0.35 ROF)

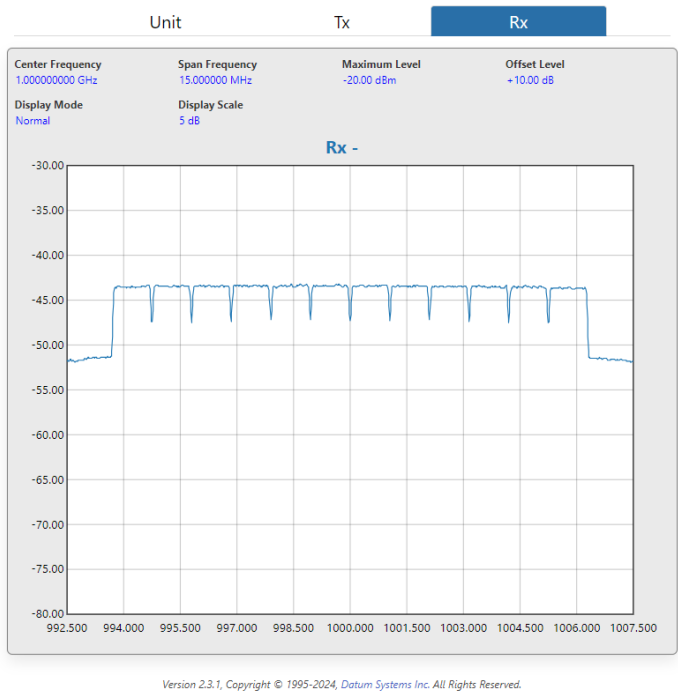


Figure 1-10 (12 Carriers 1Msps Each with 0.05 ROF)

For example, in a typical 36MHz transponder configured with 1Msps carriers with a 0.4 roll-off there could be twenty-five (25) carriers. If the 1Msps carrier roll-off was changed to 0.08, there now could be 32 carriers – a 28% increase in transponder throughput. Table 1-1 Bandwidth Savings vs. Carrier ROF shows the relative improvement gained in a bandwidth limited 36MHz transponder by changing the carrier Alpha settings from 0.4.

Table 1-1 Bandwidth Savings vs. Carrier ROF

Carrier Alpha	0.4	0.35	0.3	0.25	0.2	0.15	0.1	0.05	0.02
# of 1Msps Carriers	25	26	27	28	30	31	32	34	35
% Improvement from 0.4 Alpha	0	4	8	12	20	24	28	36	40

1.5.6 Smart Carrier Cancelling Overview (Release on Request)

Smart Carrier is a patent pending second-generation advanced carrier canceller which allows two carriers to occupy the same transponder spectrum. Our advanced carrier canceller not only provides excellent performance but is different from the other cancellers. Our canceller is a baseband canceller instead of an IF canceller. The canceller is easy to setup and requires no additional cabling. Smart Carrier Cancelling is compatible with all Datum Systems Modulation/FEC types and is well suited to be used with Sharp Roll-Off factors down to 2%. Our technique provides a Shannon Capacity improvement of ~ 2 dB, which is a ~50 % increase in the fundamental channel capacity as shown in Figure 1-11 Smart Carrier Bandwidth Savings of 50%.

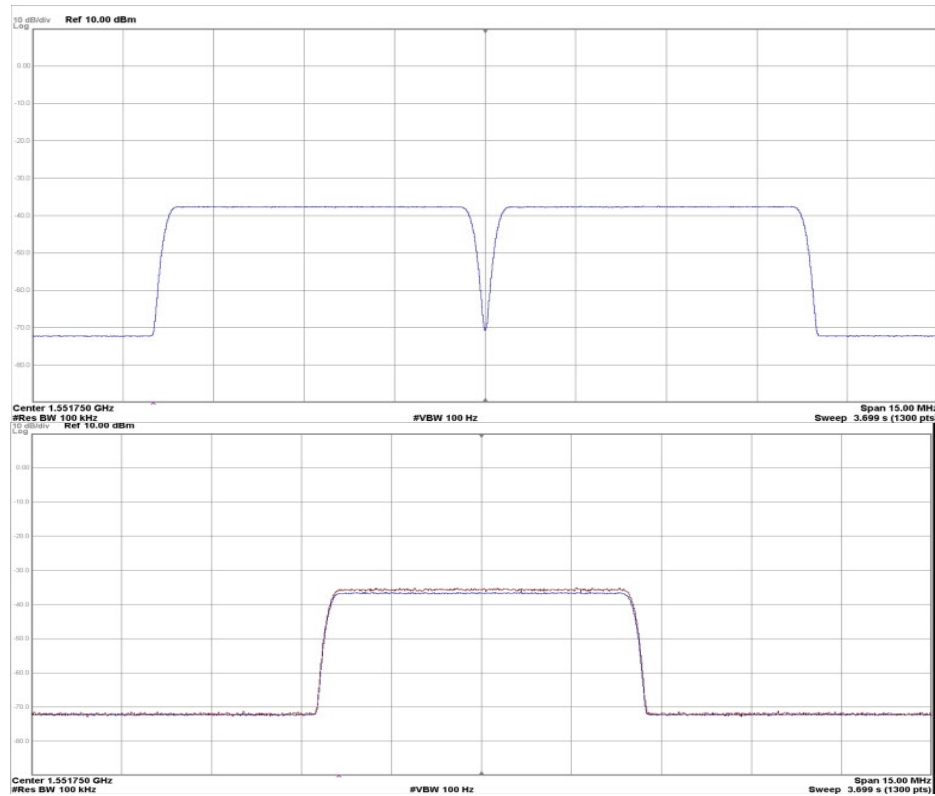


Figure 1-11 Smart Carrier Bandwidth Savings of 50%

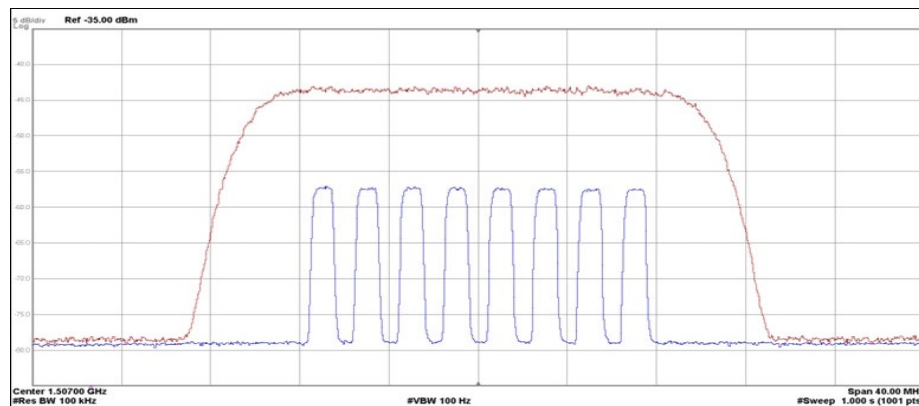


Figure 1-12 P2MP Hub Carrier Bandwidth Savings

1.5.7 Modem Control Channels (MCC + AUPC + ACM)

The MCC enables in-band communications for protocols such as ACM, AUPC, Remote Modem Control and remote communications with 3rd party equipment.

The MCC channel is provided without the need to set-up a separate multiplexer or demultiplexer function. The channel provides a 1kbps – 1Mbps overhead control channel in each direction to allow the modems at both ends of the link to communicate and pass important valuable information and data.

1.6 Forward Error Correction (FEC) Options

The standard M7XC contains the hardware for multiple waveforms such as LDPC-Gen2 and for DVB-S2X FEC types with basic functional capability for all standard operating networks.

1.6.1 LDPC-Gen2 FEC

The LDPC-Gen2 FEC performance improvement achieved is the highest and most flexible of any specialized FEC technology to date, outperforming Viterbi and TPC across all modes. For example, the LDPC-Gen2 at rate 1/2 can operate at a sustained Eb/No of +1.5 dB with an error rate less than 10^{-9} .

1.6.2 DVB-S2X

Refer to Appendix G DVB-S2X Operation for more information.

1.7 1:1 Redundancy Function

NOTE: Both Primary and Backup modems must be at the same hardware and firmware versions for redundancy to operate properly.

More information on the set-up and use of the 1:1 redundancy functions and modes is given in Section 3.9

2 Installation

2.1 Installation Requirements

The Datum Systems M7XC Compact Satellite Modem is primarily designed for installation in an integrated satellite terminal and requires 1.3" (H) x 3" (W) x 5" (D) of space (excluding cabling). The M7XC is shipped with a mating connector for connection to DC power. An optional AC to DC external power converter module designed to accept a 3-wire AC line cord is available on request. Installation and connection to the AC power line must be made in compliance to local or national wiring codes and regulations. The M7XC may be placed on a table or other stable flat surface is required for testing.

⇒ **CAUTION!!!** There are no user-serviceable parts or configuration settings located inside the M7XC modem chassis.

⇒ **CAUTION!!!** Before initially applying power to the modem, it is a good idea to disconnect the transmit output from the satellite ground station equipment. This is especially true if the current modem configuration settings are unknown, where incorrect settings could disrupt existing communications traffic.

2.2 Unpacking

The M7XC was carefully packed to avoid damage and should arrive complete with the following items for proper installation:

- M7XC Modem Unit
- DC power mating connector
- AC to DC Adapter (Optional)
- Installation and Operation Manual plus other information on CD

2.2.1 Removal and Assembly

If using a cutting knife or blade to open the carton, exercise caution to ensure that the blade does not extend into the carton, but only cuts the tape holding the carton closed. Carefully unpack the unit and ensure that all the above items are in the carton. If the Prime AC power available at the installation site requires a different power cord/AC connector, arrangements to receive the proper power cord will be necessary prior to proceeding with the installation.

The M7XC is shipped fully assembled and does not require removal of the covers for any purpose during the installation. All hardware configuration is under software control. The type of Feature Sets, FEC Options and other configurable options are available via the Web Browser.

2.3 Mounting Considerations

When mounting the unit in an integrated terminal, adequate ventilation must be provided. The ambient temperature in the enclosure must stay between -40°C to $+60^{\circ}\text{C}$ and held constant for best equipment operation. The air available to the terminal should be clean and relatively dry. Modem units should not be placed immediately above high heat or EMF generator to ensure the output signal integrity and proper receive operation.

Do not mount the modem in an unprotected outdoor location where there is direct contact with rain, snow, wind, or sun. The modem is designed for enclosed applications that can be indoor or outdoor.

Tools and accessories required for mounting the modem:

- Six (6) 6x32 screws
- Screwdriver

The following interface connections must be available at the mounting location:

- Prime DC power
- 50Ω Transmit IF cable with SMA male connector
- 50Ω Receive IF cable with SMA male connector
- 50Ω External Reference cable with SMA male connector (Optional)
- RJ-45 cables to mate with the modem's two Ethernet interfaces

2.4 Modem Connections

Connections to the modem are located on both ends of the unit. The connector definitions in the following sections describe the connectors on the modem. Any connection interfacing to the modem must be done with the appropriate mating connector. Refer to Figure 2-1 Baseband Connections for the location of the baseband connections and Figure 2-2 IF / L-Band Connections for the IF connections.

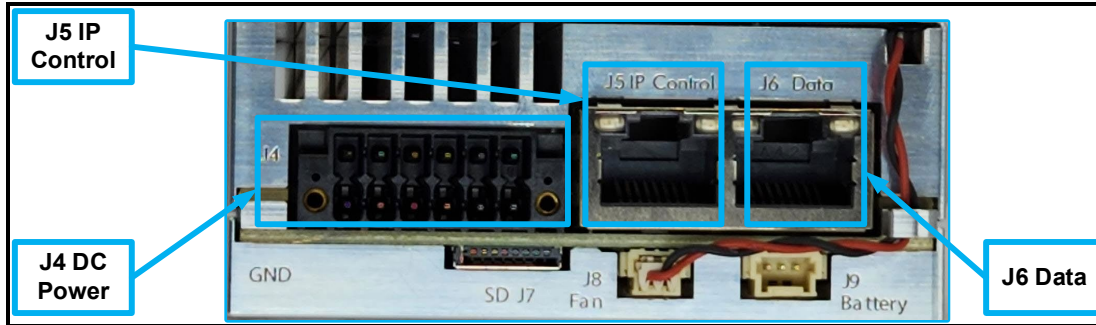


Figure 2-1 Baseband Connections

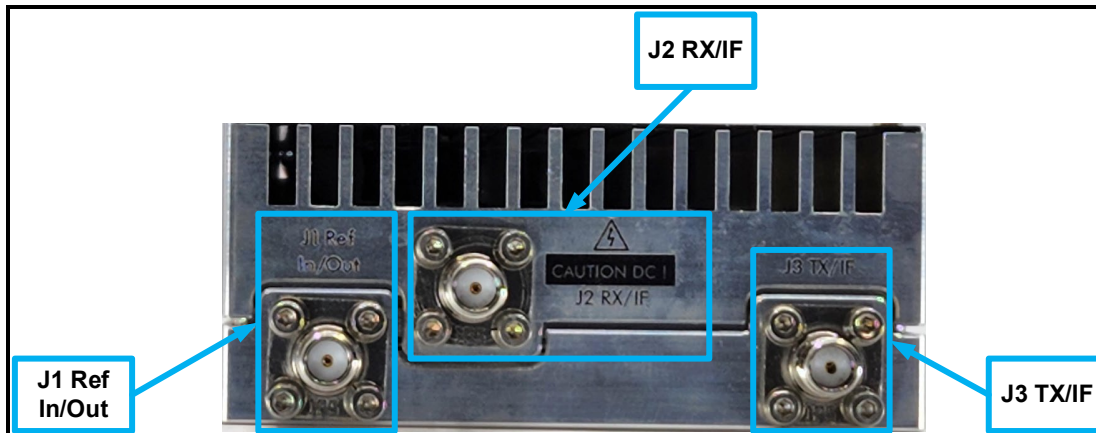


Figure 2-2 IF / L-Band Connections

2.4.1 Reference Input / Output - SMA (f) (J1)

The M7XC J1 Reference In/Out SMA (f) connector accepts 10 MHz or 50 MHz external reference for signals generated by the modem, or the modem can provide 10 MHz or 50 MHz output that is based on the internal OCXO (1 x 10⁻⁸ Stability).

2.4.2 L-Band Demodulator L-Band RX / IF Input - SMA (f) (J2)

The M7XC J2 IF Input is a 50-ohm SMA (f) connector that is L-Band from 950 to 2250 MHz.

2.4.3 L-Band Modulator L-Band TX / IF Output - SMA (f) (J3)

The M7XC J3 IF Output is a 50-ohm SMA (f) connector that is L-Band from 950 to 2250 MHz.

2.4.4 Multi-Function Connector Pinout (J4)

The multi-function connector provides pins (Figure 2-3 Multi-Function Connector Pinout J4), (Table 2-1 Multi-Function Connector Pinout (J4)) for DC Power, RS-232 Control & Monitor, and Alarm contact closures.

The M7XC input power (+10 to +36 VDC) is on Pin 2 with the Main Power Ground on Pin 1.

The modem has a configurable form-A dry contact alarm relays on Pins 3 & 4. The actuation of this alarm relay is configurable via the alarm control interfaces.

NOTE: By convention, “NO” means Normally Open which is the non-powered, Alarm State.

The analog monitor output on Pin 9 is configurable via the control interfaces to select Receive Eb/No, Receive AGC voltage, or Transmit output power.

The analog monitor input on Pin 5 is configurable via the control interfaces to select the input voltage detection.

The RS-232 control interface is on Pins 6, 8, and 10.

Pins 11 and 12 are provided to reset the IP address on the modem. Connecting these two pins will reset the IP address to the factory default setting of **192.168.0.222**.

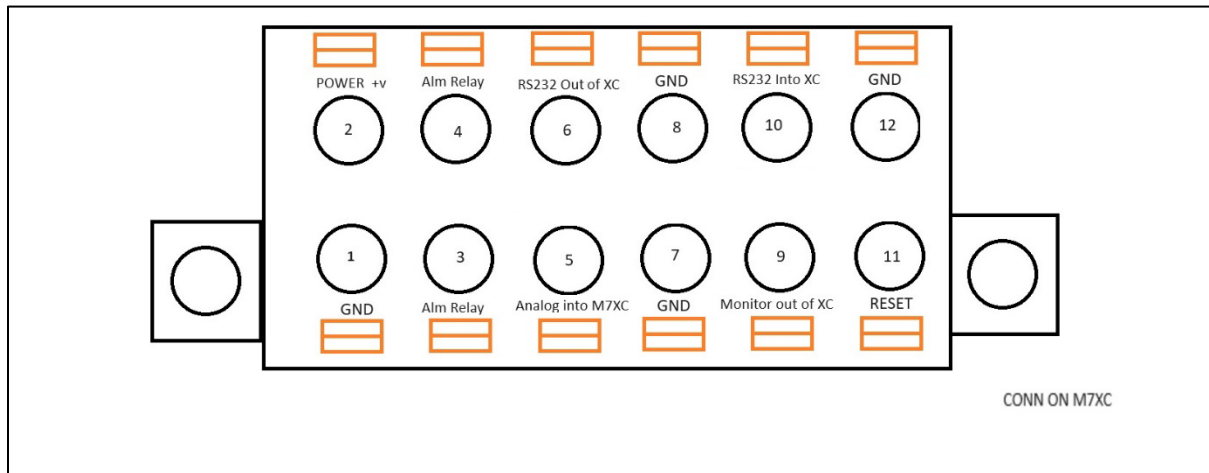


Figure 2-3 Multi-Function Connector Pinout J4

Table 2-1 Multi-Function Connector Pinout (J4)

Pin #	Name	Description	Direction
1	Ground	Main Power Ground	Ground
2	Power	Main Power +10 to +36 VDC	Input
3	Alarm	Form Relay A - NO on Alarm	
4	Alarm	Form Relay B - NO on Alarm	
5	Monitor	Analog Monitor Input 0 to +5V	Input
6	Transmit	RS-232 Transmit	Output
7	Ground	Monitor Common	Ground
8	Ground	RS-232 Common	Ground
9	Monitor	Analog Monitor Output 0 to +5V	Output
10	Receive	RS-232 Receive	Input
11	Reset	IP Reset to Factory	Input
12	Ground	IP Reset Common	Ground

2.4.5 SD Flash Slot (J7)

The M7XC provides a SD Flash memory slot for configuration and re-boot if necessary.

2.4.6 Fan Connections (J8)

The M7XC provides a connection for an internal or external fan. The M7XC modem has a fan or fan less option when ordering. The fan less option M7XC must be mounted to a cold plate or heat sink to properly function without overheating.

2.4.7 Battery Connections (J9)

The M7XC provides an input connection for an optional external Lithium battery to maintain a real time clock. All user settings and encryption keys do NOT require the external battery to retain user settings as they are stored in non-vol memory.

2.4.8 GigE IP Control RJ-45 (J5)

The 1000BaseT IP Control RJ-45 (J5) connection provides remote control of the modem thru a Web Browser page or SNMP. Refer to Table 2-2 GigE Data Interface Connector Pinout (J6) for the pinout of the RJ-45 connector.

2.4.9 GigE Data Interface RJ-45 (J6)

The 10/1001000BaseT Ethernet Data Interface RJ-45 (J6) connection provides data interface to the Local Area Network (LAN) via Layer 2 Switched Bridge traffic. Refer to Table 2-2 GigE Data Interface Connector Pinout (J6) for the pinout of the RJ-45 connector.

Table 2-2 GigE Data Interface Connector Pinout (J6)

Pin #	Signal Name	Use	Direction
1	TX D1+	Tx Data+ Pair A	I/O
2	TX D1-	Tx Data- Pair A	I/O
3	RX D2+	Rx Data+ Pair B	I/O
4	BI D3+	Bi Data+ Pair C	I/O
5	BI D3-	Bi Data- Pair C	I/O
6	RX D2-	Rx Data- Pair B	I/O
7	BI D4+	Bi Data+ Pair D	I/O
8	BI D4-	Bi Data- Pair D	I/O

2.4.10 Physical Setup

This section will describe a simple physical connection set up to get the user started.

1. Plug both of the RJ-45 cables into your modem (J5 IP Control, J6 Data)
2. Connect the Ext Ref SMA cable to your modem (J1 Ref In/Out)

3. Connect the RX SMA cable to your modem (J2 RX/IF)
4. Connect the TX SMA cable to your modem (J3 TX/IF) after verification that TX Output power is OFF
5. Plug the Multi-Function connector (DC Power) into your modem (J4)
6. The modem/fan should power on and the PWR (Blue) LED should be illuminated
7. Open Web Browser and Login into the modem using the default username and password using the directions in section 3.3
8. Configure the modem using the directions in section 3.3

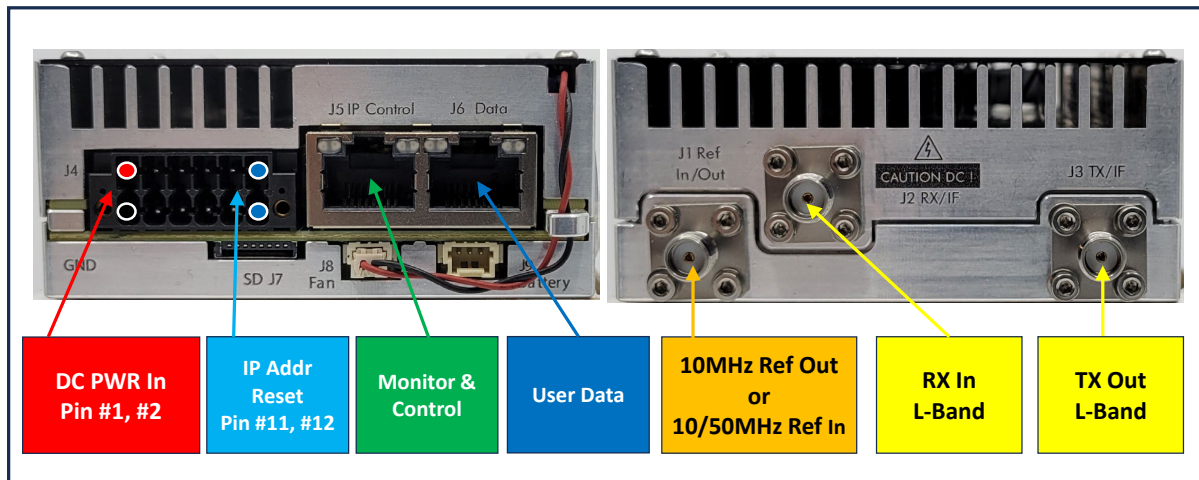


Figure 2-4 Modem Physical Connections

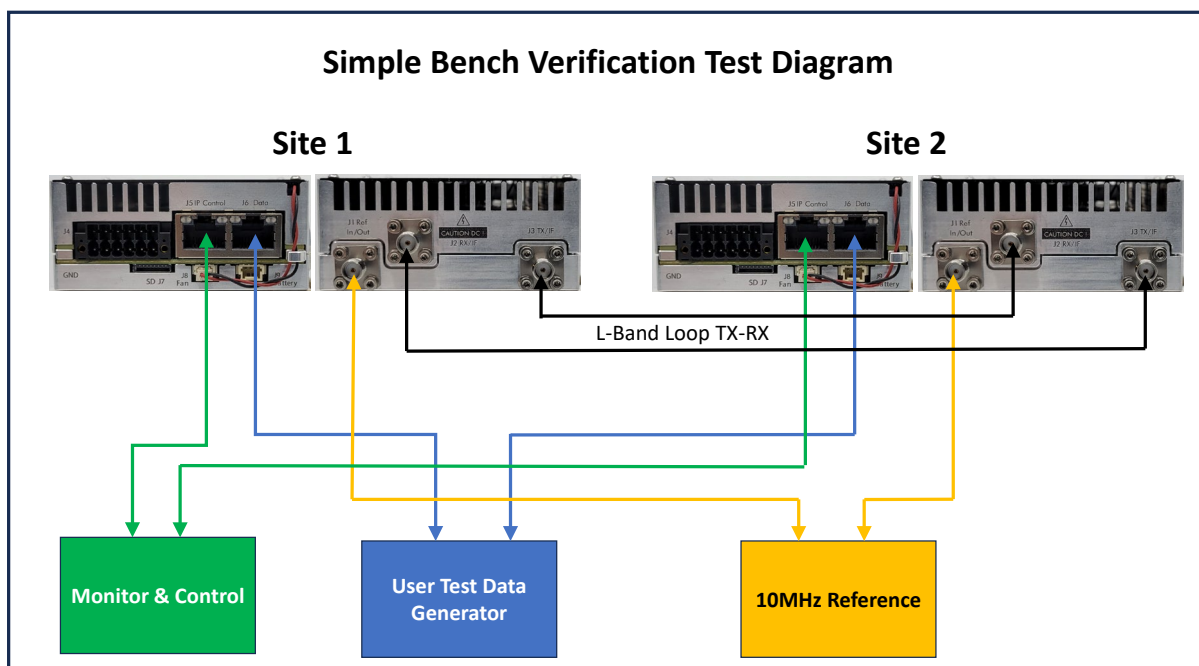


Figure 2-5 Simple Bench Verification Test Diagram

3 Operation

3.1 Indicators

There are seven (7) LED indicators (Figure 3-1 LED Indicators) on the side of the M7XC that give a quick visual indication of the status of the unit. These indicators are:

- RCV LOCK: Green – indicates the demodulator has locked to the incoming carrier
- RCV ERRS: flashing Yellow – indicates that there are frame errors in the link
- LNB PWR: Blue – when optioned, indicates the LNB Power is active on the RX IF connector (J2)
- XMT ON: Green – indicates a carrier is active on the TX IF Connector (J3)
- ALARM: Red – indicates a summary alarm for the unit
- ON-LINE: Green – indicates the on-line/off-line status when configured in redundancy. Note the LED is green if it is stand-alone.
- PWR: Blue – indicates power is applied to the unit

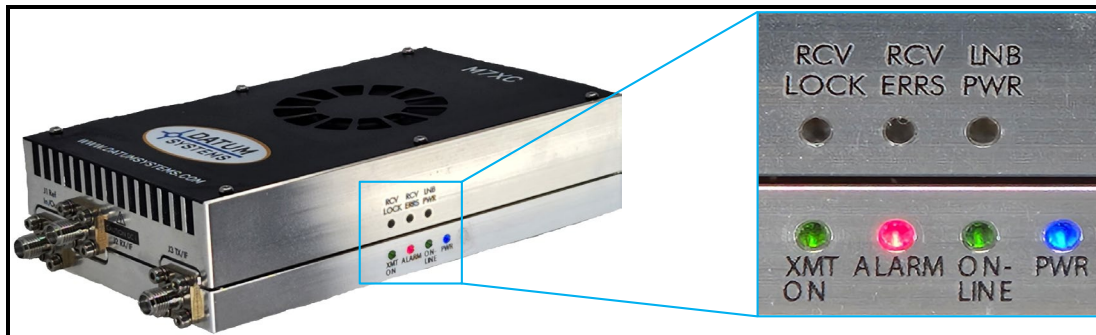


Figure 3-1 LED Indicators

3.2 Operating Procedures

Operation of the Datum Systems M7XC Compact Satellite Modem consists of controlling the unit's operating parameters, monitoring its status and responses thru one of the control interfaces. There are two possible control methods for the modem:

- Web Browser as discussed in Section 3.3.
- SNMP v1, v2c, and v3 as discussed in Appendix B SNMP Operation.

These methods may be used separately or together to monitor and control the modem. Each of these interfaces and their respective methods are discussed separately below in the sections or Appendices noted above.

Additional operating procedures are presented later in this section on using some of the unique features of the M7XC that would not normally be set-up during installation. These include items such as the storing and recalling configuration information, AUPC, the analog monitor output, redundancy, and automatic recovery.

3.3 Web Browser Operation

Initiate the M7XC Web Browser function by opening an Internet Browser (Microsoft Edge, Google Chrome, Firefox or similar) on the computer and typing in the IP address of the M7XC in the text bar of the browser. The default IP address of the M7XC modem is **192.168.0.222**.

Note: This is the *ONLY* IP address assigned to the unit when shipped from the factory.

If the IP address is unknown, the use of a program like Wireshark may help in finding the IP address. Power cycle the M7XC modem and look for the ARP message that indicates the IP address of the M7XC modem. At that point, set the computer IP address in the same address range. Open the Web Browser, set the unique IP address using the Unit page > IP Control tab > IP address. All IP Control interface parameters can be set in this tab. **Alternatively, the IP address may be RESET to the factory default by shorting pins 11 and 12 together on the multi-function connector (J4) for at least 30 seconds.**

After boot-up, the operator will be prompted (Figure 3-2 User Login) to enter the username and password. The default Username is **Admin**, and the default Password is **Datum**.

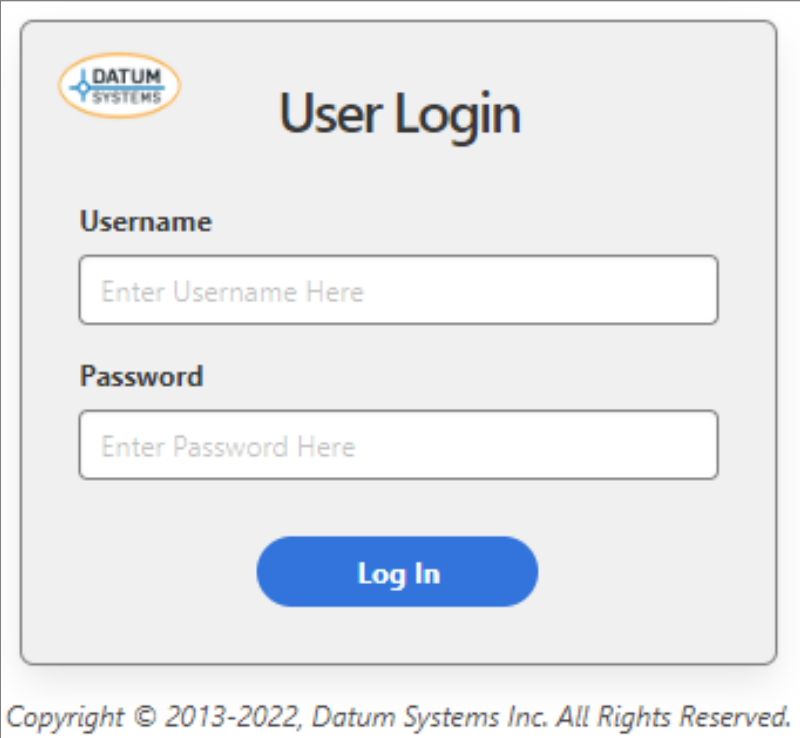


Figure 3-2 User Login

NOTE: Login security may be configured in the Web Browser >Unit >HTTP page. Available configuration items are Full Access Name, Full Access Password, Read Access Name, Read Access Password. Once this parameter is active, the operator will be prompted to enter the username and password.

3.3.1 Page Sections

There are four (4) main sections on all M7XC web pages (refer to Figure 3-3 Web Browser Sections).

- Tabs
- Status
- Page Menu
- Parameter Window

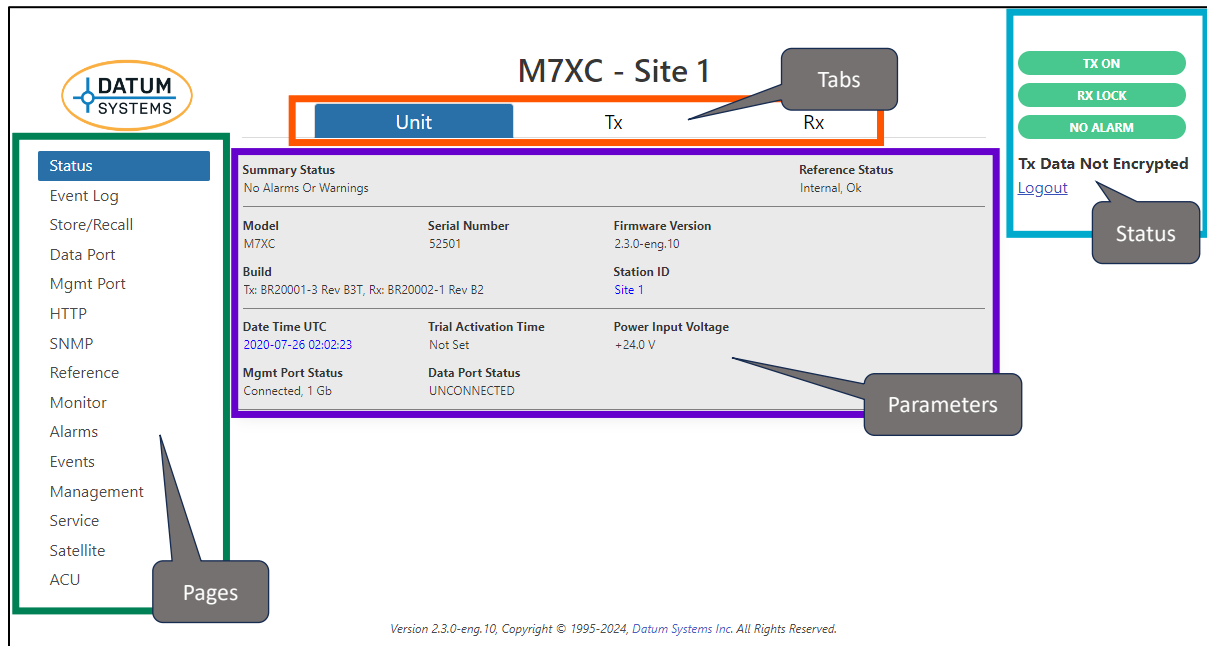


Figure 3-3 Web Browser Sections

Throughout this manual, reference to Web Browser sections will follow the following designation. >Tab >Page >Parameter. For example, if the user is looking for the Station ID the designation would be >Unit >Status >Station ID. The setting of the parameter would be identified in “quotes” ex: >Unit >Status >Station ID “Site 1”.

The main Window provides the user a summary status of the M7XC. Active TX, RX and Alarm functional status are shown with color-coded text (Green, Yellow, Red). The Alarm section on the right screen will display the current statuses of the (Unit, TX, or RX).

Clicking on Logout will revert back to the User Login page.

The Tab Menu (Figure 3-4 Web Browser Tab Menu) allows the operator to navigate between the modem functions - Unit, TX, and RX. The Tab Window is divided into pages for the desired function.

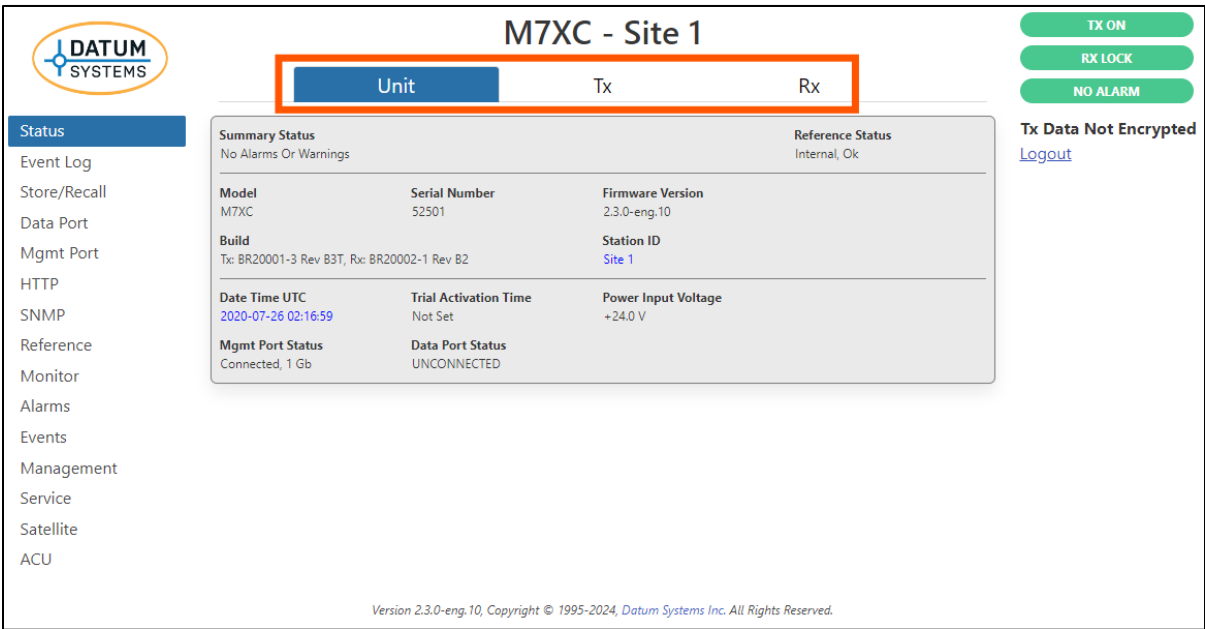


Figure 3-4 Web Browser Tab Menu

The page selection (Figure 3-5) is the section of the M7XC Web Browser that allows detailed status and modification of every parameter of the M7XC. The M7XC Web Browser allows the user to make all the desired changes and execute them at one time. When all of the changes are done, the operator must click the "Set Config" button for any change to be accepted. Clicking on the "Cancel" button or closing the browser will cancel any pending changes.

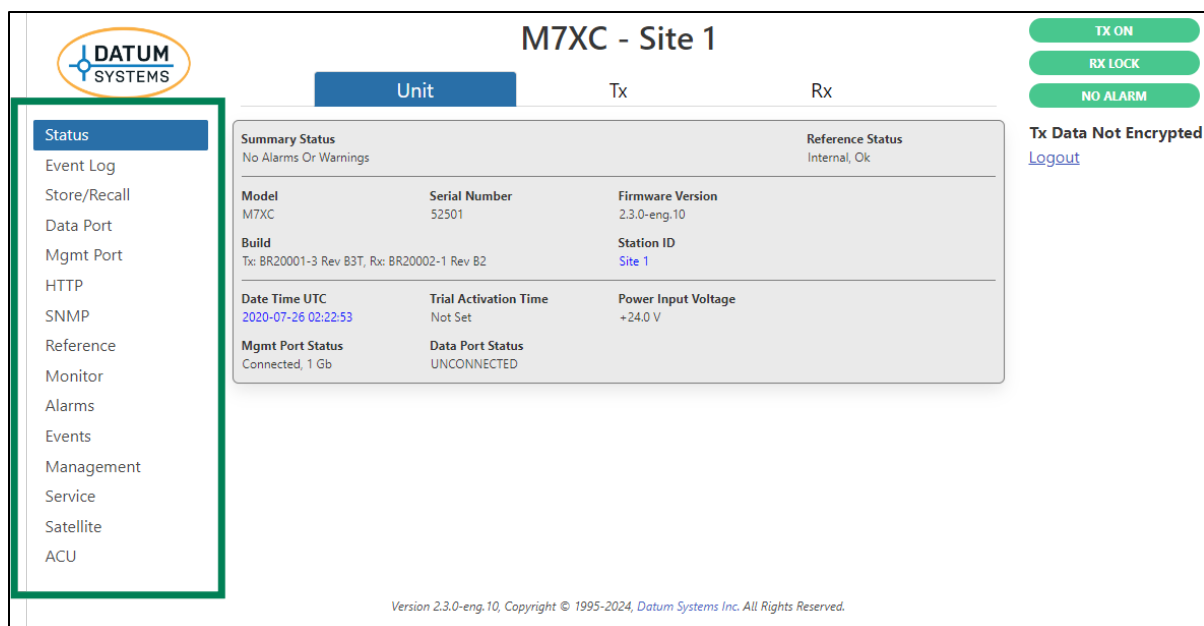


Figure 3-5 Page Selection

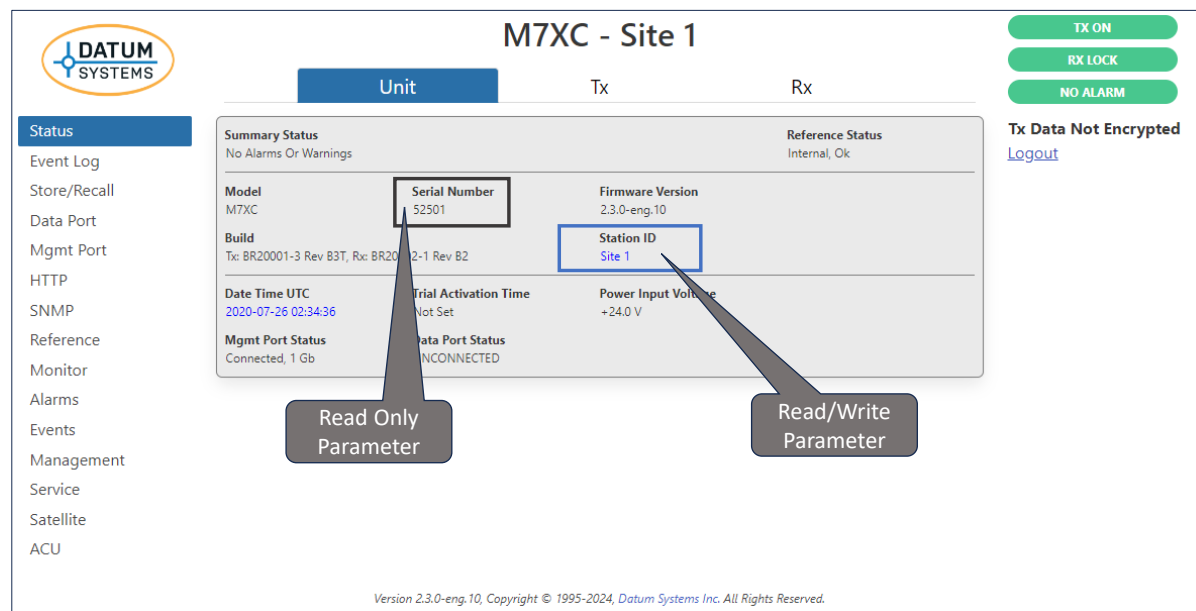


Figure 3-6 Read/Write and Read Only Parameters

User parameters can be read/write or read only. A user parameter shown in blue can be overwritten by manual entry or a drop-down window (parameter dependent). All items in grey are read only parameters and are status only.

3.4 M7XC Web Browser Unit Tab and Pages

3.4.1 >Unit >Status Page

The Unit Status page (Figure 3-7 Unit Status Page) will be the first page that is displayed when the User logs in. The Unit Status page provides the user a snapshot of the current M7XC status. The status window displays the Summary, Reference, WAN & IP Control Port Status, Voltage and System information. The status window also displays Date & Time and Station ID that can be edited accordingly.

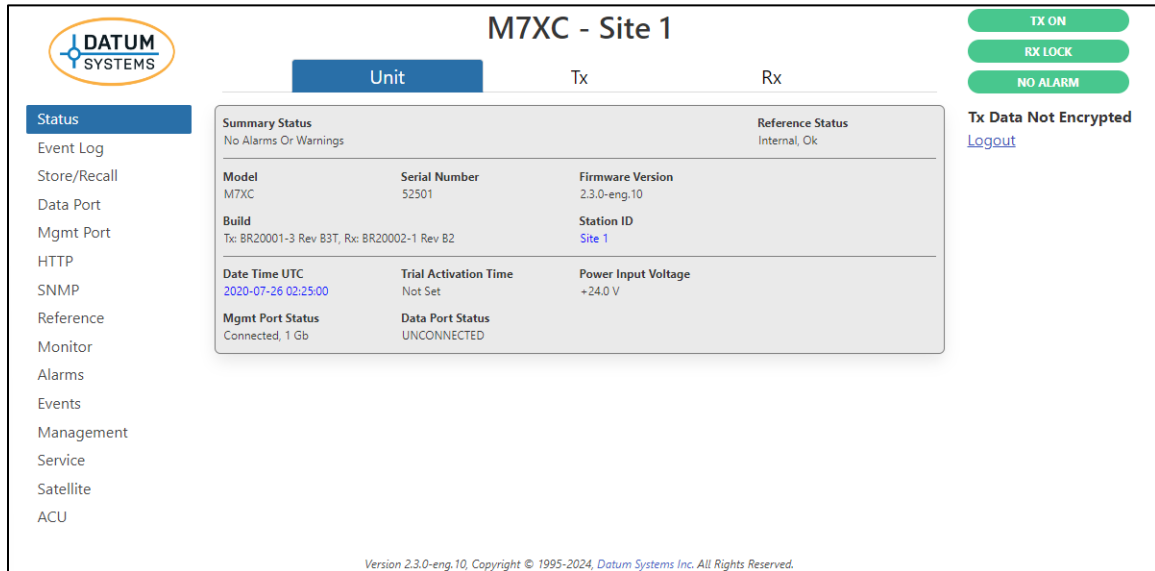


Figure 3-7 Unit Status Page

The Unit “Status” page (Figure 3-7 Unit Status Page) is used to configure the common parameters of the M7XC. The Date Time and Station ID parameters are Set in the “Unit” page.

- **Summary Status:** Displays current status of the modem
- **Reference Status:** Displays current status of the modem reference
- **Model:** Displays model of the unit you are connected to
- **Serial Number:** Serial number of the unit assigned at Datum’s Factory
- **Firmware Version:** Current running FW on modem
- **Build:** Hardware version numbers of major HW components
- **Station ID:** User entry / name of modem or site
- **Date Time UTC:** User entry date / time. NOTE: If you do not have a battery connected to the modem the date will reset when power is lost. See section 2.4.7 Battery Connections (J9)
- **Trial Activation Time:** Displays the remaining time of the trial features in the modem if a trial activation key has been entered
- **Power Input Voltage:** Displays prime input power to the modem

- **Management Port Status:** Displays the connection status and speed of the management port (J5)
- **Data Port Status:** Displays the connection status and speed of the data port (J6)

3.4.2 >Unit >Event Log

The Unit Event Log will log all UNIT, TX and RX parameters, parameter changes and status of the modem per the users programming. Virtually all status and configuration changes can be logged to the event log and exported to a file for offline storage or analysis. See Figure 3-8 Unit Event Log.

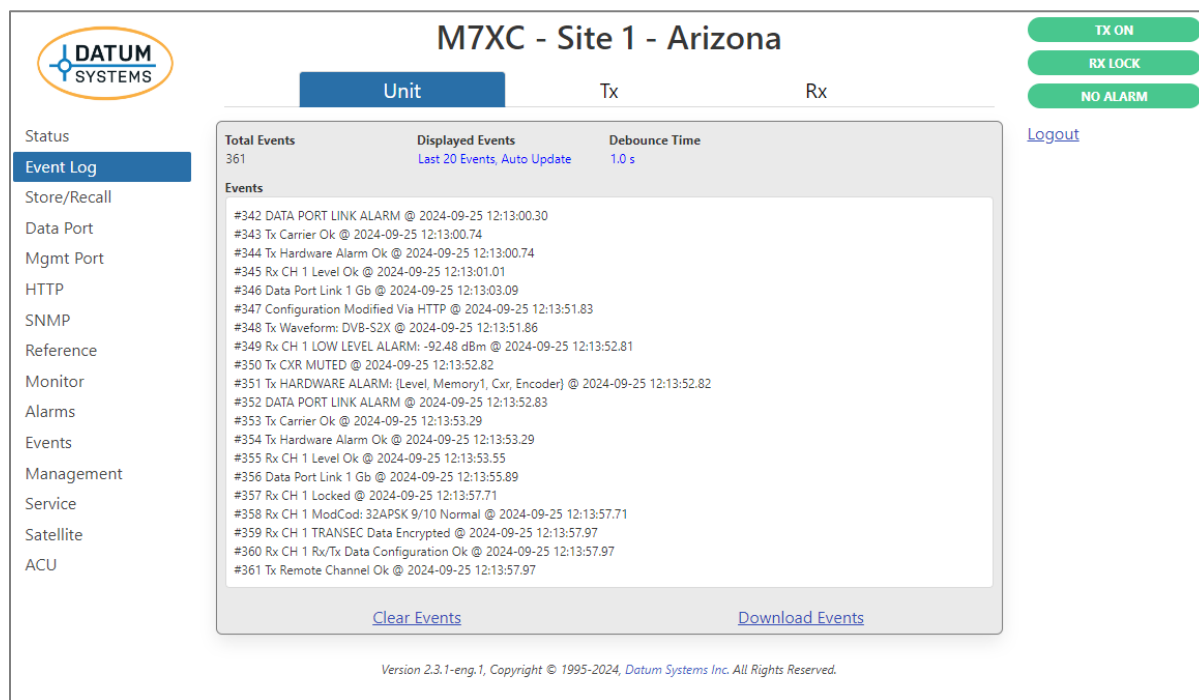


Figure 3-8 Unit Event Log

- **Total Events:** Number of events recorded since last “Clear Events” reset. NOTE there is a maximum of 2000 stored events available before the buffer is full.
- **Displayed Events:** Selects how many events to show on the GUI and if the events should continually update or be frozen until manually updated
- **Debounce Time:** User selectable between 0.5 and 600 seconds. This is the delay before an event is displayed.
- **Clear Events:** Clears all events in the buffer
- **Download Events:** Downloads an event log to a *.txt file for offline view / save.

3.4.3 >Unit >Store/Recall

The Unit “Store/Recall” tab (Figure 3-9 Unit Store/Recall) provides the ability to save known M7XC modem configurations to be recalled later.

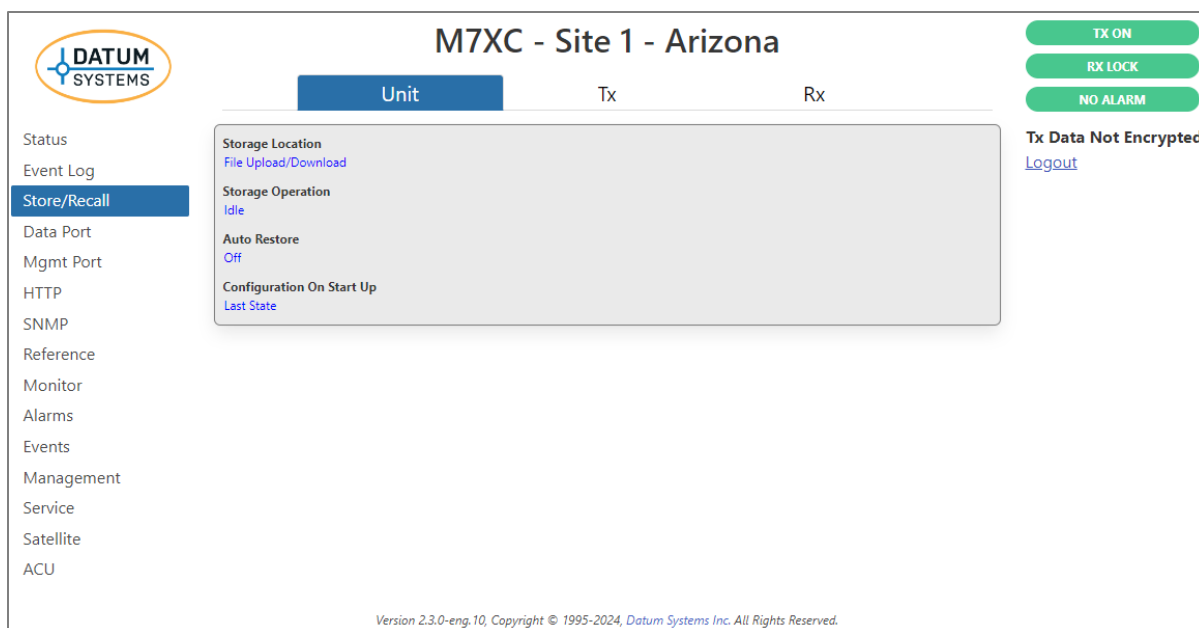


Figure 3-9 Unit Store/Recall

- **Storage Location:** The configuration file can be stored or recalled from up to 99 locations saved within the modem itself or to a file on a PC. The selection of "File Upload/Download" will save or recall a configuration from a file on a PC.
- **Storage Operation:** Set operation of the modem to perform:
 - Idle: do nothing
 - Recall All: Load all user parameters from the saved location
 - Recall Unit Only: Load only the >Unit parameters and overwrite
 - Recall TX/RX Only: Load only the >TX and >RX parameters and overwrite
 - Recall TX Only: Load only the >TX parameters and overwrite
 - Recall RX Only: Load only the >RX parameters and overwrite
 - Store: Save current configuration to the location in "Storage Location"
- **Auto Restore:** Set a configuration based on loss of Demod carrier (Off, Restore All, Restore Unit Only, Restore TX & RX Only, Restore TX Only, Restore RX Only). Default is "Off" where the unit will not change the parameters or reload the modem on loss of RX lock. Auto Restore can be used to daisy-chain multiple configurations in a mobile or multi-beam or multi-satellite environment. It can be configured to attempt to acquire "N" different configurations before re-starting the initial acquisition attempt. To see more information on this configuration see 5.9 Appendix I: M7XC Demod Unlock Daisy Chain Operation
- **Configuration On Start-Up:** Behavior on power-up (Default = Last State). You can select a different load configuration on startup from up to the 99 saved locations. This can be useful if the modem is in a roaming environment and may have unknown settings on startup.

3.4.4 >Unit >Data Port

The Unit “Data Port” tab (Figure 3-10 Unit Data Port Figure 3-9 Unit Store/Recall) provides the ability to configure the user data port.

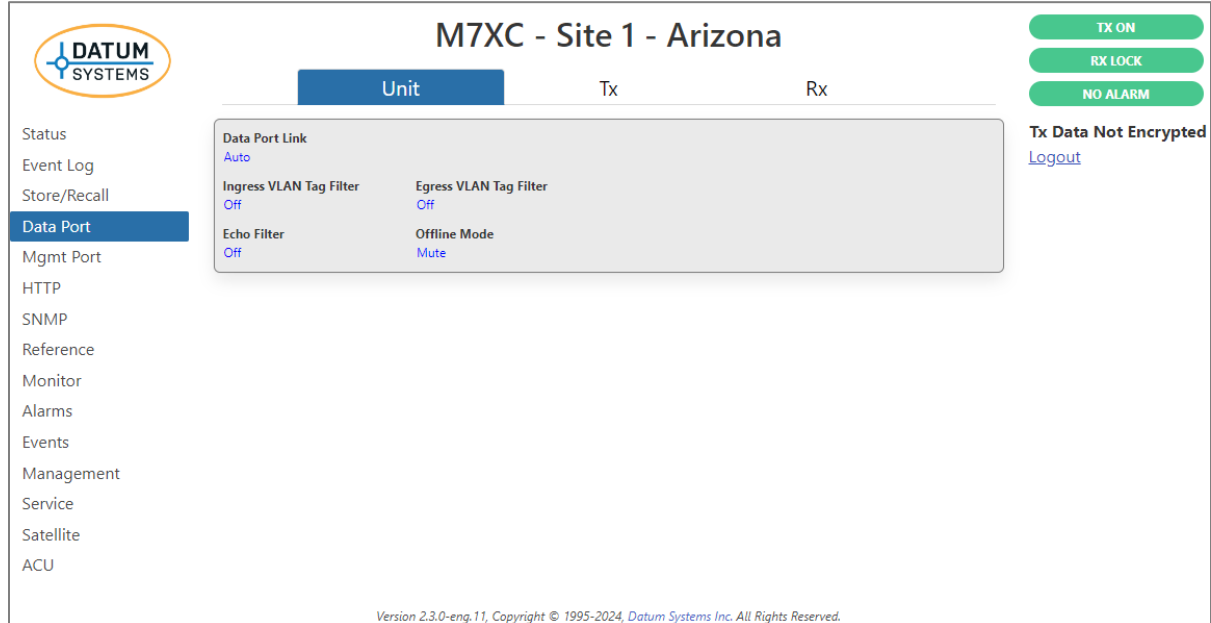


Figure 3-10 Unit Data Port

- **Data Port Link:** Select the link speed of the Ethernet data port of the modem. Selections are 10 Mb, 100 Mb, 1 Gb. Default = “Auto”
- **Ingress VLAN Tag Filter:** Set operation of the modem in case of VLAN operation.
 - Off: do nothing, all VLAN tags are passed onto the WAN / Satellite
 - Pass Match, Strip Tag: Only traffic with VLAN tags that match the “VLAN ID” are accepted. Once accepted, the matching outmost VLAN tag is removed before the Ethernet frame is sent onto the WAN / Satellite
 - Pass, Append Tag: All traffic is accepted. Once accepted, a VLAN tag with the user defined “VLAD ID” is pushed onto the Ethernet frame before the Ethernet frame is sent onto the WAN / Satellite
 - Pass Match, Pass Tag: Only traffic with VLAN tags that match the “VLAN ID” are accepted. Once accepted, the Ethernet frame is sent onto the WAN / Satellite
 - Pass Range Match, Pass Tag: Only traffic with VLAN tags that are within the user defined “VLAN Tag Range Start” and “VLAN Tag Range End” are accepted. Once accepted, the Ethernet frame is sent onto the WAN / Satellite
- **Echo Filter:**

- “Off” = pass all ingress traffic on the LAN Data Port to the WAN / Satellite.
- “Learn Tx, Filter Rx” = learn the destination MAC of all ingress traffic on the LAN Data Port and do not allow the learned MAC addresses to egress the modem’s LAN port from the WAN / Satellite with the learned MAC destination addresses. This can prevent Ethernet loops. Note: all learned address will age out in 2 seconds covering most satellite loops. There are 1023 address the modem can learn within the 2 second time frame.
- *Offline Mode*: When connected in a 1:1 redundancy configuration “Mute” will keep the link active but no traffic will egress the modem. “Unlink” will appear to any connected device as if the Ethernet connection is down or disconnected.

3.4.4.1 >Unit >Data Port (In “Admin” Mode)

NOTE: When the modem is set for “Admin” mode ON under >Unit>Management. The >Unit>Data Port GUI will look slightly different allowing for the user to set the modem into different modes of data loopback. The following additional configuration settings are enabled while in “Admin” mode.

The reason for the extra precaution is that most switches and routers will NOT ACCEPT the same traffic coming from a physical port that the data was sent on. The traffic will get dropped or possibly cause the switch or router to hang-up. For this reason, placing the modem in “Admin” mode is used as a special precaution for this test mode.

It may be necessary to perform an Ethernet Loopback on a Datum M7XC satellite modem for test purposes or to test the integrity of the Modem’s Ethernet PHY and interface connector. To perform this test, an operator can log into the modem as an administrator and set the following user configuration.

User Configuration:

Goto >Unit>Management and set the “Admin” selection as “On”

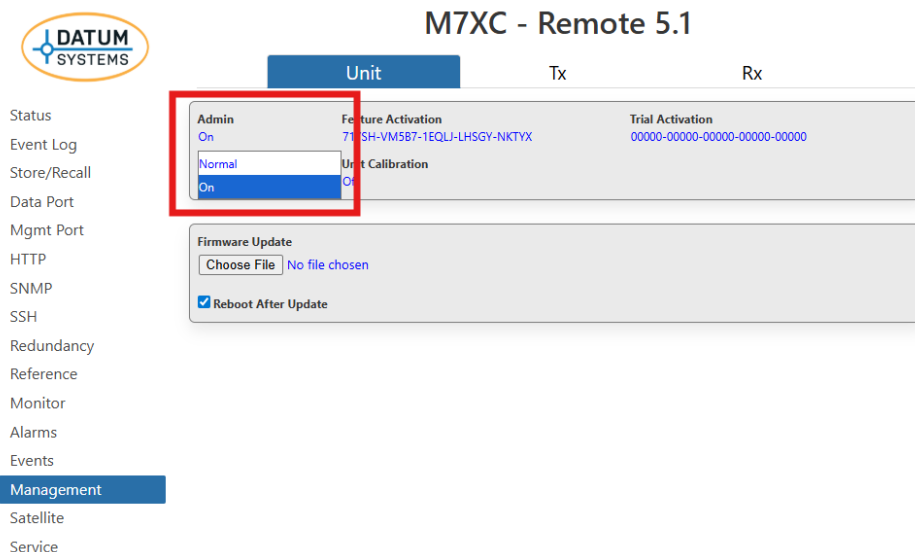


Figure 3-11 >Unit>Data Port (In "Admin" mode)

Once in Admin mode there are two loopback settings that are allowable: 1) Ingress/Egress Loopback 2) Rx/Tx Internal Data Loopback. These settings are found under >Unit>Data Port. NOTE: These fields are NOT SEEN if you are not in Admin Mode (see above).

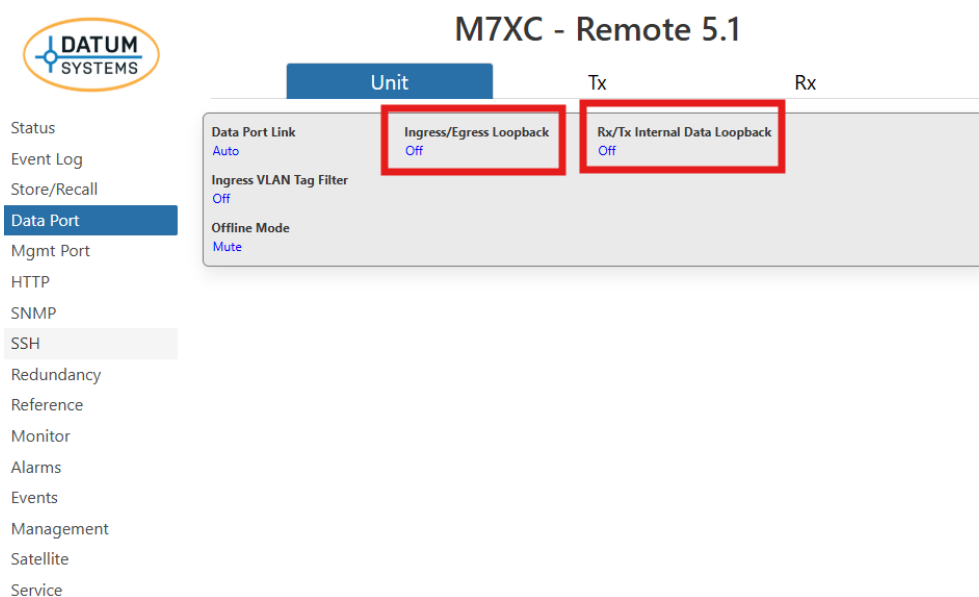


Figure 3-12 >Unit>Data Port Loopback Modes

The loopback modes are available independently and can be set one by one or both simultaneously. The Ingress/Egress Loopback is similar to a terrestrial loopback in that all traffic sent to the modem on the data port will be returned from the modem to the LAN on the same Data Port. The Rx/Tx Internal Data Loopback acts similar to a RF loopback but occurs after the demodulator, decoder and de-framer so it must be valid data. See the below block diagram that explains the loopback status.

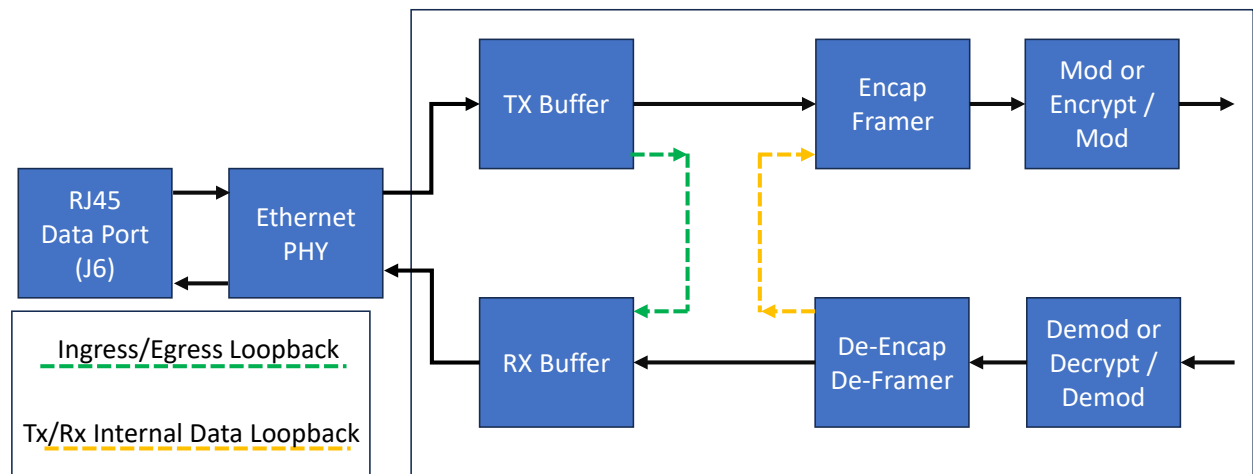


Figure 3-13 Loopback Operation Block Diagram

3.4.5 >Unit >Mgmt Port

The Unit “Mgmt Port” tab (Figure 3-14 Unit Mgmt Port) provides the ability to set the IP access information for the Web Browser and the M&C settings.

Figure 3-14 Unit Mgmt Port

- **DHCP:** Set DHCP access to On or Off (Default = Off). A setting of “Off” requires that the user set the IP address of the modem directly. If DHCP is set to “On” the modem will get it’s IP address from a DHCP server that it is directed to.
- **Address:** When DHCP is set to “Off” enter the unique Unit IP Address information the format is IPv4 IP address xxx.xxx.xxx.xxx/xx

- **VLAN ID:** Optional setting, select VLAD ID of the modem from 2 - 4094
- **Gateway:** Optional setting of gateway router if management PC is not on the same subnet of the modem. Format is IPv4 IP address xxx.xxx.xxx.xxx
NOTE: The gateway router must be on the same subnet as the modem
- **MCC Bridge:** Set this to “On” to enable in-band communications between local and remote modems and to enable in-band functions such as ACM and AUPC. A selection of “Off” will disable in-band communications. See section TX>MCC to set the in-band communications overhead DR.
- **Bridge IP Filter:** WHEN >Unit>Mgmt Port>MCC Bridge is set to “On” A selection of “Off” in the Bridge IP Filter will allow all IP traffic sent to the Mgmt Port of the modem to be sent over the WAN / satellite link. Note: you will not see the user selection of “Bridge IP Address Range Start” or “Bridge IP Address Range End”. A selection of “On” allows the user to filter destination IP addresses that are allowed to be sent over the WAN / satellite link to only the addresses that fall within the “Bridge IP Address Range Start” and “Bridge IP Address Range End”.
- **Bridge IP Address Range Start:** The start of the allowable destination IP addresses that the modem will allow to be sent over the WAN when ‘MCC Bridge’ is set to “On”
- **Bridge IP Address Range End:** The end of the allowable destination IP addresses that the modem will allow to be sent over the WAN when ‘MCC Bridge’ is set to “On”
- **MAC Address:** Displays unit MAC Address of the management port

3.4.6 >Unit >HTTP

The Unit “HTTP” tab (Figure 3-15 Unit HTTP) provides the ability to set the access control for the Web Browser.

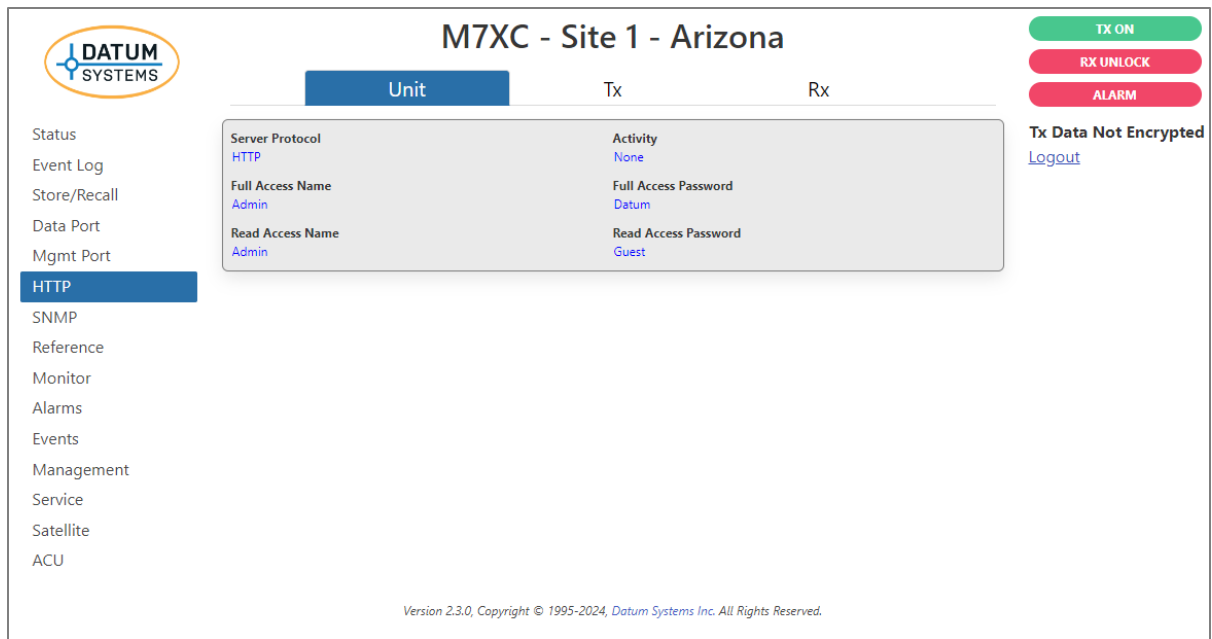


Figure 3-15 Unit HTTP

- **Server Protocol:** Control access mode to the Web Browser (HTTP, HTTPS)
- **Activity:** Set the side “Online LED” to blink when there is HTTP port activity (None, Flash Online)
- **Full Access Name:** HTTP full access username (Factory Default = Admin)
- **Full Access Password:** HTTP full access password (Factory Default = Datum)
- **Read Access Name:** HTTP read only access username (Factory Default = Admin)
- **Read Access Password:** HTTP read only access password (Factory Default = Guest)

3.4.7 >Unit>SNMP

The Unit “SNMP” tab (Figure 3-16 Unit SNMP) provides the ability to set the parameters for the SNMP Agent.

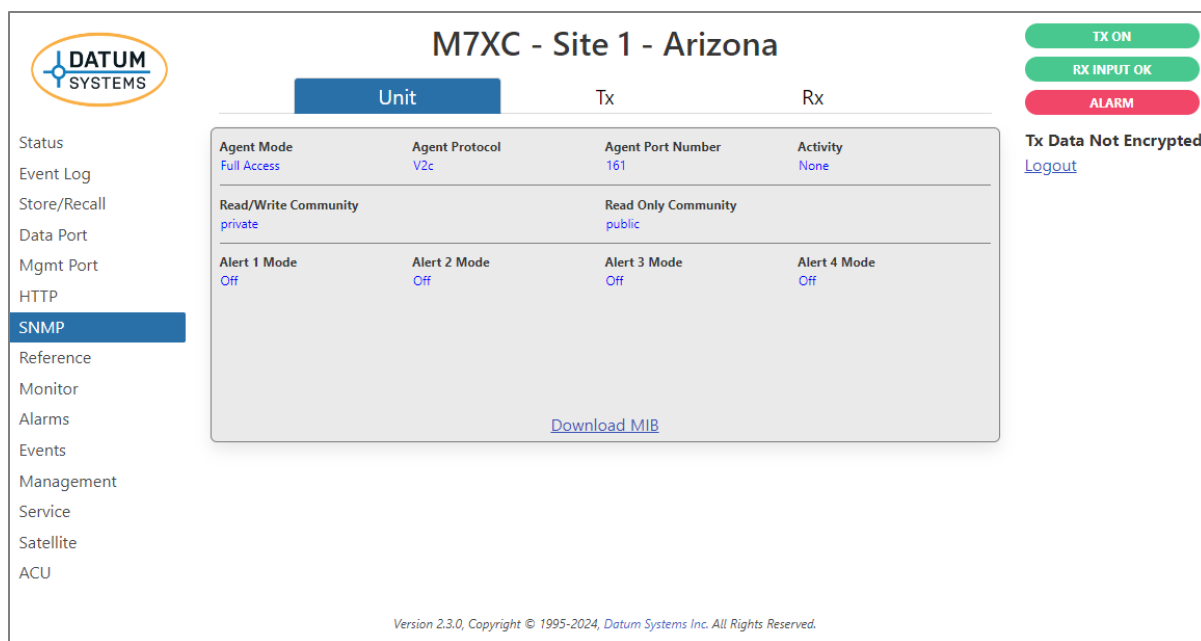


Figure 3-16 Unit SNMP

- *Agent Mode*: Control access and mode to the SNMP Agent (Off, Full Access, Read Only)
- *Agent Protocol*: Set the SNMP Server type (V1, V2c)
- *Agent Port Number*: Set the Internet socket port number for the SNMP server (default = 161)
- *Activity*: Set the side “Online LED” to blink when there is HTTP port activity (None, Flash Online)
- *Read-Only Community*: Set the Read-Only community name for authentication for SNMP traffic (default = public)
- *Read/Write Community*: Set the Read/Write community name for authentication for SNMP traffic (default = private)
- *Alerts*: Send notifications or information to servers and communities.

NOTE: For full detail on SNMP configuration and operation please see “M7XC Appendix B SNMP Operation”.

3.4.8 >Unit >Reference

The Unit “Reference” tab (Figure 3-17 Unit Reference) provides the ability to set the parameters for the system reference on J1.



Figure 3-17 Unit Reference

- **Source:** Set Frequency and SR clock source.
 - Internal, Ref Out Off: Internal reference is used by the modem with no output on J1 SMA reference.
 - Internal, Ref Out On: Internal reference is used by the modem. Modem will also output (10MHz or 50MHz) reference signal on J1 SMA.
 - External Input: External reference on J1 SMA is required by the modem. Reference can be either (10MHz or 50MHz) on J1 SMA.
- **Frequency:** The output or input reference frequency depending on “Source” selection above. Options are 10MHz or 50MHz.
- **Fine Tune:** When Internal Reference is selected, Internal reference fine adjustment. Only available if “Source” is set to internal reference (-/+2000, 0 = Calibration Value)

3.4.9 >Unit >Monitor

The Unit “Monitor” tab (Figure 3-18 Unit Monitor) can monitor and supply analog voltage inputs and outputs to the modem for special purposes. The modem can measure an input of 0.0 to 5.0VDC and output a voltage associated with the strength of various signals. See Figure 2-3 Multi-Function Connector Pinout J4. This can be useful for antenna controllers or 3rd party equipment that requires an analog voltage for modem status.

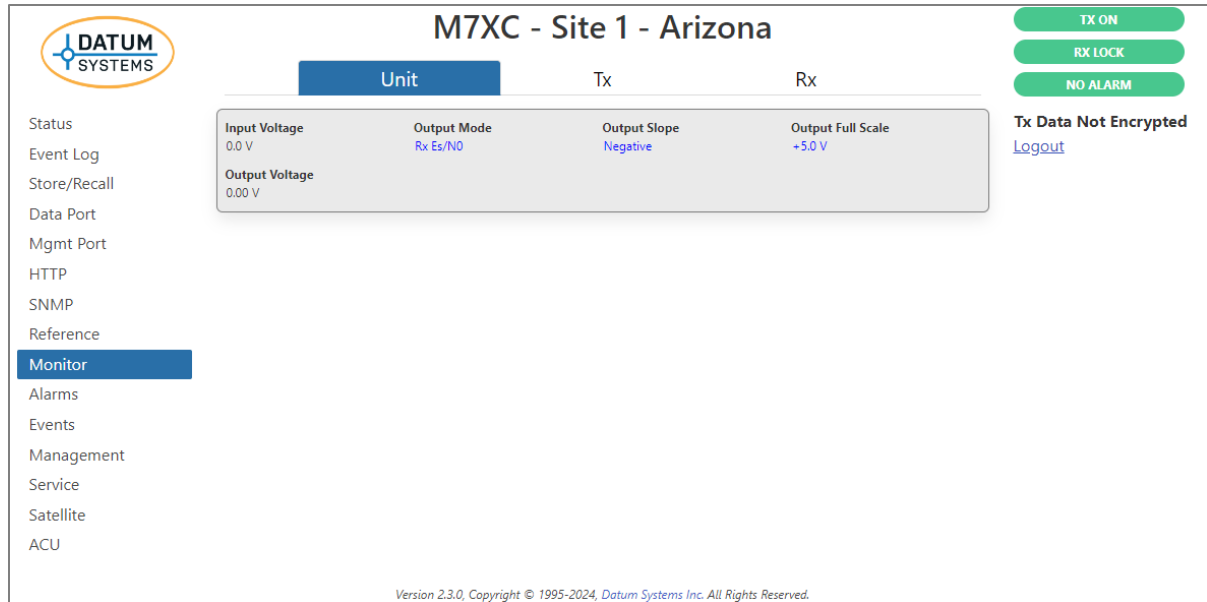


Figure 3-18 Unit Monitor

- **Input voltage:** The voltage reading by the modem as measured between GND and Pin #5 on J4. (Acceptable levels are 0.0 to 5.0 VDC)
- **Output Mode:** Selects source of analog output (Off, TX Level, TX ACM, RX Level, RX Es/No)
- **Output Slope:** Slope selection determines if the increase in “Output Mode Source” will increase or decrease the voltage between 0.0VDC and the “Output Full Scale” voltage.
- **Output Full Scale:** Selectable between +1.0 and +5.0 VDC. Will determine what the maximum output voltage is in association with the max level of the Output Mode.

3.4.10 >Unit >Alarms

Different parameters of the modem can be configured to set or clear an alarm. Alarms can be configured to perform multiple functions or combinations of functions. Below is a list of the possible functions of an alarm:

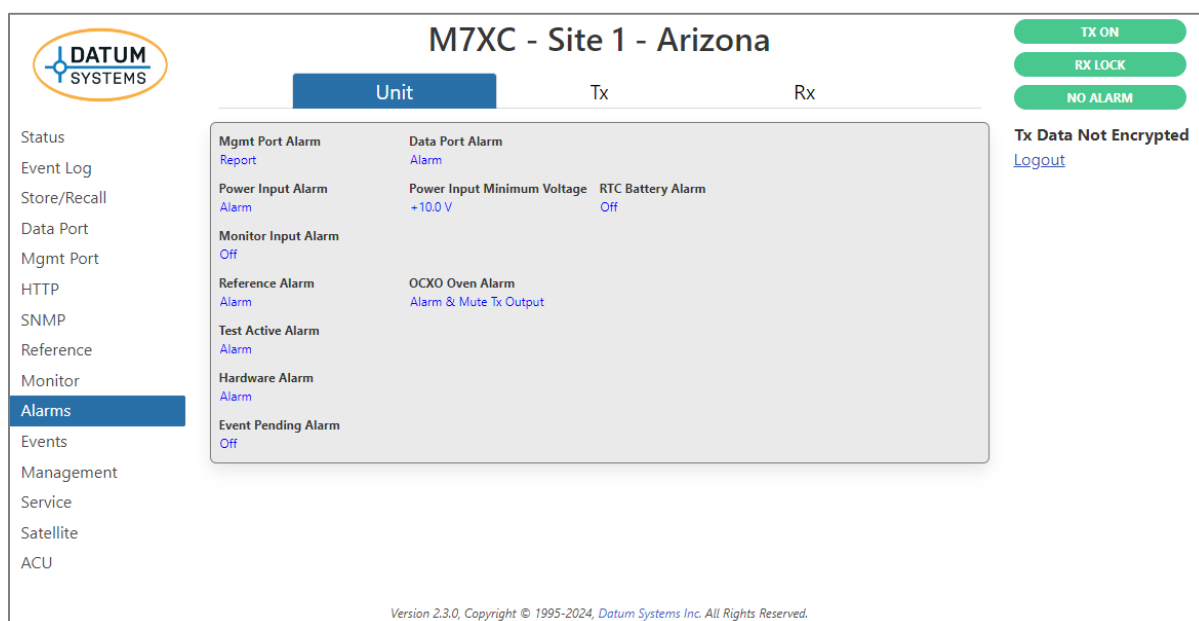
Table 3-1 Alarm Settings Description

Alarm Setting	Description when Active
Off	Will not impact modem GUI status (top right corner of GUI), will not log Alarm to the Event Log, will not send a SNMP trap, notification or inform, will not change the modem TX operation, will not cause 1:1 redundancy switchover, will not cause form C contact closure if configured
Report	Will not impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log , will not send a SNMP trap, notification or inform, will not change the modem TX operation, will not cause 1:1 redundancy switchover, will not cause form C contact closure if configured
Report, Mute Tx Output	Will not impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log , will not send a SNMP trap, notification or inform, will change the modem TX operation , will not cause 1:1 redundancy switchover, will not cause form C contact closure if configured
Alarm	Will impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log, will send a SNMP trap, notification or inform , will not change the modem TX operation, will cause 1:1 redundancy switchover , will not cause form C contact closure if configured
Alarm & Mute Tx Output	Will impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log, will send a SNMP trap, notification or inform, will change the modem TX operation, will cause 1:1 redundancy switchover , will not cause form C contact closure if configured
Alarm & Relay Output	Will impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log, will send a SNMP trap, notification or inform , will not change the modem TX operation, will cause 1:1 redundancy switchover, will cause form C contact closure if configured
Alarm & Relay Output & Mute Tx Output	Will impact modem GUI status (top right corner of GUI), will log Alarm to the Event Log, will send a SNMP trap, notification or inform, will change the modem TX operation, will cause 1:1 redundancy switchover, will cause form C contact closure if configured

Table 3-2 Alarm Settings Comparison

Alarm Setting	Event Log	Status GUI	SNMP Trap	TX Mute	1:1 Switch	Form C
Off						
Report	✓					
Report, Mute Tx Output	✓			✓		
Alarm	✓	✓	✓		✓	
Alarm & Mute Tx Output	✓	✓	✓		✓	
Alarm & Relay Output	✓	✓	✓		✓	✓
Alarm & Relay Output & Mute Tx Output	✓	✓	✓	✓	✓	✓

Unit Alarms pertain the modem as a whole. The following are a list of the unit alarms that can be triggered as shown in Figure 3-19 Unit Alarms. Note that not all Unit Alarms can have all options for Alarm settings.



M7XC - Site 1 - Arizona

Unit Tx Rx

Alarms

- Mgmt Port Alarm: Report
- Data Port Alarm: Alarm
- Power Input Alarm: Alarm
- Power Input Minimum Voltage: +10.0 V
- RTC Battery Alarm: Off
- Monitor Input Alarm: Off
- Reference Alarm: Alarm
- OCKO Oven Alarm: Alarm & Mute Tx Output
- Test Active Alarm: Alarm
- Hardware Alarm: Alarm
- Event Pending Alarm: Off

TX ON
RX LOCK
NO ALARM

Tx Data Not Encrypted
[Logout](#)

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Figure 3-19 Unit Alarms

- **Management Port Alarm:** Triggers if an Ethernet connection is not detected on the Management Port J5
- **Data Port Alarm:** Triggers if an Ethernet connection is not detected on the Data Port J6
- **Power Input Alarm:** Triggered if the power input is lower than the user selection of Power Input Minimum Voltage
- **Power Input Minimum Voltage:** User selection between +10VDC and +36VDC
- **RTC Battery Alarm:** Triggered if the Real Time Clock battery is low or missing

- **Monitor Input Alarm:** If selection is other than “Off” will be triggered with the Monitor Input Sense and Monitor Input Threshold are met.
 - Monitor Input Sense: Active High or Active Low (triggers at voltage above or below Monitor Input Threshold)
 - Monitor Input Threshold: User setting from 0.5 to 50.0 VDC as measured on J4 Pin 5 and GND
- **Reference Alarm:** Triggers when an external reference is selected but missing or out of range
- **OCXO Oven Alarm:** Triggers when the Internal Oscillator is out of spec
- **Test Active Event:** Triggers when the modem is in a test mode
- **Hardware Alarm:** Triggers when hardware fault is detected
- **Event Pending Alarm:** Triggers when event buffer is full

3.4.11 >Unit >Events

Events are less critical than alarms but may provide valuable information to the user. Unit events can be logged, traps sent or status ignored depending on the Event setting. The following are the Event setting options:

Table 3-3 Event Settings Description

Event Setting	Description when Active
Do Not Log	An Event will not be logged into the Event Log, an alert or trap will not be sent after the debounce time, an alert or trap will not be sent immediately
Log	An Event will be logged into the Event Log , an alert or trap will not be sent after the debounce time, an alert or trap will not be sent immediately
Log & Send Alert	An Event will be logged into the Event Log, an alert or trap will be sent after the debounce time , an alert or trap will not be sent immediately
Log & Send Alert Now (No Debounce)	An Event will be logged into the Event Log, an alert or trap will be sent immediately

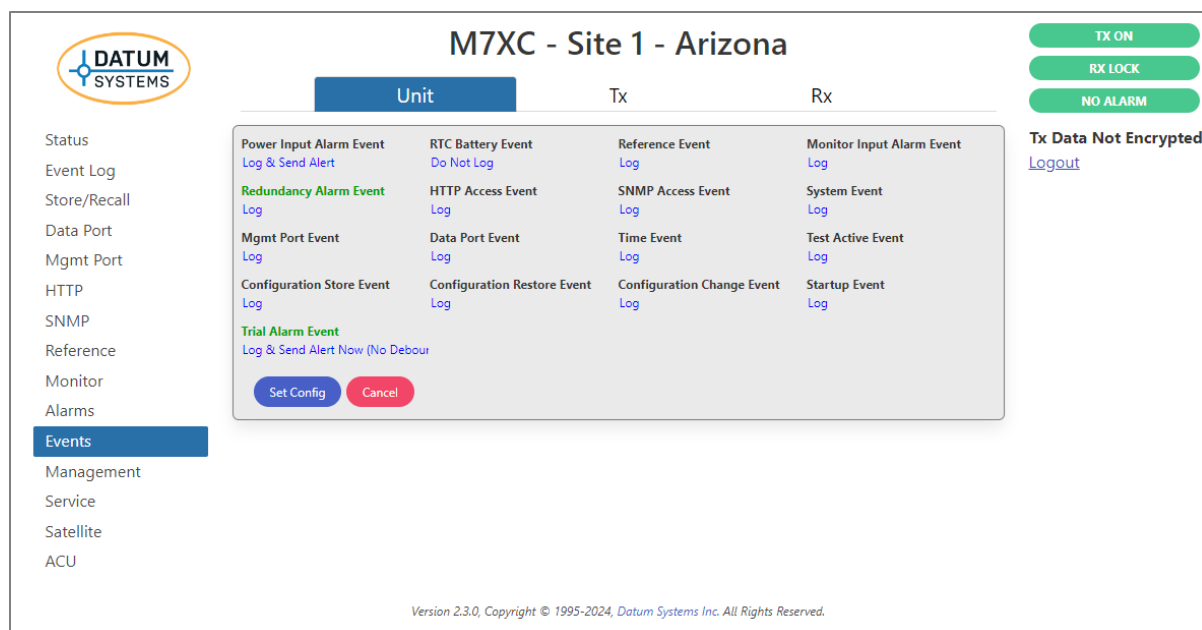


Figure 3-20 Unit Events

- **Power Input Alarm Event:** Triggers when power to the unit was lost or was re-initiated
- **RTC Battery Event:** Triggers when a battery source was removed or replaced
- **Reference Event:** Triggered if the external reference was removed or replaced
- **Monitor Input Alarm Event:** Triggers when the analog monitor voltage exceeds or drops below the user settings on the J4 connector
- **Redundancy Alarm Event:** Triggers when a switchover from prime to backup occurs
- **HTTP Access Event:** Triggers when a HTTP session is initiated with the modem
- **SNMP Access Event:** Triggers when a SNMP session is initiated with the modem
- **System Event:** Triggers when a change to the system configuration has taken place (change of any user parameter)
- **Management Port Event:** Triggers when a connection is made to the management port
- **Data Port Event:** Triggers when a connection is made to the data port
- **Time Event:** Triggers when a SNMP session begins or ends with the modem

3.4.12 >Unit >Management

The >Unit >Management page is used for special functions when the fundamental operation of the modem is to be changed. In this page a user can re-boot the

modem, add features, add trial features, clear special encryption keys or load different versions of firmware for the modem to operate with.

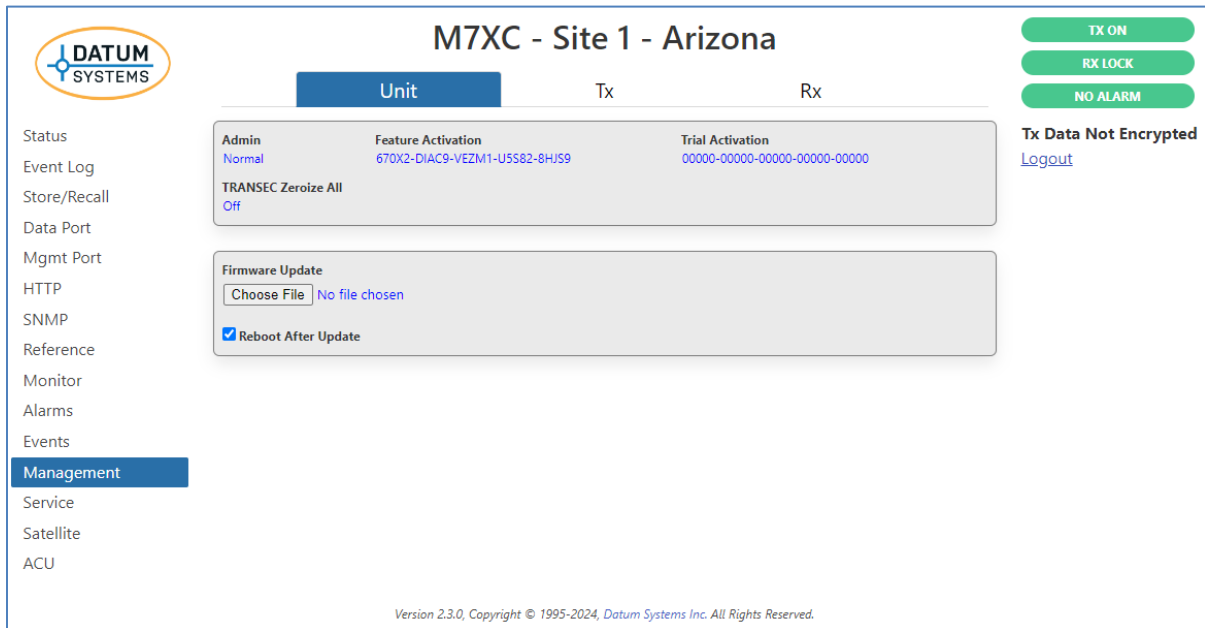


Figure 3-21 Unit Management

- **Admin:** (Normal, On) In “Normal” mode the user had access feature updates, feature trials, encryption key zeroize (clear) and the ability to upload new firmware to the modem. When selected as Admin “On” the user can do all of the above plus re-booting the modem or re-calibrating the modem. Admin mode also allows the user to set data loopback operation under >Unit>Data Port page when enabled.
- **Feature Activation:** The feature activation code is an encrypted key that unlocks all of the features that have been paid for on the modem. The 25-character key is unique to the specific modem and cannot be used on other modems. When new features are purchased, a new key will be sent to the user and that key can be entered in this field to upgrade modem capabilities.
- **Trial Activation:** If a trial of a software feature is requested and approved a Trial Activation key will be sent to the user. Once the trial activation key is entered and accepted by the modem the trial activation time will begin to decrement.
- **Firmware Update:** Modem firmware can be upgraded or downgraded by selecting a FW release from a file on a PC. Once a FW file has been selected the user can click “Upload Firmware and make a choice to have the unit re-boot after upload (strongly suggested) or stay running without reboot. **NOTE: DO NOT CHANGE TABS, PAGES OR GUI SCREENS UNTIL THE FW LOAD IS COMPLETED OR THE PROCESS WILL STOP.**

3.4.13 >Unit >Service

The >Unit >Service page is a read only information page, offering status on temperature, voltages and other parameters.

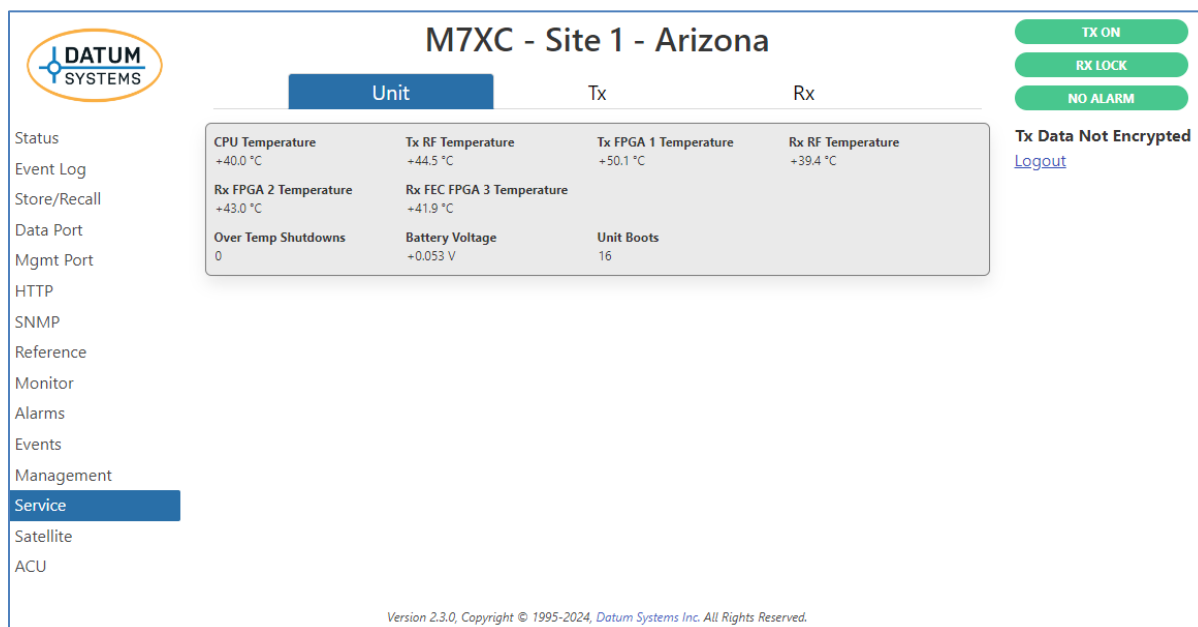


Figure 3-22 Unit Service

- **Temperature Reading:** Readout of the temperature of various parts or sections of the modem.
- **Over Temp Shutdowns:** Number of times the modem has been automatically shut down due to temperature issues
- **Battery Voltage:** Averaged reading of the (optional) battery for the Real Time Clock (RTC)
- **Unit Boots:** Total number of times that modem has been booted up

3.4.14 >Unit >Satellite

The >Unit >Satellite page is used when communicating to an Antenna Controller (ACU) via OpenAMIP™ protocol. The modem can upload a large database of satellite TLE information and select a satellite to communicate with. This information in conjunction with information provided by the Antenna Controller allows satellite terminals to function in a multi-orbit, multi-satellite, multi-beam environment either in a fixed, on the halt or on the move environment.

For more information on setting up communications with an antenna or Antenna Controller (ACU) please see Appendix G Open AMIP.

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Unit Tx Rx

TLE Database File
ses.txt Find Catalog Number 0 Search Operation Idle

TLE Orbital Elements

SES-17
1 49332U 21095A 24022.23800295 -.00000269 00000+0 0 00000+0 0 9993
2 49332 0.0121 70.4291 0001260 223.3603 205.8531 1.00269810 9053

Name	Catalog Number	Classification
SES-17	49332	Unclassified
Launch Year	Launch Number of the Year	Piece Of The Launch
2021	95	A
Epoch Year	Epoch Day	Revolution Number At Epoch
2024	22.23800295	905
Mean Motion	1st Derivative Motion	2nd Derivative Motion
1.00269810	-0.00000269	0.00000E0
Inclination	Right Ascension	Argument Of Perigee
0.0121 *	70.4291 *	223.3603 *
Eccentricity		Mean Anomaly
0.0001260		205.8531 *

[Upload TLE Database File](#)

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Tx Data Not Encrypted
[Logout](#)

TX ON
RX LOCK
NO ALARM

Status
Event Log
Store/Recall
Data Port
Mgmt Port
HTTP
SNMP
Reference
Monitor
Alarms
Events
Management
Service
Satellite
ACU

Figure 3-23 Unit Satellite

3.4.15 >Unit >ACU

Once a satellite is selected from the >Unit >Satellite TLE database the >Unit >ACU page will be visible (Figure 3-24 Unit ACU). The communications between the modem and ACU will pass modem to ACU and ACU to modem parameters to keep the terminal updated on all configuration elements. For more information on setting up communications with an antenna or Antenna Controller (ACU) please see Appendix G Open AMIP.

M7XC - Site 1 - Arizona

Unit Tx Rx

Status
Off

Operation	OpenAMIP Version	Send Command to ACU
Off	1.17	Idle
IP Protocol	IP Address	IP Port
TCP	192.168.0.224	9102
Tx Cable Loss	Min Skew	Max Skew
0.00 dB	0.000 *	+90.000 *
Tx Polarization	Rx Polarization	Regulation Type
Left-Hand Circular	Left-Hand Circular	FCCGEO
Satellite Longitude	Satellite Latitude Variance	Satellite Polarization Skew
0.000 *	0.000 *	0.000 *
Hunt Frequency	Hunt Bandwidth	
1.250000000 GHz	1.000000 MHz	
GPS Location		
Not Valid		

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Tx Data Not Encrypted
[Logout](#)

TX ON
RX LOCK
NO ALARM

Status
Event Log
Store/Recall
Data Port
Mgmt Port
HTTP
SNMP
Reference
Monitor
Alarms
Events
Management
Service
Satellite
ACU

Figure 3-24 Unit ACU

3.5 M7XC Web Browser TX Tab and Pages

The TX pages are used to configure the transmit parameters of the M7XC. The IF, Symbol, Data Rates, and other transmit parameters are controlled in the “TX” pages.

3.5.1 >TX >Status

The TX “Status” tab (Figure 3-25 TX Status) displays the TX Status of the transmit parameters and remote receive levels.

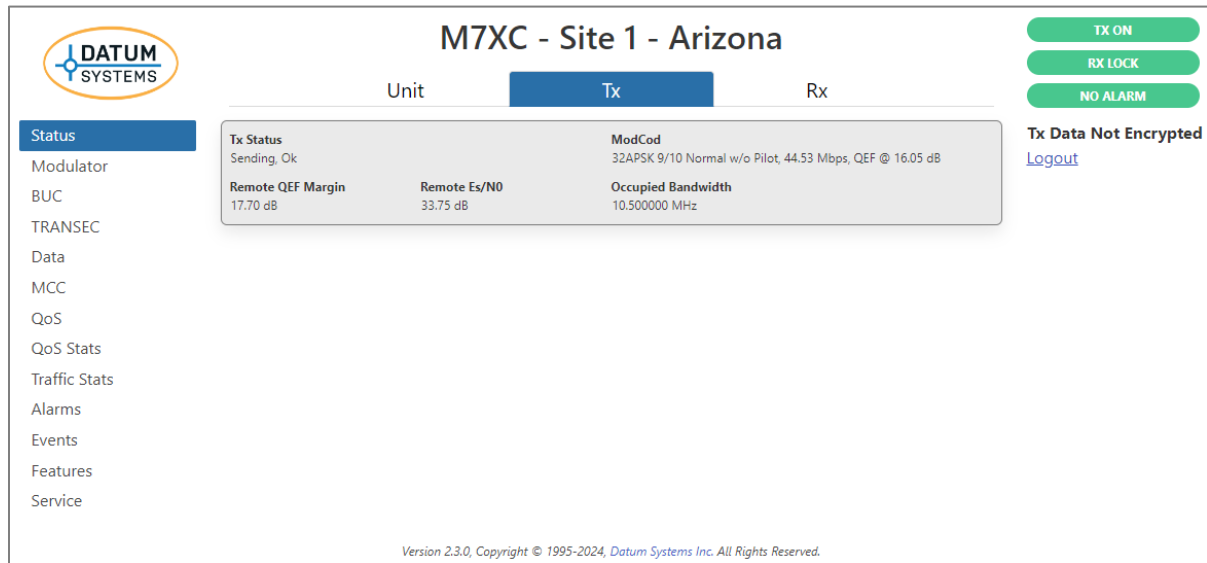


Figure 3-25 TX Status

- **TX Status:** Displays current status of the transmit path of the modem. Normal operation status will declare “Sending, Ok” or “Tx OUTPUT OFF”. If there is a fault in the TX path or TX parameters it will be displayed here
- **MODCOD:** Displays current MODCOD being transmitted
- **Remote QEF Margin:** Displays current QEF Margin as reported by the (far side) demod Es/No for the MODCOD being received. The margin is based on the MODCOD QEF Es/No and does not take into account any ACM or Hysteresis margin added. NOTE: A reading of -10dB indicates the remote communications with the far side modem is not operational (check MCC settings).
- **Remote Es/No:** Displays current remote (far side) demod Es/No level as reported over the MCC channel if the MCC channel is configured properly. NOTE: A reading of -10dB indicates the remote communications with the far side modem is not operational (check MCC settings).
- **Occupied Bandwidth:** Displays the current Occupied Bandwidth calculated as $\text{Occupied BW} = \text{TX Symbol Rate} * (1 + \text{ROF})$. Ex: The Symbol Rate is 10Msps and the ROF is 5% therefore the Occupied BW = $10\text{Msps} * (1 + 0.05) = 10.5\text{MHz}$

3.5.2 >TX >Modulator

The >TX >Modulator page (Figure 3-26 TX Modulator) provides the ability to set the parameters for the Modulator.

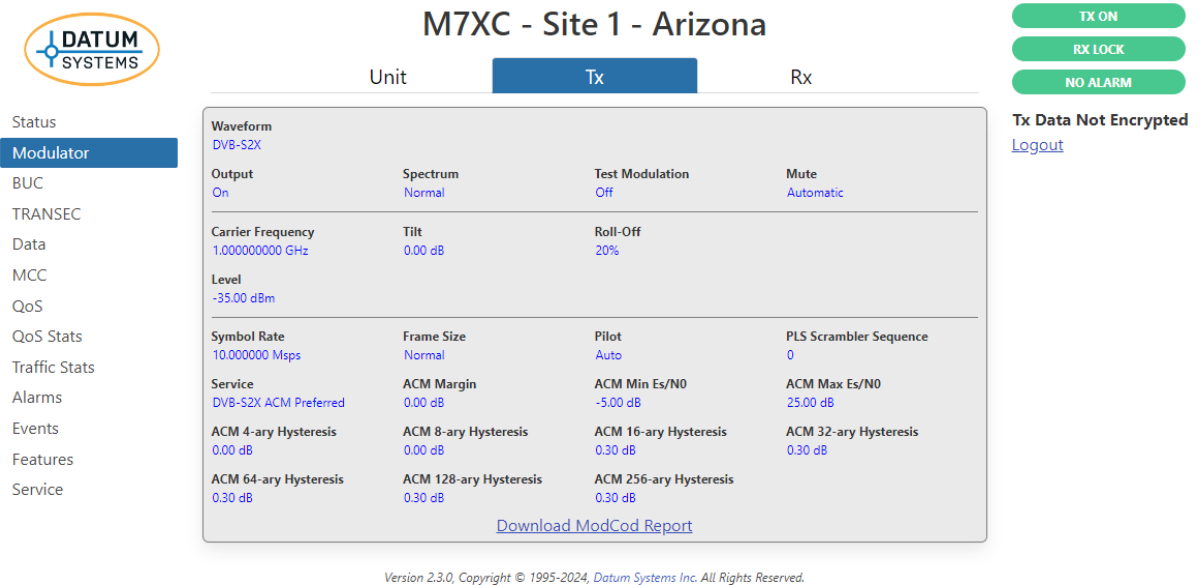


Figure 3-26 TX Modulator

- **Waveform:** Set the transmit waveform (Tx Off, DVB-S2X, MIL 165B DVB-S2X, LDPC-Gen2, Segmented LDPC-Gen2, DSSS)
- **Output:** Set the transmit Output function (Off, On) a setting of “Off” is carrier muted.
- **Spectrum:** Set the transmit Spectrum function (Normal, Inverted)
- **Test Modulation:** Set the transmit Test function (Off, Pure Cxr, Alt 1/0, Lower Sideband, Upper Sideband, QPSK PN Sequence). A setting of “Off” is for normal operation.
- **Mute:** Set the transmit Mute function (Automatic, Confirm, Manual, Manual & Power Loss).
 - Automatic: Will keep the >Tx >Modulator Output setting regardless of changes to Tx Settings
 - Confirm: Will prompt to keep the >Tx >Modulator Output “On” when a change is made to Tx Settings
 - Manual: Will automatically mute the >Tx >Modulator Output on any changes to Tx Settings
 - Manual & Power Loss: Will automatically mute the >Tx >Modulator Output setting regardless of changes to Tx Settings and/or if the modem is power cycled
- **Carrier Frequency:** Set the transmit center frequency
- **Tilt:** Set the transmit Tilt function to match transponder slope, setting of +/- 2dB per 100MHz is used for the full carrier BW

- **Roll-Off:** Set the transmit Roll-Off factor (35%, 30%, 25%, 20%, 15%, 10%, 5%, 2%) NOTE: Waveform dependent.
- **Level:** Set the transmit output Level (-35dBm to +5dBm)
- **Symbol Rate:** Set the transmit output Symbol Rate (Waveform dependent)
- **Frame size / Block Size:** (Waveform dependent) configures the size of the FEC Frame or Block. In DVB-S2 modes “Normal” = 64k, “Short” = 16k, other waveforms declare block size.
- **Scrambler:** Set the transmit scrambler type. Scrambler or Scrambler Sequence number will be displayed depending on waveform. It is critical that the far side demod MATCH the TX scrambler or scrambler sequence number.
- **Service (without ACM):** Set the transmit Service (Waveform dependent): (LDPC-Gen2, DVB-S2, DVB-S2X). **Note to use ACM you must choose a service with ACM in the title.**
- **MODCOD:** In a service WITHOUT ACM: Set the transmit MODCOD

Note when a service WITH ACM is chosen a screen such as (Figure 3-27 TX Modulator with ACM Service) will be displayed similar to the below:

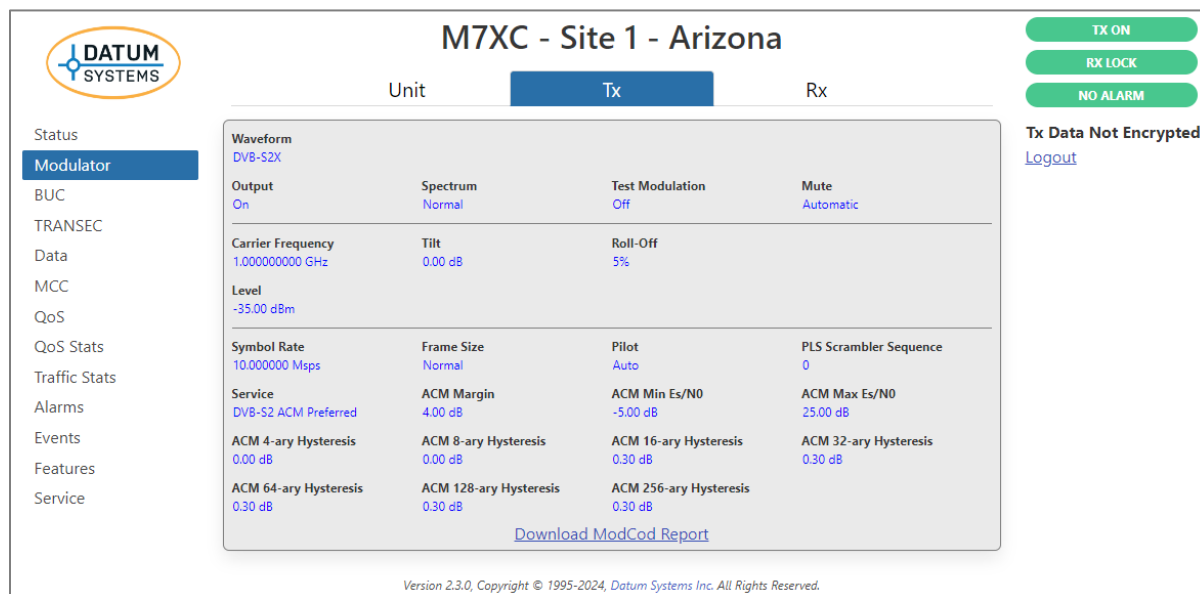


Figure 3-27 TX Modulator with ACM Service

- **Service (with ACM):** Sets the modulator Tx to run MODCODs per the ACM parameters (not a specific MODCOD). Choices are:
 - **DVB-S2 ACM Preferred:** Chooses only DVB-S2 MODCODs that are increasingly spectrally efficient for the ACM parameters chosen
 - **DVB-S2X ACM Preferred:** Chooses only DVB-S2X MODCODs that are increasingly spectrally efficient for the ACM parameters chosen but will NOT use any “-L” or linear MODCODs

- **DVB-S2X ACM Preferred-L:** Chooses only DVB-S2X MODCODs that are increasingly spectrally efficient for the ACM parameters chosen INCLUDING “-L” or linear MODCODs
- **LDPC-Gen2 ACM:** Chooses only LDPC-Gen2 MODCODs that are increasingly spectrally efficient for the ACM parameters chosen.
- **ACM Margin:** Requires additional Es/No to switch UP to a MODCOD over the standard MODCOD QEF point. Configurable from 0.0dB to 0.5dB
- **ACM Min Es/No:** The minimum Es/No allowed to be reported from the Remote far side demod to the local Tx modulator. Has the effect of limiting the lowest MODCOD available. Choosing a higher value will keep the link at a higher user rate at the cost of dropping the link in poor conditions.
- **ACM Max Es/No:** The maximum Es/No allowed to be reported from the Remote far side demod to the local Tx modulator. Has the effect of limiting the highest MODCOD available. Choosing a low value will prevent the use of higher order MODCODs such as 128 / 256 QAM but at the cost of lower user datarate.
- **ACM X-ary Hysteresis:** In some satellite links the linearity of the ground equipment or satellite may not be sufficient to run higher order MODCODs without additional margin. For this reason the user can add additional margin on a per modulation ARY basis. You must use equal or higher margin as the modulation increases in bits/symbol. Ex. You can set 8-ary hysteresis to 0.2dB additional margin and 16-ary to 0.4dB but you could not set 8-ary hysteresis to 0.2dB additional margin and 16-ary to 0.1dB.
- **Download ModCod Report:** The ACM algorithm will remove or “prune” MODCODs that are not spectrally more efficient for an equal or lower Es/No. The remaining non-pruned MODCODs can and will be different depending on the user settings for XX-ary hysteresis. To see exactly what MODCODs will be used in the ACM operation click on the “Download ModCod Report” to get a .TXT file similar to (Figure 3-28 Sample ModCod Report) below:

```

Service: DVB-S2X ACM Preferred
Frame Size: Normal
Pilot: Auto
Margin (M): 0.00 dB
  4-ary Hysteresis (H): 0.00 dB
  8-ary Hysteresis (H): 0.00 dB
 16-ary Hysteresis (H): 0.30 dB
 32-ary Hysteresis (H): 0.30 dB
 64-ary Hysteresis (H): 0.30 dB
128-ary Hysteresis (H): 0.30 dB
256-ary Hysteresis (H): 0.30 dB

```

No.	ModCod	Efficiency	QEF	QEF+M	QEF+M+H
1	QPSK 13/45	0.567805479	-2.03	-2.03	-2.03
2	QPSK 9/20	0.889135118	0.22	0.22	0.22
3	QPSK 1/2	0.988858110	1.00	1.00	1.00
4	QPSK 11/20	1.088581102	1.45	1.45	1.45
5	QPSK 3/5	1.188304094	2.23	2.23	2.23
6	QPSK 2/3	1.322253001	3.10	3.10	3.10
7	QPSK 3/4	1.487473069	4.03	4.03	4.03
8	QPSK 4/5	1.587196060	4.68	4.68	4.68
9	QPSK 5/6	1.654662973	5.18	5.18	5.18
10	8PSK 3/5	1.739569253	5.50	5.50	5.50
11	8PSK 23/36	1.896173352	6.12	6.12	6.12
12	8PSK 2/3	1.935658286	6.62	6.62	6.62
13	8PSK 25/36	2.062148456	7.02	7.02	7.02
14	8PSK 13/18	2.145136007	7.49	7.49	7.49
15	16APSK 26/45	2.281645181	7.51	7.51	7.81
16	16APSK 3/5	2.370042971	7.80	7.80	8.10
17	16APSK 28/45	2.458440761	8.10	8.10	8.40
18	16APSK 23/36	2.524739104	8.38	8.38	8.68
19	16APSK 2/3	2.637200737	8.97	8.97	9.27
20	16APSK 25/36	2.745733579	9.27	9.27	9.57
21	16APSK 13/18	2.856230816	9.71	9.71	10.01
22	16APSK 3/4	2.896320268	10.21	10.21	10.51
23	16APSK 7/9	3.077225292	10.65	10.65	10.95
24	16APSK 4/5	3.165623082	11.03	11.03	11.33
25	16APSK 5/6	3.300184162	11.61	11.61	11.91
26	32APSK 32/45	3.510191571	11.75	11.75	12.05
27	32APSK 11/15	3.620536398	12.17	12.17	12.47
28	32APSK 3/4	3.703295019	12.73	12.73	13.03
29	32APSK 7/9	3.841226054	13.05	13.05	13.35
30	32APSK 4/5	3.866246814	13.64	13.64	13.94
31	32APSK 5/6	4.119540230	14.28	14.28	14.58
32	64APSK 11/15	4.338659320	14.81	14.81	15.11
33	64APSK 7/9	4.603122130	15.47	15.47	15.77
34	64APSK 4/5	4.735353535	15.87	15.87	16.17
35	64APSK 5/6	4.936639118	16.55	16.55	16.85
36	128APSK 3/4	5.163247863	17.73	17.73	18.03
37	128APSK 7/9	5.355555556	18.53	18.53	18.83
38	256APSK 32/45	5.593162393	18.59	18.59	18.89
39	256APSK 3/4	5.900854701	19.57	19.57	19.87

Figure 3-28 Sample ModCod Report

NOTE: The ModCod Report will display the ACM Service running, the Frame Size used, the status of Pilot Symbols used (Auto or On), The ACM margin configured, the per X-ARY Hysteresis configured, a count of all MODCODs used, what the MODCOD is, it's spectral efficiency, the Threshold QEF point (without any additional margin), the QEF point plus the ACM margin and the QEF + ACM Margin + Hysteresis Margin.

The QEF + ACM Margin + Hysteresis Margin will be the “switch UP” point for that MODCOD. Meaning the far side modem must report a value equal to or higher than QEF + ACM Marg + Hysteresis Marg before it will switch to that MODCOD. The modem will “switch DOWN” when the far side modem reports a value below the QEF + ACM Margin. Not the QEF + ACM + Hysteresis.

3.5.3 >TX >BUC

The TX “BUC” tab (Figure 3-29 TX BUC) provides the ability to set parameters of the transmit RF conversion equipment to display the RF frequency transmitted based on the IF frequency setting. This is helpful to the operator to confirm the transmit carrier is in the proper transponder location.

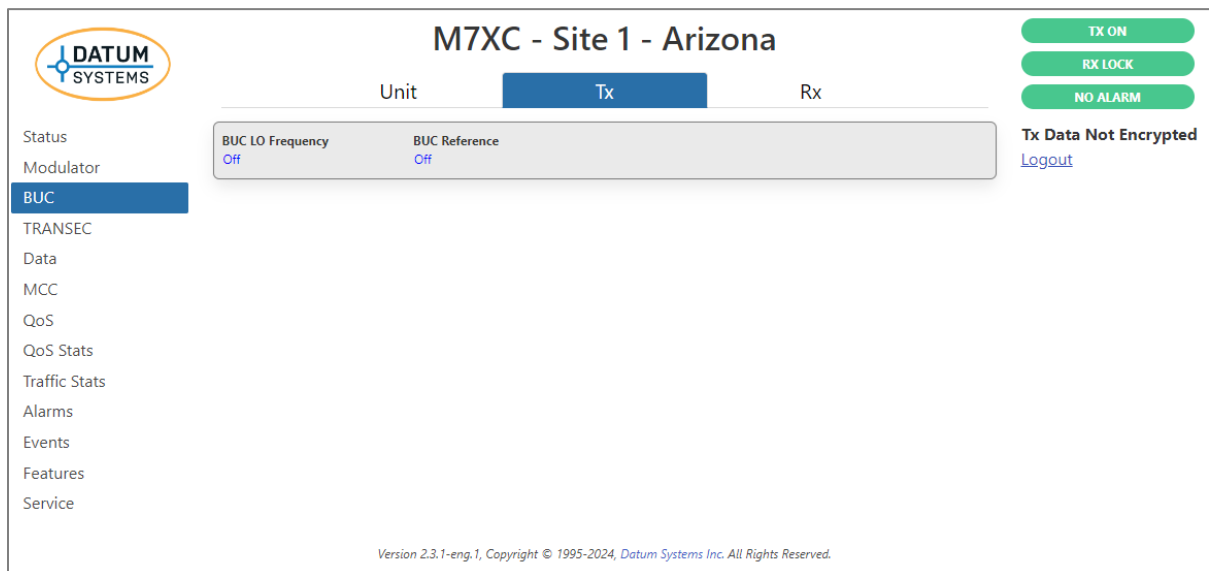


Figure 3-29 TX BUC

- **BUC LO Frequency (GHz):** Configuration of the BUC LO frequency to calculate and display the RF frequency in the TX Modulator tabs. If set to zero (0) then the RF frequency display is the same as the IF frequency and “Off” will be shown
- **BUC Reference:** Set the configuration of the BUC Reference (Off, On). When set to “On” the reference sent to the BUC will be the reference configured on the >Unit >Reference >Frequency section (3.4.8 >Unit >Reference).

3.5.4 >TX >TRANSEC (Optionally Equipped)

The optional TRANSEC operation of the modem will encrypt / decrypt the user traffic. When Encryption is enabled, all user traffic going over the WAN is encrypted including the User data, IP headers, Layer 2.5 headers, Ethernet headers and GSE encapsulation headers. When TRANSEC is enabled on the TX side, the far side RX must have a matching TRANSEC decryption setting to land the traffic.

Prior to any TRANSEC operation (when installed) the TRANSEC screen will look like the following (Figure 3-30 TX TRANSEC Not Initiated).

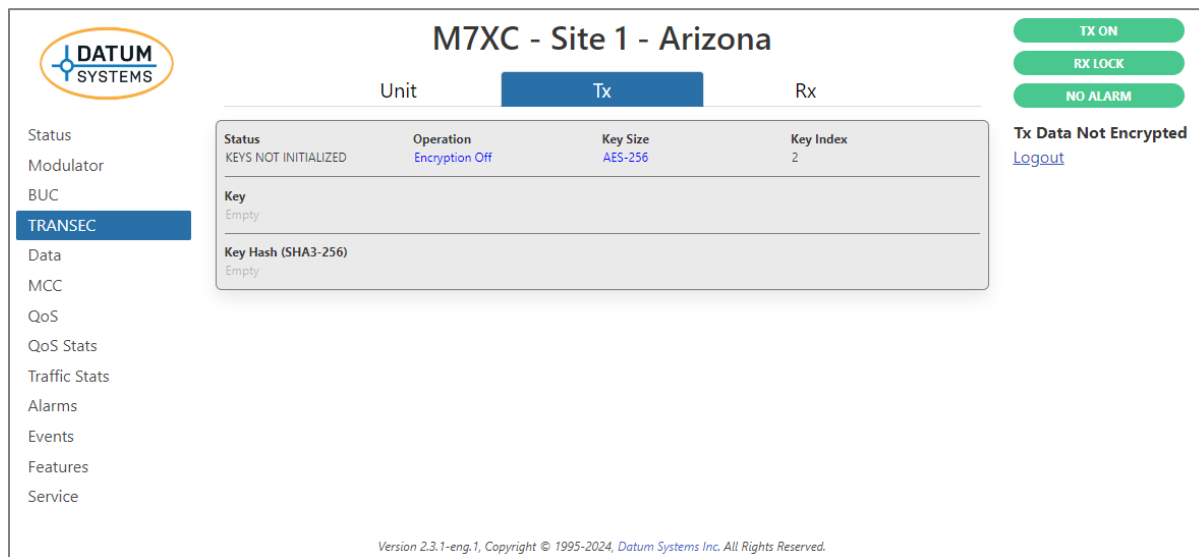


Figure 3-30 TX TRANSEC Not Initiated

The user should see the >TX >TRANSEC >Status “KEYS NOT INITIALIZED” and >Operation “Encryption Off”. This indicates that the TRANSEC module has not been configured.

- **Status:** “KEYS NOT INITIALIZED” indicates that a key with the proper length matching the >Key Size has not been entered. “Data Encrypted” indicated TRANSEC is configured and operating.
- **Operation:** (Encryption Off, Encryption On, Generate Random Key)
 - Encryption Off: Will disable TRANSEC and will display “Tx Data Not Encrypted” in the top right status. This is due to the modem knowing that TRANSEC has been installed but is not enabled.
 - Encryption On: Encryption is Enabled
 - Generate Random Key: Can be used to generate an Encryption Key that is the proper length for the >Key Size selected. NOTE: This is only optional and a proper length Key can be manually entered.
- **Key Size:** (AES-128, AES-192, AES-256) will set the key length with the larger numbers indicating longer (and more secure) keys.
- **Key Index:** Displays the rolling Key changes that occur automatically, there are over 64,000 unique keys that the modem will roll through before any key is repeated. Keys are rolled once a set number of frames have been transmitted.

The key index will automatically synchronize with the far side decryption if the user entered >Tx >TRANSEC >Key matches the >Rx >TRANSEC >Key.

Once the user has turned “Encryption On” and entered the proper length >Key as well as matched the settings on the far side >RX >TRANSEC page the screen should look like the below Figure 3-31 TX TRANSEC Proper Configuration.

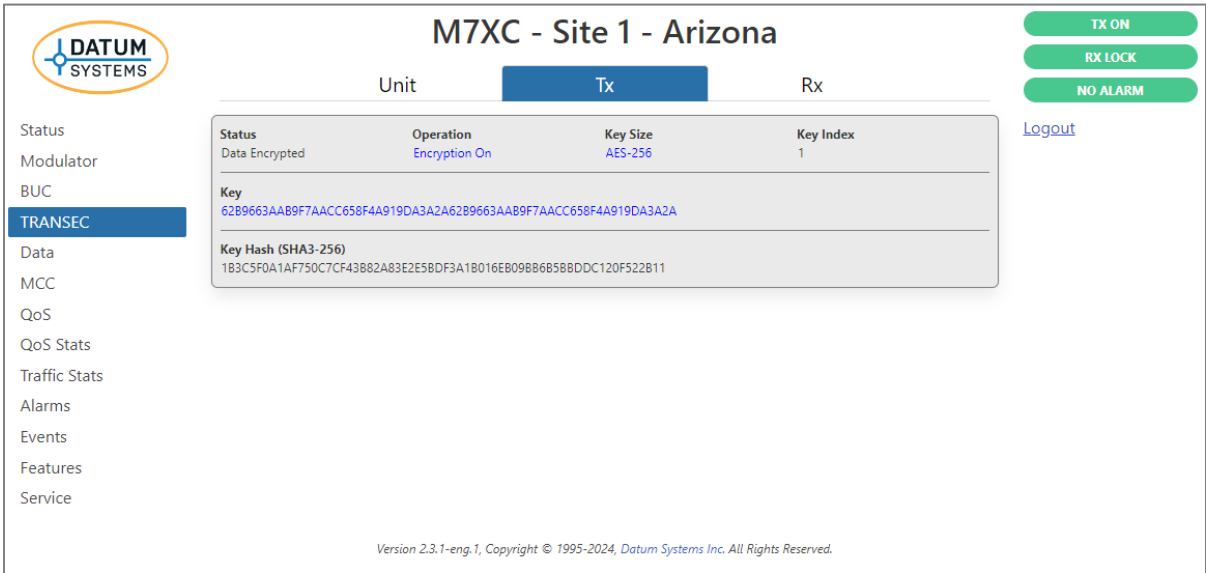


Figure 3-31 TX TRANSEC Proper Configuration

- *Key Hash (SHA3-256)*: Displays the SHA3-256 standards-based hash of the user entered Key. When logging in as a read only user, only the Key Hash can be viewed for verification that the far side demod has the matching Key Hash as the actual Key will not be visible.

3.5.5 >TX >Data

The Tx Data page is used to configure Tx framing and special message framing (Figure 3-32 Tx Data).

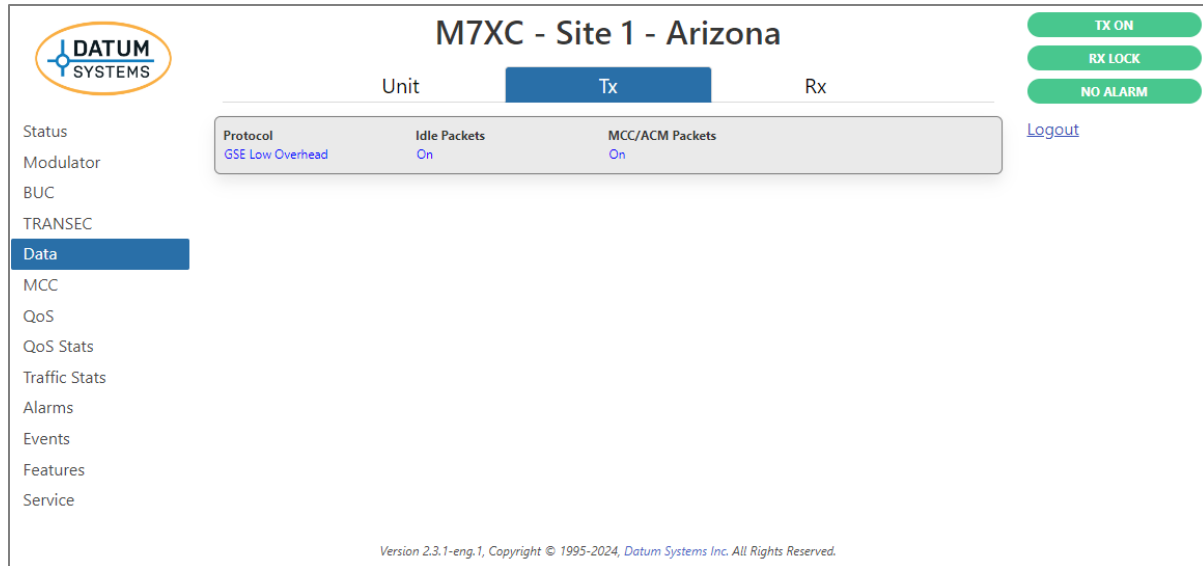


Figure 3-32 Tx Data

- **Protocol:** (waveform dependent) Defines the IP Encapsulation method used by the framer (GSE DVB, GSE Low Overhead, GSE CT). The lowest overhead framing is "GSE Low Overhead". Other framing modes are available for compatibility purposes with other vendors or standards only. "GSE Low Overhead" is default.
- **Idle Packets:** (On, Off) When On, the modem will send IP packets filled with pseudo random data when the user data buffer is empty. This is done to keep the carrier spectrum random. "On" is default.
- **MCC/ACM Packets:** (On, Off) When On, the modem enable an In-Band channel between the near side and far side modems for internal communications. This communications channel is REQUIRED for ACM operation and to get remote demod Es/No & Es/No margin. Default is "On" When "Off" the modem will not use the OH channel and the link will be slightly more efficient. The Overhead channel has a configurable OH data rate see (section 3.5.6 >TX >MCC) on how to configure the OH channel.

3.5.6 >TX >MCC

The TX "MCC" tab (Figure 3-33 TX MCC) provides the ability to set the parameters of the Send and Receive Modem Communications Channel (MCC) between the local and remote modems.

See Appendix D MCC Operation for details and examples.

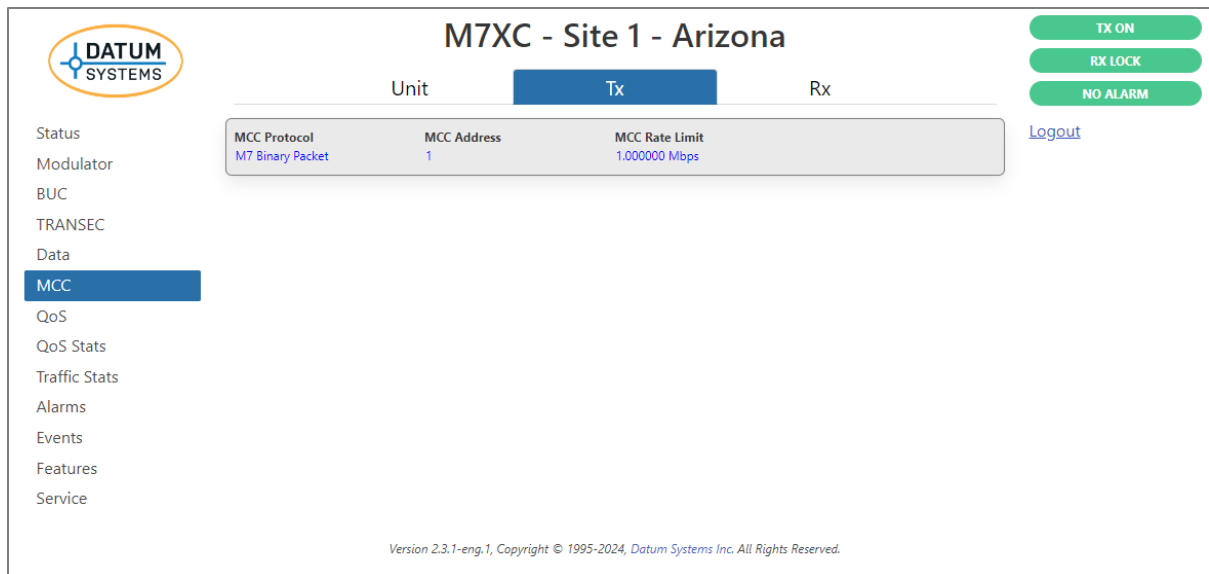


Figure 3-33 TX MCC

- **Protocol:** Sets the MCC channel protocol (Default = M7 Binary Packet)
- **Send Address:** Sets the address of the Send MCC port (1 to 255, 255 = Global). In a point to multi-point network, or MESH network, take care to configure each modulator's TX MCC Address to a unique number.
- **MCC Rate Limit:** Sets the maximum rate of the MCC Overhead channel (1.0 kbps - 1.0 Mbps)

3.5.7 >TX >QoS

The M7XC typically ingests bursty or variable rate data on the Local Area Network (LAN) side of the Ethernet interface and has a fixed rate (or in the case of ACM an adaptive fixed rate) data load transmission to the satellite (WAN) side. In any QoS Mode other than "Off", QoS provides a type of flow control between the LAN and the WAN. Buffering is provided within the internal Ethernet interface to store data until it can be transmitted to the WAN.

Packet Drain and Drop Selection

QoS Mode

The QoS mode defines the modems TX buffer drain and drop preference. QoS mode can be configured as "Off", "WRED" (Weighted Random Early Detection), "Strict Priority", or "Custom". These settings will determine the procedure for draining the buffer to the WAN and dropping packets when the QoS buffer has reached the point of congestion. The default setting is WRED (Figure 3-34 QoS Mode).

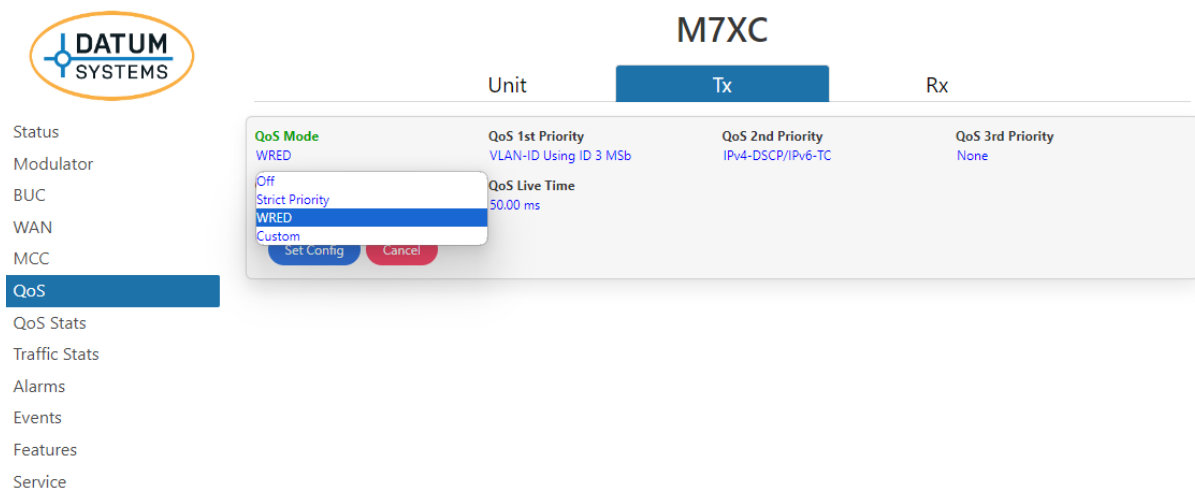


Figure 3-34 QoS Mode Selection

- **Off:** All packets are treated with equal priority and all packets fill a single queue/classification. No re-ordering of packets into queues is performed, draining is first in / first out. If the buffer is full, all packets that arrive from the LAN to the WAN will be dropped until the buffer is drained enough to add more packets.
- **Strict Priority:** Packets are classified into their appropriate queue depending on the QoS classification, the QoS buffers will drain from the highest priority buffer until empty, then drain the next lower priority buffer until empty and so forth down to the priority 0 buffer. If a packet arrives that is classified in a higher priority than a currently draining QoS buffer, that buffer will immediately stop draining until all higher priority buffers are empty (**Error! Reference source not found.**). A side effect of “Strict Priority” drain is that lower priority QoS buffers often fill and do not have the chance to drain, thus starving lower priority QoS classifications. Any packet in any QoS queue older than the global “Live Time” is dropped.

Table 3-4 Strict Priority Drain Algorithm

Strict Priority Drain Algorithm	
Queue Priority	Drain Operation
7 (Highest)	Drain until empty
6	If no data in Queue 7: Drain until empty
5	If no data in Queue 6-7: Drain until empty
4	If no data in Queue 5-7: Drain until empty
3	If no data in Queue 4-7: Drain until empty
2	If no data in Queue 3-7: Drain until empty
1	If no data in Queue 2-7: Drain until empty
0 (Lowest)	If no data in all other Queues: Drain until empty

- WRED: Packets are classified into their appropriate queue depending on the QoS classification, each QoS classification buffer queue uses a strict drain algorithm based on the buffer priority. See below (Table 3-5 WRED Drain Algorithm) where Queue Priority 7 will drain 33 packets for every 1 packet drained in the lowest Queue Priority “0”. Any packet in any QoS queue older than the global “Live Time” is dropped.

Table 3-5 WRED Drain Algorithm

WRED Drain Algorithm	
Queue Priority	Drain Weighting
7 (Highest)	33 (Most Packets Drained)
6	25
5	17
4	12
3	6
2	3
1	1
0 (Lowest)	(Least Packets Drained)

- Custom: Packets are classified into their appropriate queue depending on the QoS classification. Queues are drained based on the queue weight, with higher weighting draining faster than lower weighting. Each queue can have a unique Live Time (0.1ms to 650ms) before a packet from that queue is dropped, see (Figure 3-35 TX QoS Custom Page)

QoS Mode	QoS 1st Priority	QoS 2nd Priority	QoS 3rd Priority
Custom	VLAN-PCP	VLAN-PCP	VLAN-PCP
QoS Default Priority	QoS Live Time	QoS Queue 0 Live Time	QoS Queue 0 Weight
0	50.00 ms	50.00 ms	Off
QoS Queue 1 Live Time	QoS Queue 1 Weight	QoS Queue 2 Live Time	QoS Queue 2 Weight
50.00 ms	Off	50.00 ms	Off
QoS Queue 3 Live Time	QoS Queue 3 Weight	QoS Queue 4 Live Time	QoS Queue 4 Weight
50.00 ms	Off	50.00 ms	Off
QoS Queue 5 Live Time	QoS Queue 5 Weight	QoS Queue 6 Live Time	QoS Queue 6 Weight
50.00 ms	Off	50.00 ms	0.00 ms
QoS Queue 7 Live Time	QoS Queue 7 Weight		
50.00 ms	Off		

Figure 3-35 TX QoS Custom Page

Packet Classification

QoS 1st, 2nd, 3rd and Default Priority

The QoS 1st, 2nd and 3rd Priority settings define methods of classification of a packet into one of 8 QoS queues. Once a packet is ingested by the modems Ethernet interface for transmission to the WAN, the modem will attempt to classify the packet into a QoS classification queue (this is true of all QoS Modes other than “Off”). Each Tx Queue is 16 MB deep. There are 8 Tx QoS Queues each with a different priority level. The highest priority queue is queue ‘7’ and the lowest priority is queue ‘0’.

In all QoS modes other than “Off” classification will attempt to classify a packet based on QoS 1st Priority rules. If successful, no other attempts are made to further classify the packet. If a classification cannot be made by QoS 1st Priority then the modem will attempt to classify the packet by using the rules in QoS 2nd Priority and finally QoS 3rd Priority. If none of the QoS 1-3 Priorities can classify the packet, the packet will be placed in the “QoS Default Priority”. The Default Classification Priority can be set to fill any queue from 0- 7 with default being ‘0’. See (Figure 3-36 QoS Priorities and Figure 3-37 QoS Priorities Logic Diagram).

M7XC - Site 1 - Arizona

Unit Tx Rx

QoS Mode WRED	QoS 1st Priority VLAN-ID Using ID 3 MSb	QoS 2nd Priority IPv4-DSCP/IPv6-TC	QoS 3rd Priority None
QoS Default Priority 0	QoS Live Time 100.00 ms		

TX ON
RX LOCK
NO ALARM
[Logout](#)

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Figure 3-36 QoS Priorities

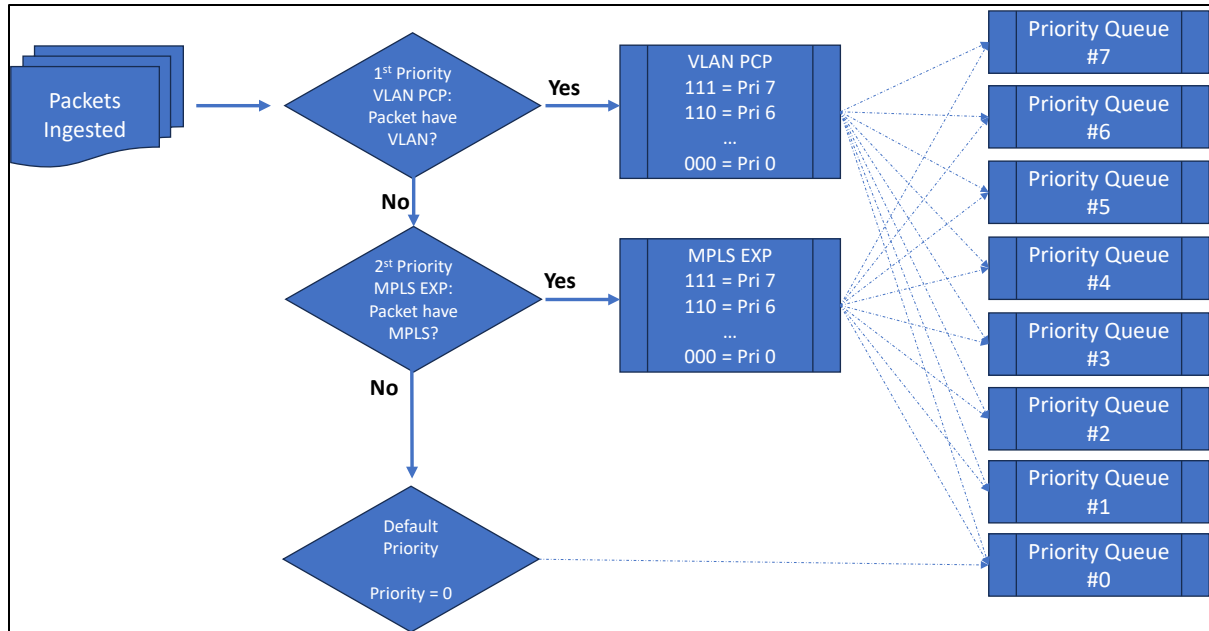


Figure 3-37 QoS Priorities Logic Diagram

- **QoS 1st Priority:** Set the 1st priority preference for packet classification using the priority tags (VLAN-PCP, VLAN-ID Using ID 3MSb, MPLS-EXP, MPLS-EXP Using IP Only, IPv4-DSCP/IPv6-TC)
- **QoS 2nd Priority:** Set the 2nd priority preference for packet classification using the priority tags (None, VLAN-PCP, VLAN-ID Using ID 3MSb, MPLS-EXP, MPLS-EXP Using IP Only, IPv4-DSCP/IPv6-TC)
- **QoS 3rd Priority:** Set the 3rd priority preference for packet classification using the priority tags (None, VLAN-PCP, VLAN-ID Using ID 3MSb, MPLS-EXP, MPLS-EXP Using IP Only, IPv4-DSCP/IPv6-TC)
 - **None:** No classification is used (only applicable to QoS 2nd, 3rd and Default Priority).
 - **VLAN-PCP:** If the “EtherType” is 0x8100 (VLAN 802.1Q) or 0x88A8 (Stacked VLAN 802.1Q) the OUTERMOST VLAN tag is used to determine the QoS queue priority. The Priority Code Point (PCP) value on the outmost VLAN tag is used to determine the QoS queue using the following:

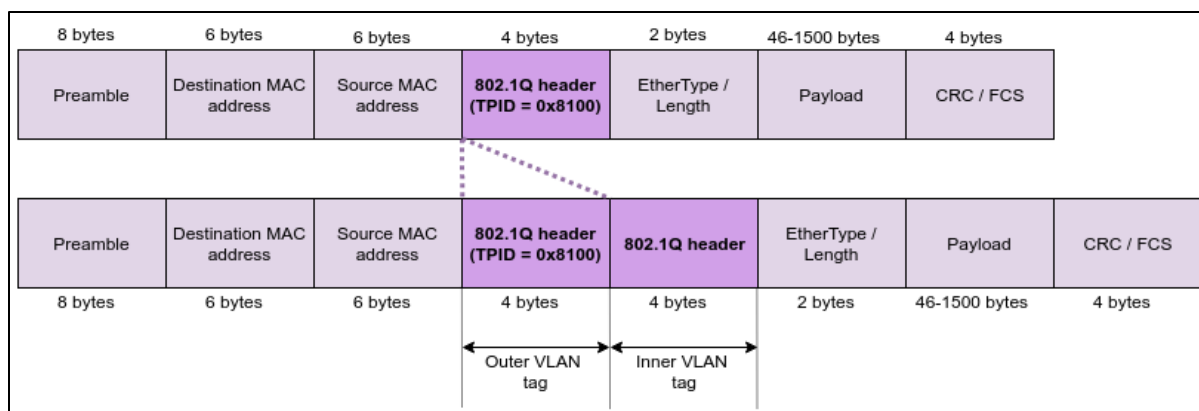


Figure 3-38 Ethernet Frame with Stacked VLAN Tags

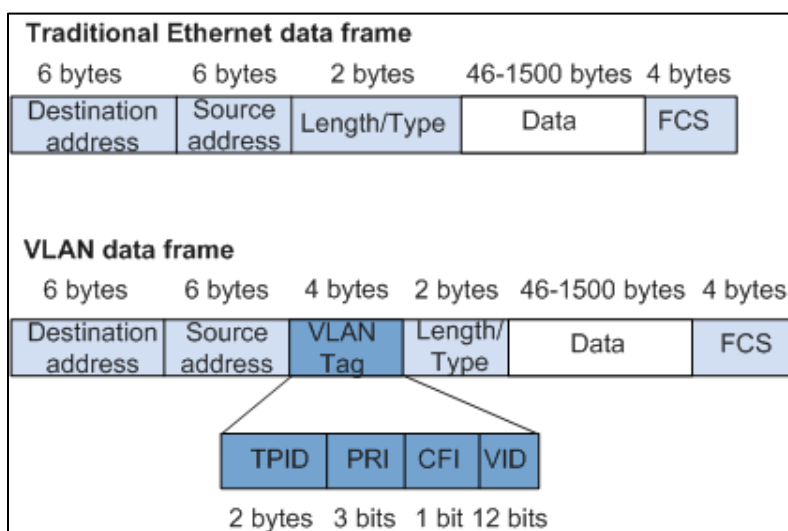


Figure 3-39 Ethernet Frame with VLAN Tag

Table 3-6 VLAN PCP map to Queue Priority

PCP Value	Priority
000	0 (Lowest)
001	1
010	2
011	3
100	4
101	5
110	6
111	7 (Highest)

- **VLAN-ID Using ID 3MSb:** If the “EtherType” is 0x8100 (VLAN 802.1Q) or 0x88A8 (Stacked VLAN 802.1Q) the OUTERMOST VLAN ID will be used. The three Most Significant Bits (MSB) in the TPID tag are used:

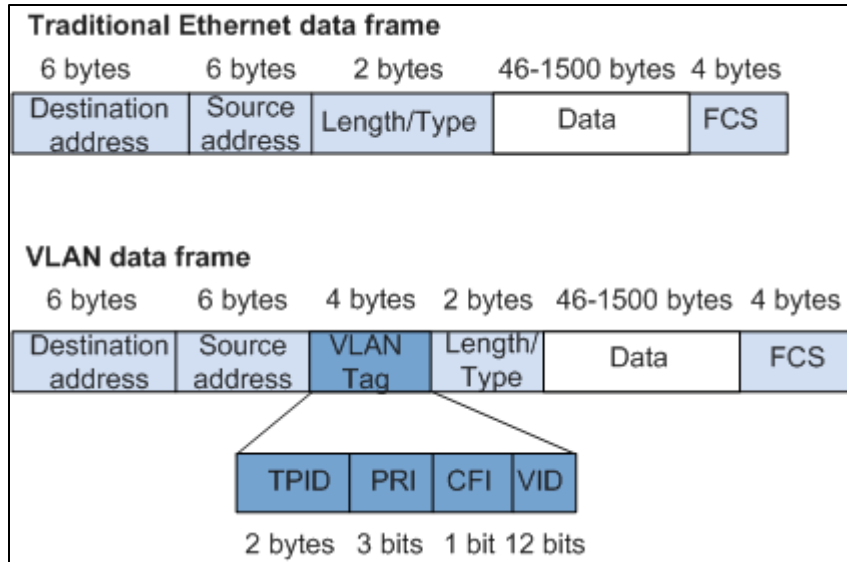


Figure 3-40 VLAN ID / TPID

Table 3-7 VLAN ID map to Queue Priority

TPID MSB	Priority
000	0 (Lowest)
001	1
010	2
011	3
100	4
101	5
110	6
111	7 (Highest)

- **MPLS-EXP:** If the “EtherType” is 0x8847/0x8848 MPLS Unicast/Multicast the EXP or CoS “Traffic Class” value on the outmost MPLS tag is used to determine the QoS queue using the following:

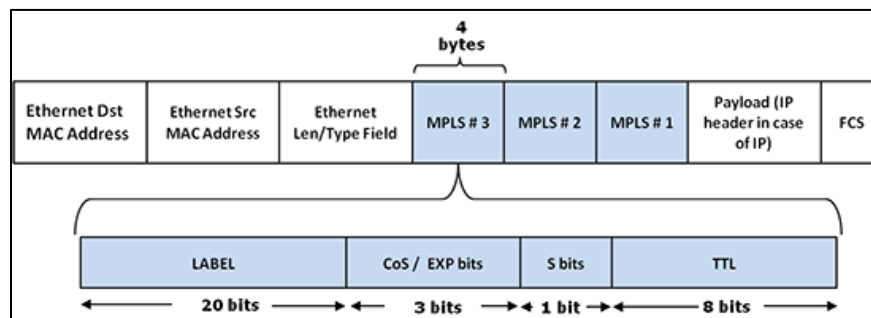


Figure 3-41 Ethernet Frame with MPLS Tag

Table 3-8 MPLS EXP/CoS map to Queue Priority

MPLS CoS / EXP Bits	Priority
000	0 (Lowest)

001	1
010	2
011	3
100	4
101	5
110	6
111	7 (Highest)

- **MPLS-EXP Using IP Only:** Ethernet Frames with MPLS stacks replace the Ethertype field. The MPLS payload may be IP or any type of packet. If it is known beforehand that the all MPLS packets contain IP packets, you can choose this to allow the classifier to dig into the MPLS payload for DSCP /TC values. In this case the MPLS label stack and EXP fields are ignored.
- **IPv4 DSCP/IPv6 TC:** If the traffic is IP (with or without VLAN or MPLS tags) the priority is set by IPv4 or IPv6 TOS/TC bits. The 6 MSB bits are used in the IPv4 TOS or IPv6 TC header to represent DSCP services (Figure 3-42 IPv4 & IPv6 DSCP Bits). There is a fixed DSCP to Queue map:

Table 3-9 IPv4 and IPv6 DSCP map to Queue Priority

IPv4 DSCP 6 bit value (decimal)	IPv6 DSCP 6 bit value (decimal)	Priority
0-7	0-7	0 (Lowest)
8-15	8-15	1
16-23	16-23	2
24-31	24-31	3
32-39	32-39	4
40-47	40-47	5
48-55	48-55	6
56-63	56-63	7 (Highest)

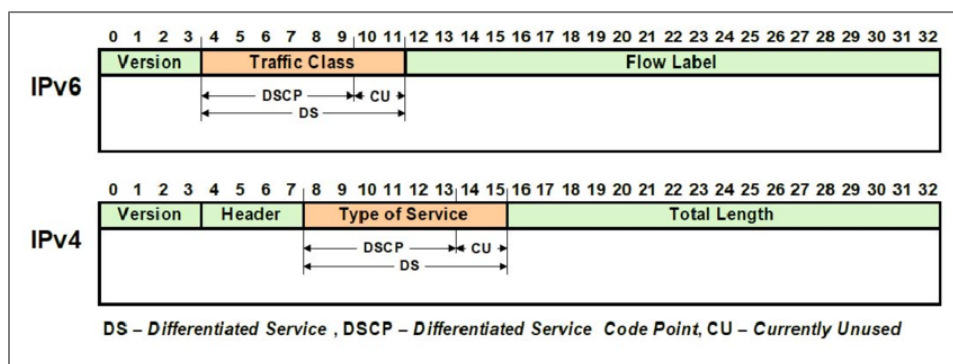


Figure 3-42 IPv4 & IPv6 DSCP Bits

- **QoS Live Time:** In Strict Priority or WRED mode operation QoS Live Time is a global parameter that effects all QoS priority buffers. It can be set between 0.1ms and 650msec. If a buffered packet exceeds the time set for the QoS Live Time the packet is dropped.

3.5.8 >TX >QoS Stats

TX QoS Stats tab provides the user with active statistics for each of the QoS buffer queues. These stats are only available when the QoS mode is set to Custom. The TX QoS Stats tab is shown in Figure 3-43 TX QOS Mode Custom Stats Page.

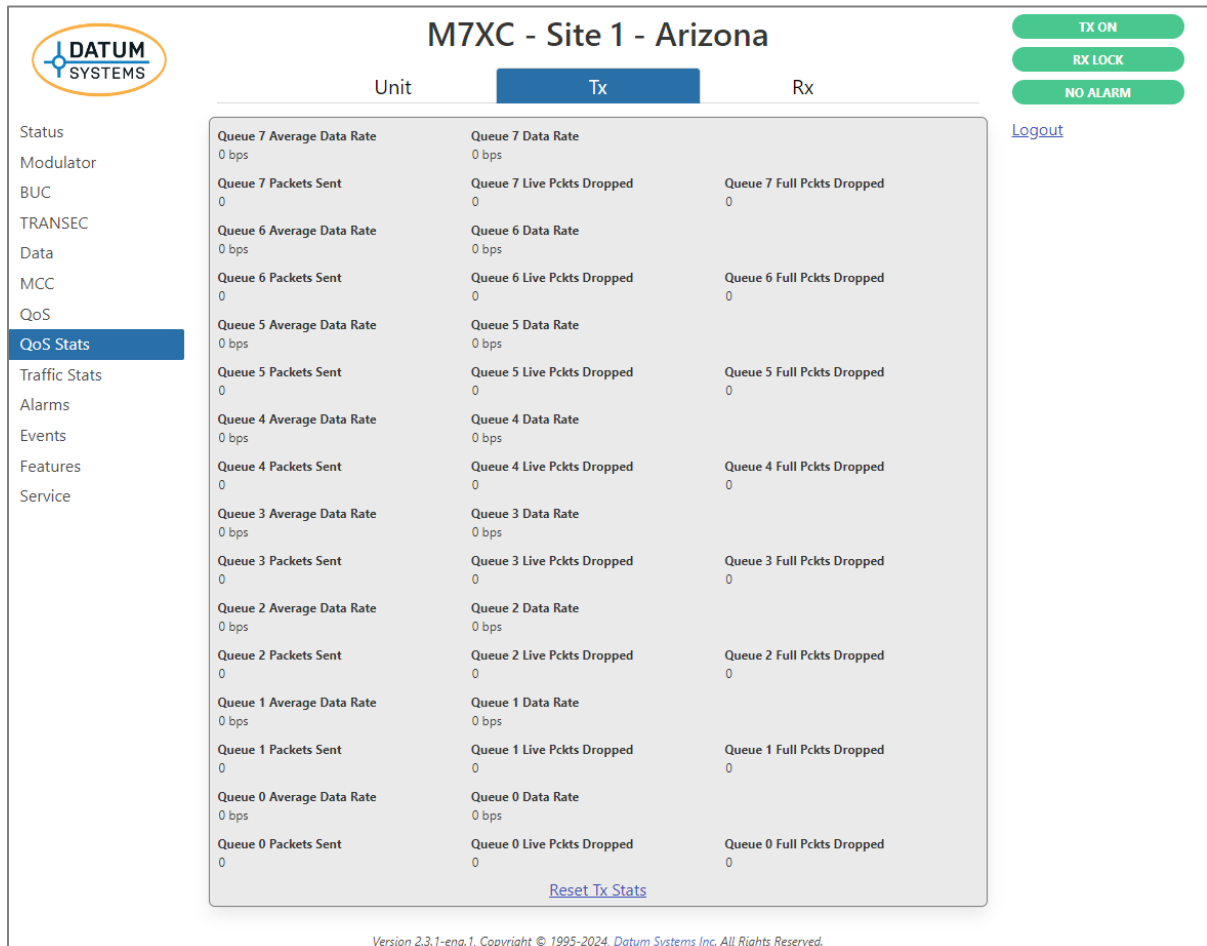


Figure 3-43 TX QOS Mode Custom Stats Page

The QoS Stats report Live Pkts Dropped (packets dropped due to exceeding the time to live in the queue) and Queue Full Pkts Dropped (dropped due to 16MB queue for that priority being full) for each Tx QoS Queue. A large burst of packets can be ingested as long as they are drained before the live time expires and they do not overflow the 16 MB queues.

3.5.9 >TX >Traffic Stats

The TX “Traffic Stats” tab (Figure 3-44 TX Traffic Stats) displays the TX Traffic Stats of the transmit Data and Packet parameters.

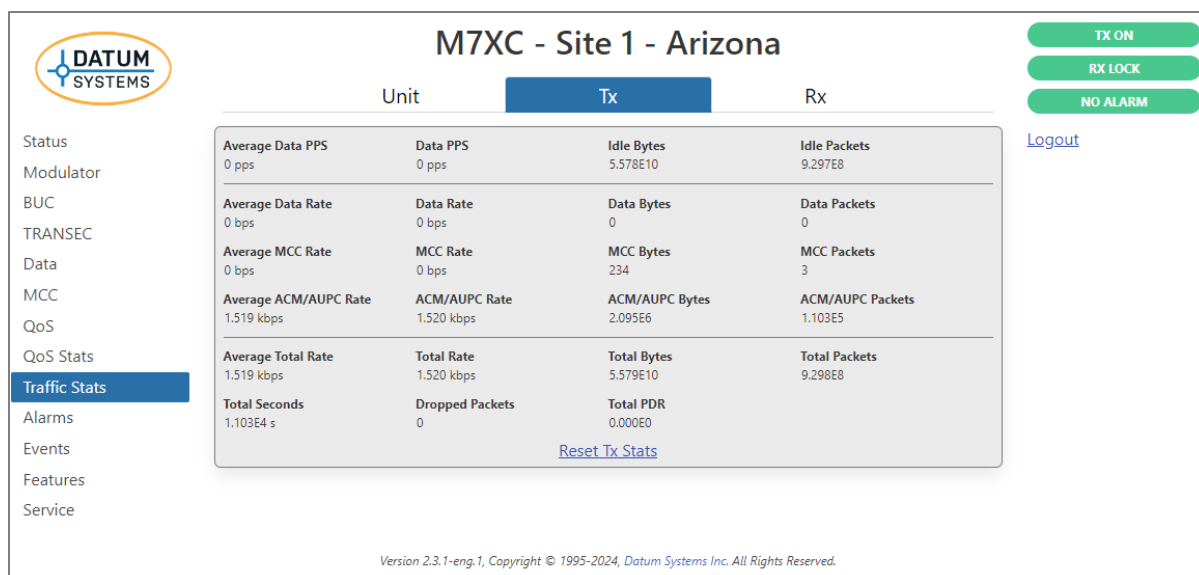


Figure 3-44 TX Traffic Stats

- **Average Data PPS:** Displays avg user traffic PPS since last Reset
- **Data PPS:** Displays current statistics of the user traffic PPS
- **Idle Bytes:** Displays current statistics of the transmit Idle Bytes
- **Idle Packets:** Displays total transmit Idle Packets since last Reset
- **Average Data Rate:** Displays Average Data Rate since last Reset
- **Data Rate:** Displays current statistics of the transmit Data Rate
- **Data Bytes:** Displays total transmit Data Bytes since last Reset
- **Data Packets:** Displays total transmit Data Pkts since last Reset
- **Average MCC Rate:** Displays Avg MCC OH rate since last Reset
- **MCC Rate:** Displays current statistics of the transmit MCC Rate
- **MCC Bytes:** Displays total transmit MCC Bytes since last Reset
- **MCC Packets:** Displays total transmit MCC Pkts since last Reset
- **Average ACM/AUPC Rate:** Avg MCC OH rate since last Reset
- **ACM/AUPC Rate:** Current statistics of the transmit MCC Rate
- **ACM/AUPC Bytes:** Total transmit MCC Bytes since last Reset
- **ACM/AUPC Packets:** Total transmit MCC Pkts since last Reset
- **Average Total Rate:** Average Total Data Rate over the WAN since last Reset
- **Total Rate:** Current Total Data Rate over the WAN
- **Total Bytes:** Total transmitted bytes over the WAN since last Reset
- **Total Packets:** Total TX Pkts over the WAN since last Reset
- **Total Seconds:** Elapsed time of the traffic statistics since last Reset
- **Dropped packets:** Total Pkts dropped due to congestion on the TX path since Reset
- **Total PDR:** Total Pkts Dropped / Total Pkts targeted to be sent over the WAN since Reset
- **Reset TX Stats:** Restart all Averages and Totals for all TX Traffic Stats

3.5.10 >TX >Alarms

The TX “Alarms” tab (Figure 3-45 TX Alarms Page) provides configurable action to be taken by the M7XC when a TX alarm condition occurs.



Figure 3-45 TX Alarms Page

- *Carrier Alarm*: Action if Carrier TX is Off
- *MCC Remote Channel Alarm*: Action if no MCC messages are received from the far side remote modem
- *ACM Low Es/No Alarm*: Action if the far side remote reports and Es/No lower than the user setting of >TX >Alarms >ACM Low Es/No Limit
- *ACM Low Es/No Limit*: User selection of far side remote Demod Es/No reading that is too low (-5.0 to +22.0 dB)
- *TRANSEC Alarm*: Action if TRANSEC is optioned but not enabled or not properly configured
- *Data Inactive Alarm*: Action if no traffic is seen on J6 User Traffic Port

NOTE: For a full description of what alarm configurable actions are, see (Section 3.4.10 >Unit >Alarms)

3.5.11 >TX >Events

The TX “Events” tab (Figure 3-45 TX Alarms Page) provides configurable action to be taken by the

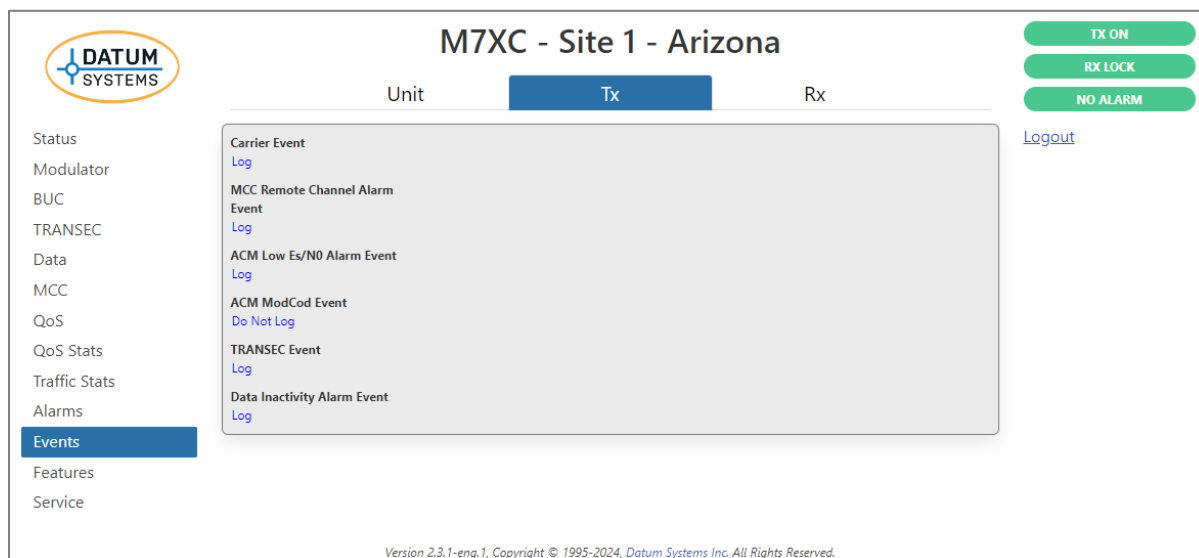


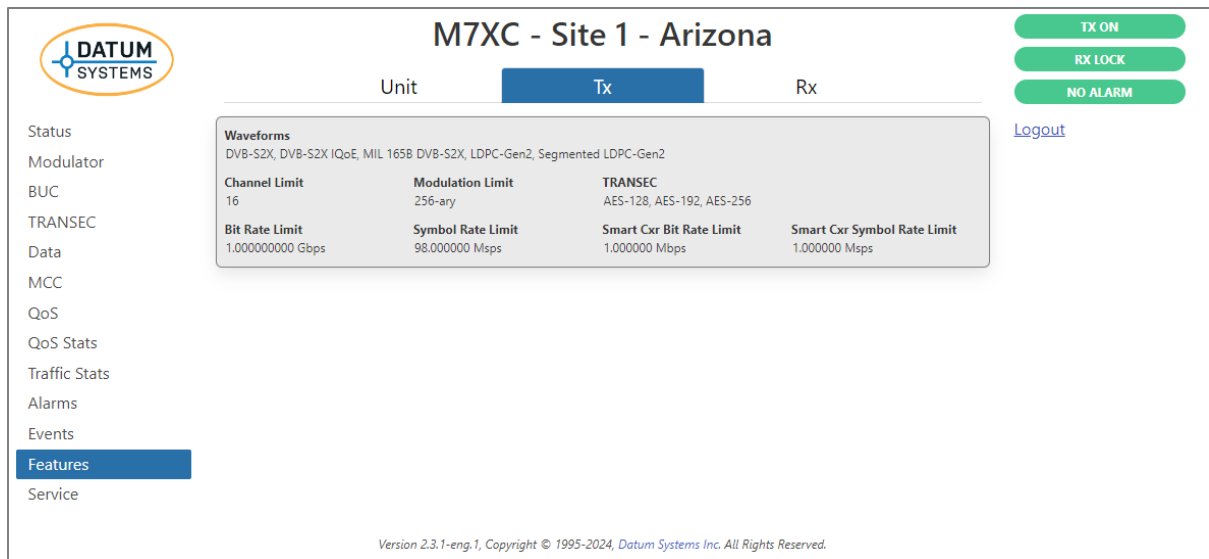
Figure 3-46 TX Events

- *Carrier Event*: Action if Carrier TX is Off
- *MCC Remote Channel Alarm Event*: Action if no MCC messages are received from the far side remote modem
- *ACM Low Es/No Alarm Event*: Action if the far side remote reports and Es/No lower than the user setting of >TX >Alarms >ACM Low Es/No Limit
- *TRANSEC Event*: Action if TRANSEC is optioned but not enabled or not properly configured
- *Data Inactive Alarm Event*: Action if no traffic is seen on J6 User Traffic Port

NOTE: For a full description of what event configurable actions are, see (Section 3.4.11 >Unit >Events)

3.5.12 >TX >Features

The TX “Features” tab (Figure 3-47 TX Features Page) displays the current feature capability of the M7XC.



M7XC - Site 1 - Arizona

Unit **Tx** Rx

Waveforms
DVB-S2X, DVB-S2X IQoE, MIL 1658 DVB-S2X, LDPC-Gen2, Segmented LDPC-Gen2

Channel Limit 16	Modulation Limit 256-ary	TRANSEC AES-128, AES-192, AES-256	
Bit Rate Limit 1.000000000 Gbps	Symbol Rate Limit 98.000000 Msps	Smart Cxr Bit Rate Limit 1.000000 Mbps	Smart Cxr Symbol Rate Limit 1.000000 Msps

TX ON
RX LOCK
NO ALARM
[Logout](#)

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Figure 3-47 TX Features Page

- **Waveforms:** Displays the selections of transmit waveforms available to the modulator
- **Channel Limit:** Displays the number of modulators or TX channels (Segmented) the unit is capable of
- **Modulation Limit:** Displays all or the maximum modulation capability of the modulator
- **TRANSEC:** displays the levels of decryption that the unit is capable of when TRANSEC is optioned
- **Bit Rate Limit:** Display of the bit rate maximum capability of the modulator
- **Symbol Rate Limit:** Displays of the symbol rate maximum of the modulator
- **Smart Cxr Bit Rate Limit:** Displays maximum bit rate of the Smart Cxr feature
- **Smart Cxr Symbol Rate Limit:** Displays maximum symbol rate of the Smart Cxr feature

3.5.12.1.1 TX Service

The TX “Service” tab (Figure 3-48 TX Service Page) displays the temperature of the major modulator system components.

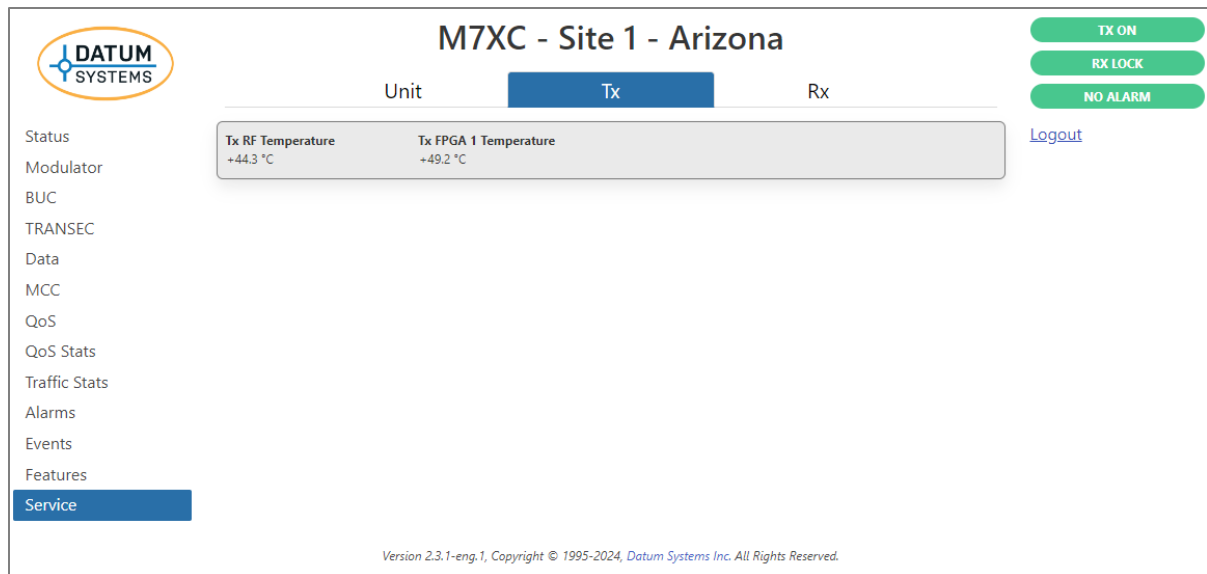


Figure 3-48 TX Service Page

- *TX RF Temperature*: Displays the current temperature of the TX RF
- *TX FPGA 1 Temperature*: Displays the current temperature of the TX FPGA

3.6 M7XC Web Browser RX Tab and Pages

The RX pages are used to configure the receive parameters of the M7XC. The IF, Symbol, Data Rates, number of active demodulators and other receive parameters are controlled in the “RX” pages.

NOTE: For use cases where the M7XC modem is configured to have more than one demodulator it is important to take note that the parameters you are looking at are specific to the particular demodulator you have highlighted in the far right Status bar. See Figure 3-49 RX GUI Example of Multi-Demod Configuration. In the below example we have clicked and highlighted demod channel #3 and all status and settings will pertain specifically to demod #3.



Figure 3-49 RX GUI Example of Multi-Demod Configuration

For most users a single modulator and single demodulator are being monitored and configured. For this reason, this manual will commonly refer to GUI pages that are for a single mod and single demod. For more information on multi-demod configuration with Point to Point, MESH or Segmented modes of operation please see the appropriate Appendix that addresses these use cases in detail.

3.6.1 >RX >Status

The RX “Status” tab (Figure 3-50 RX Status) displays the RX Status of the receive parameters of the demodulator under review.

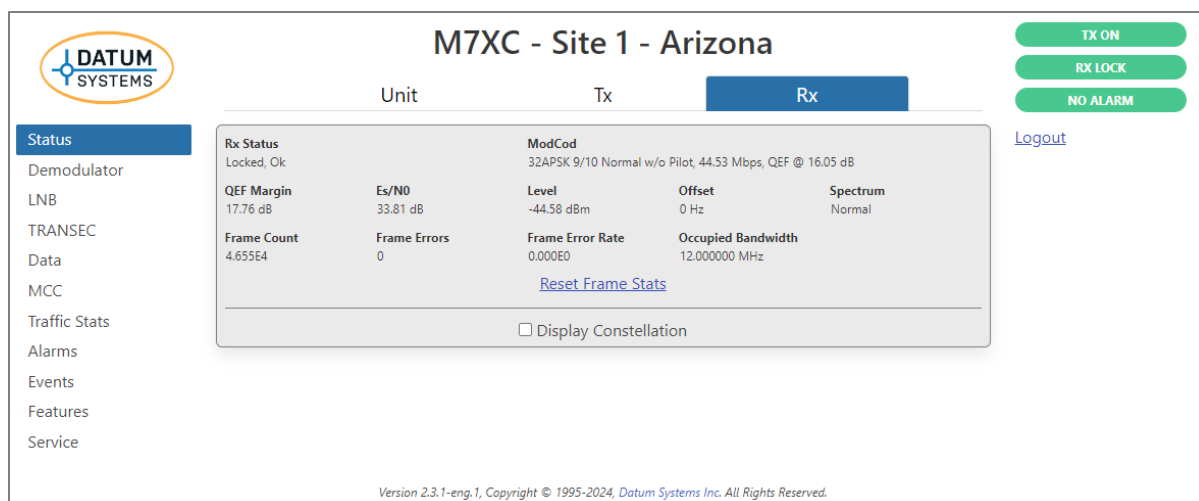


Figure 3-50 RX Status

- **RX Status:** Displays current status of the receive path of the modem. Normal operation status will declare “Locked, Ok” or “UNLOCKED”. If there is a fault in the RX path or RX configuration it will be displayed here
- **ModCod:** Displays current MODCOD being received “UNLOCKED” will be displayed if the demod is not locked.
- **QEF Margin:** Displays current QEF Margin as reported by the local demod Es/No for the MODCOD being received. The margin is based on the MODCOD QEF Es/No and does not take into account any ACM or Hysteresis margin added. NOTE: A reading of -10dB indicates the demod is unlocked.
- **Es/No:** Displays the current Es/No level as reported by the local demod. NOTE: A reading of -10dB indicates the demod is unlocked.
- **Level:** Displays the current Receive Signal Level (RSL) as reported by the local demod. The RSL will display the total input power of the receive BW it is tuned to.
- **Offset:** WHEN DEMOD IS LOCKED - Displays the difference between the expected / programmed RX frequency and the frequency the demod actually locked to
- **Spectrum:** WHEN DEMOD IS LOCKED - Displays the detected spectrum (Normal or Inverted)
- **Frame Count:** WHEN DEMOD IS LOCKED – Displays the total number of FEC or BB frames received since the “Reset Frame Stats” was set.
- **Frame Errors:** WHEN DEMOD IS LOCKED – Displays the total number of errored FEC or BB frames received since the “Reset Frame Stats” was set.
- **Frame Error Rate:** WHEN DEMOD IS LOCKED – measurement of (Frame Errors / Frame Count) since the “Reset Frame Stats” was set.
- **Occupied BW:** WHEN DEMOD IS LOCKED – Displays the calculated BW of SR * (1+ROF) for the carrier.
- **Reset Frame Stats:** Reset all counters on page

- **Display Constellation (Check Box):** When checked, the >RX >Status page will add a constellation display of the carrier the demodulator is locked to. The Constellation display can help in system debug.

A clean signal with no noise will have all symbols mapped to clearly defined locations on the X/Y plot as seen below (Figure 3-51 “Clean” 32-APSK Constellation). Symbols that are “fuzzy” or show wide circular patterns that approach other symbols indicate greater noise. Symbols that show non-circular / oval or “smeared” shaped symbols may be in a state of distortion (Amplitude or Phase Distortion) indicated that something in the link is in saturation.

The waveform of the demodulated carrier will be shown in the bottom right corner of the constellation.

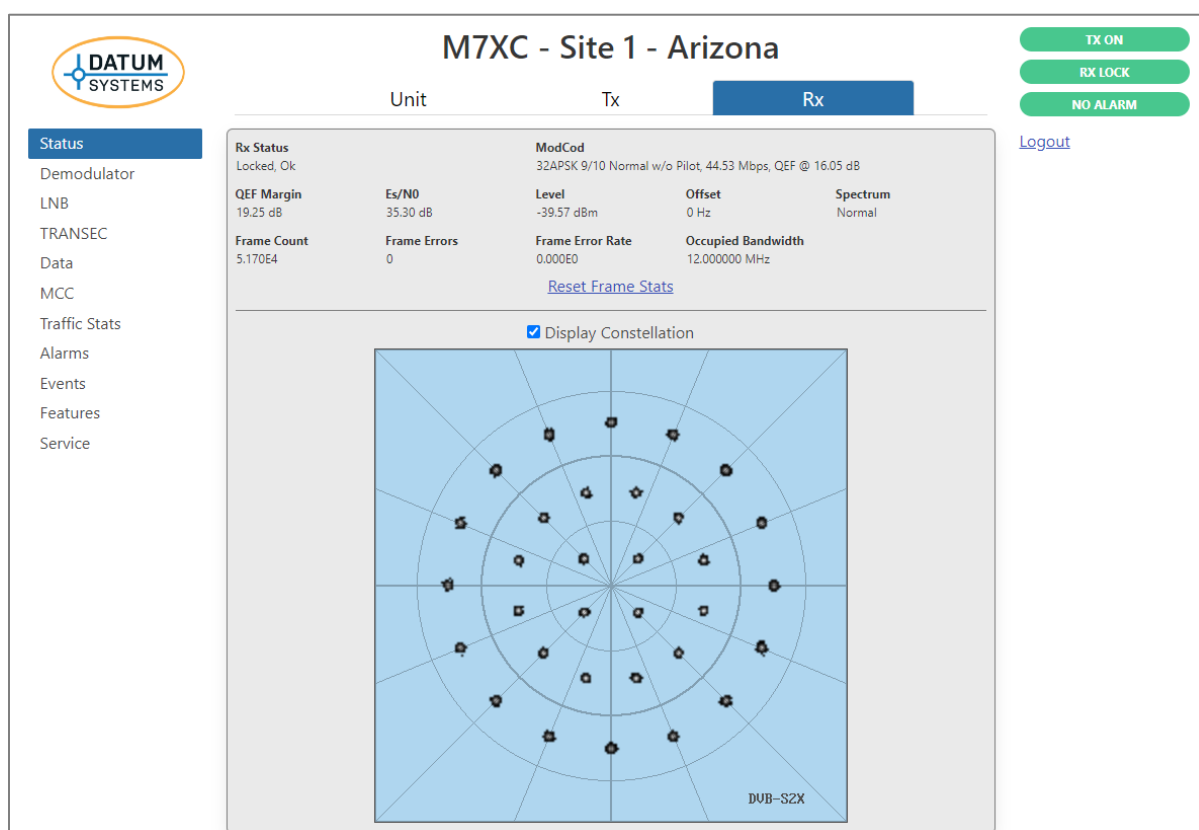


Figure 3-51 “Clean” 32-APSK Constellation

3.6.2 >RX >Demodulator

The RX demodulator page is used to set the basic configuration of the demod. See Figure 3-52 RX Demodulator.

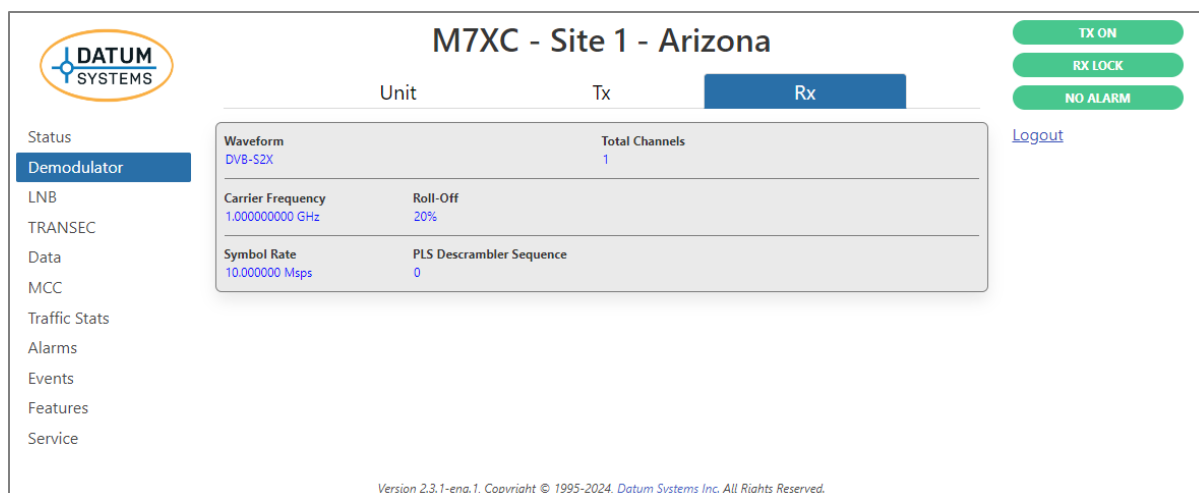


Figure 3-52 RX Demodulator

- **Waveform:** Current waveform expected by demod (Rx Off, DVB-S2X, MIL 165B DVB-S2X, LDPC-Gen2, Segmented LDPC-Gen2, DSSS, Spectrum Analyzer) must match far side TX waveform (except for Spectrum Analyzer setting).
- **Total Channels:** If optioned, user setting of number of demods active (1 – 16).
- **Carrier Frequency:** User setting of center frequency of expected carrier.
- **Roll-Off:** User setting of Roll-Off (ROF) of expected carrier (should match far side TX ROF)
- **Symbol Rate:** User setting of Roll-Off (ROF) of expected carrier. Must match far side TX Symbol Rate.
- **Block Size:** (waveform dependent) User setting of FEC Block Size of expected carrier. Must match far side TX Block Size.
- **PLS Descrambler Sequence / Scrambler:** (waveform dependent) User setting of de-scrambler. Must match far side TX scrambler.

3.6.2.1 Spectrum Analyzer Waveform Operation

The M7XC modem has a built-in spectrum analyzer function. The spectrum analyzer is a very capable and useful tool that can be used for system testing and line-up.

The analyzer has very useful functions that are commonly found in analyzers and are simple to configure. See (Figure 3-53 >RX >Demodulator >Waveform “Spectrum Analyzer”)

Note: When the “Spectrum Analyzer” is selected as the waveform for the demodulator the User Traffic will not be passed. The demodulator and decoder are disabled when in Spectrum Analyzer mode.

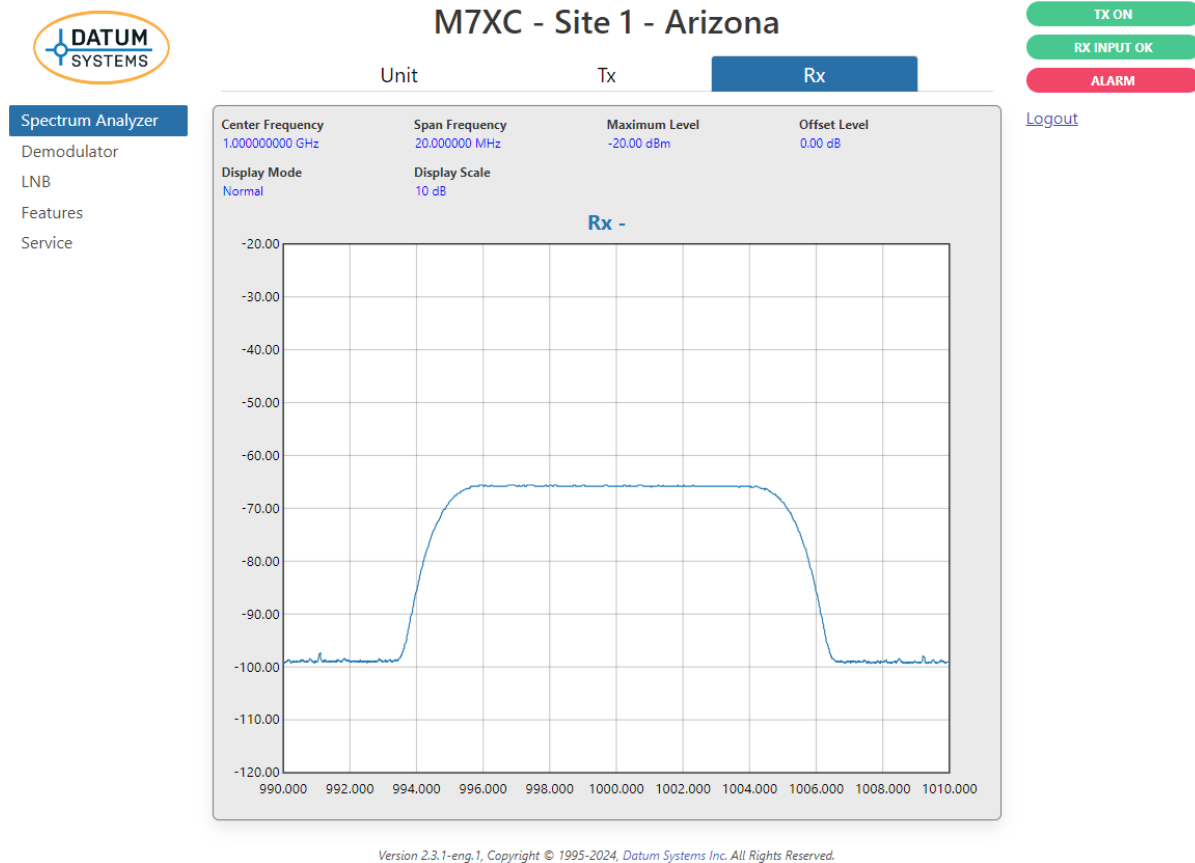


Figure 3-53 >RX >Demodulator >Waveform “Spectrum Analyzer”

- **Center Frequency:** Set the center frequency of the analyzer. Setting of 960 – 2240 MHz) allowed.
- **Span Frequency:** The bandwidth the analyzer will evaluate centered on the “Center Frequency”
- **Maximum Level:** The maximum power level the analyzer will adjust to (this will set actual power level expectation of the analyzer)
- **Offset Level:** The additional level adjustment to the analyzer GUI, does not set actual power level expectation of the analyzer
- **Display Mode:** Settings on how the spectrum is displayed or captured.
 - Normal: Standard display as shown in above figure
 - Min Hold: Will constantly analyze the spectrum and display the lowest signal value in the BW measured as well as the current signal value
 - Max Hold: Will constantly analyze the spectrum and display the highest signal value in the BW measured as well as the current signal value
 - Min & Max Hold: Will constantly analyze the spectrum and display the lowest signal value in the BW measured as well as the current signal value (see Figure 3-54 Spectrum Analyzer Min & Max Hold Example)
- **Display Scale:** Set the power scale per y-axis grid line (1, 2, 5 or 10dB per grid)
- **Clear Min & Max:** Clears the Min / Max traces and re-starts Min / Max measurements

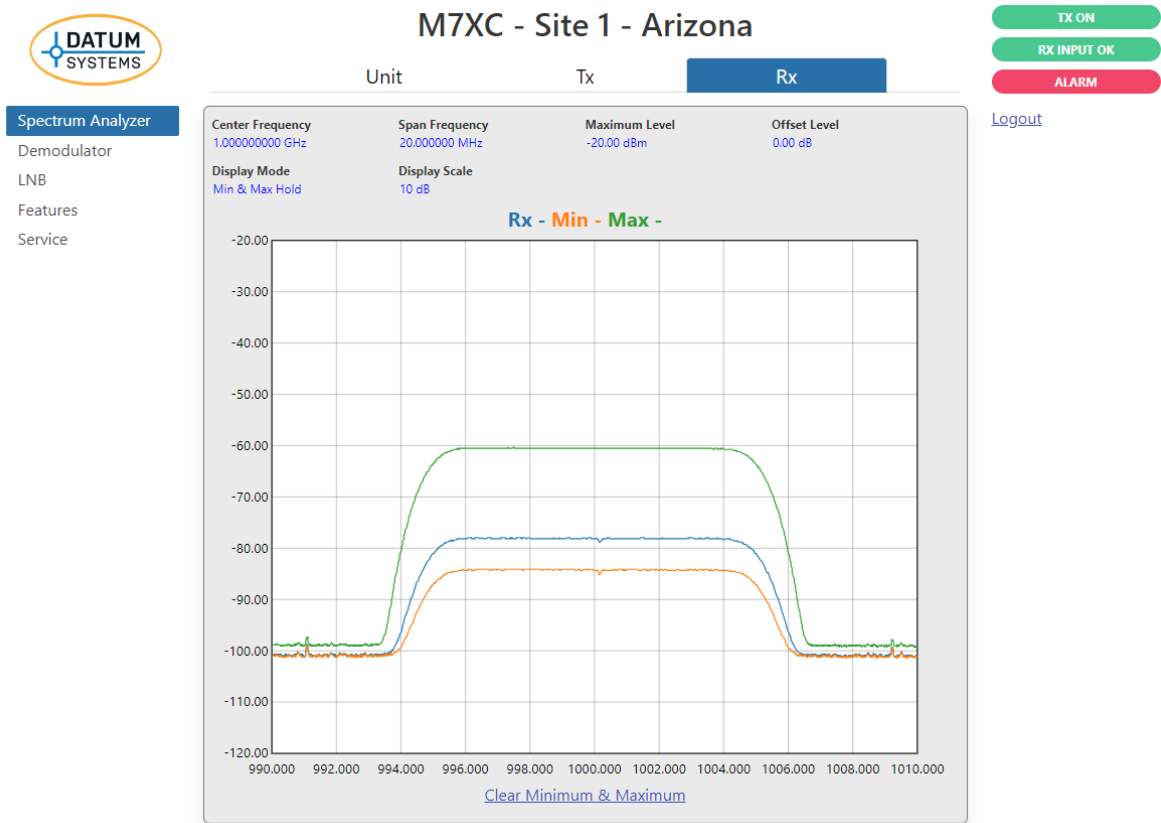


Figure 3-54 Spectrum Analyzer Min & Max Hold Example

3.6.3 >RX >LNB

The RX “LNB” tab (Figure 3-55 RX LNB Page) provides the ability to set parameters of the receive RF conversion equipment to display the RF frequency to be received based on the IF frequency setting. This is helpful to the operator to confirm that the transmit carrier is in the proper transponder location.

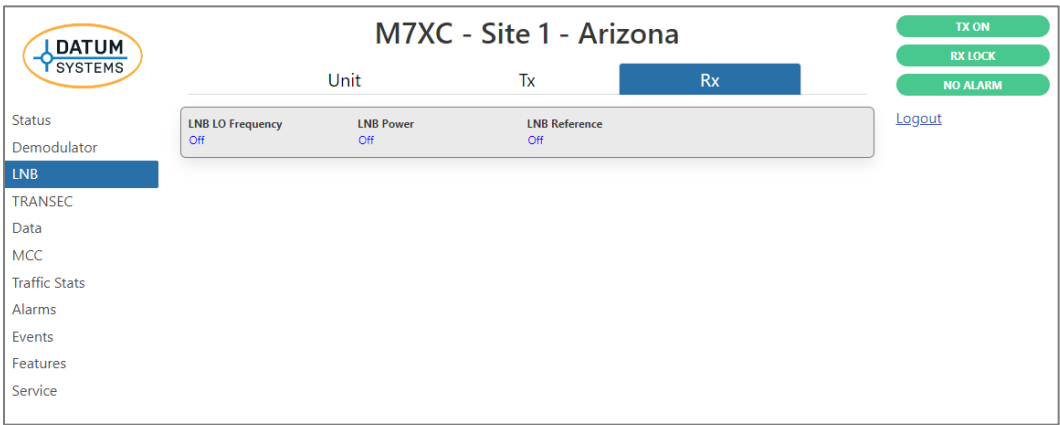


Figure 3-55 RX LNB Page

- **LO Frequency (MHz):** Configuration for the LNB LO frequency to calculate and display the RF frequency. If set to zero (0), the RF frequency displayed is the same as the IF frequency and “Off” will be shown
- **LNB Power:** Configuration for the LNB VDC Power (Off, + 13V, + 18V)
- **LNB Reference:** Configuration for the LNB Reference (Off, 10 MHz, 50 MHz)

3.6.4 >RX >TRANSEC (Optionally Equipped)

The optional TRANSEC operation of the modem will encrypt / decrypt the user traffic. Encryption effects all user traffic going over the WAN including the User data, IP headers, Layer 2.5 headers, Ethernet headers and GSE encapsulation headers. When TRANSEC is enabled on the RX side, the far side TX must have a matching TRANSEC encryption setting when transmitting.

NOTE: If data is arriving at the decoder that is encrypted but the decryption is Off or decryption key does not match you will see a >RX >Status >Rs Status “Locked, Tx/Rx CONFIG MISMATCH” alarm.

For details on setting up TRANSEC for the first time see section (3.5.4>TX >TRANSEC (Optionally Equipped))

Below is an example of a properly configured >RX >TRANSEC page that is properly passing traffic (Figure 3-56 RX TRANSEC)

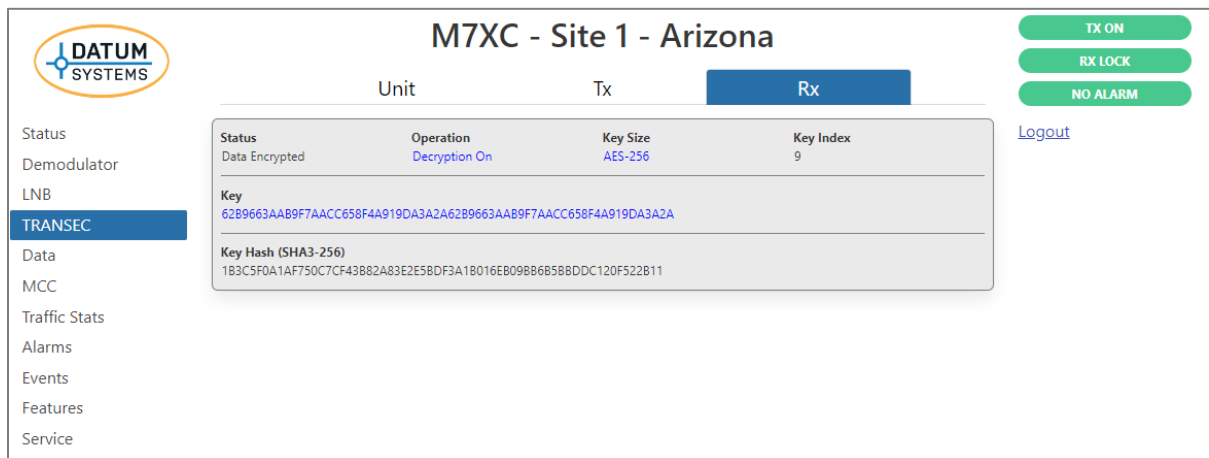


Figure 3-56 RX TRANSEC

- **Status:** “KEYS NOT INITIALIZED” indicates that a key with the proper length matching the >Key Size has not been entered. “Data Encrypted” indicated TRANSEC is configured and operating.
- **Operation:** (Decryption Off, Decryption On)
 - Decryption Off: Will disable TRANSEC on the RX path
 - Decryption On: Decryption is Enabled
- **Key Size:** (AES-128, AES-192, AES-256) will set the key length with the larger numbers indicating longer (and more secure) keys.
- **Key Index:** Displays the rolling Key changes that occur automatically, there are over 64,000 unique keys that the modem will roll through before any key is

repeated. Keys are rolled once a set number of frames have been transmitted. The key index will automatically synchronize with the far side decryption if the user entered >Tx >TRANSEC >Key matches the >Rx >TRANSEC >Key.

- **Key Hash (SHA3-256):** Displays the SHA3-256 standards-based hash of the user entered Key. When logging in as a read only user, only the Key Hash can be viewed for verification that the far side demod has the matching Key Hash as the actual Key will not be visible.

3.6.5 >RX >Data

The RX Data page (Figure 3-57 RX Data) is used to configure RX framing and special message framing.

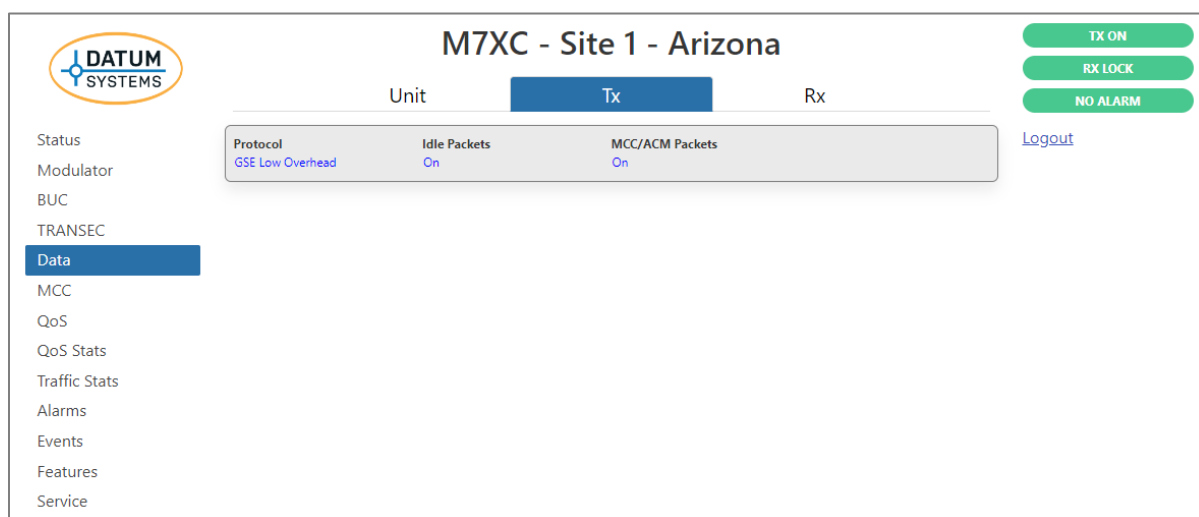


Figure 3-57 RX Data

- **Protocol:** (waveform dependent) Defines the IP Encapsulation method used by the framer (GSE DVB, GSE Low Overhead, GSE CT). The lowest overhead framing is “GSE Low Overhead”. Other framing modes are available for compatibility purposes with other vendors or standards only. “GSE Low Overhead” is default.
- **Idle Packets:** (On, Off) when “On”, the modem will send IP packets filled with pseudo random data when the user data buffer is empty. This is done to keep the carrier spectrum random. “On” is default.
- **MCC/ACM Packets:** (On, Off) when “On”, the modem enable an In-Band channel between the near side and far side modems for internal communications. This communications channel is REQUIRED for ACM operation and to get remote demod Es/No & Es/No margin. Default is “On”, when “Off” the modem will not use the OH channel and the link will be slightly more efficient. The Overhead channel has a configurable OH data rate, see (section 3.5.6 >TX >MCC) on how to configure the OH channel.

3.6.6 >RX >MCC

The RX “MCC” tab (Figure 3-58 RX MCC Page) provides the ability to set the parameters of the Send and Receive Modem Communications Channel (MCC) between the local and remote modem.

See Appendix D MCC Operation for details and examples.

M7XC - Site 1 - Arizona

Unit Tx **Rx**

MCC Mode	MCC Protocol	MCC Address	Activity
Full Access	M7 Binary Packet	1	None

Remote Tx Path	Remote Tx MCC Address
Out MCC Port	1

TX ON
RX LOCK
NO ALARM
[Logout](#)

Figure 3-58 RX MCC Page

- **MCC Mode:** Sets the MCC access mode (Read Only, Full Access), default is “Full Access”
- **MCC Protocol:** Sets the MCC channel protocol, set to match >TX >MCC Protocol setting
- **MCC Address:** Sets the modem’s MCC address (0 to 255, 255 = Global), this should match the >TX > MCC >MCC Address
- **Activity:** Set to blink the “Online” LED on the modem if MCC activity is detected (None, Flash Online)
- **Remote Tx Path:** Sets the MCC return path function (Off, Local, One Channel Only, Out MCC Port). “Off” will disable MCC messages, “Local” is primarily used for bench testing, “Out MCC Port” is default and how most networks need to be configured.
- **Remote Tx MCC Address:** Should be set to the MCC address of the far side TX modulator that is being landed by the demod

3.6.6.1.1 RX Traffic Stats

The RX “Traffic Stats” tab (Figure 3-59 RX Traffic Stats Page) displays the RX Traffic Stats of the receive Data and Packet parameters.

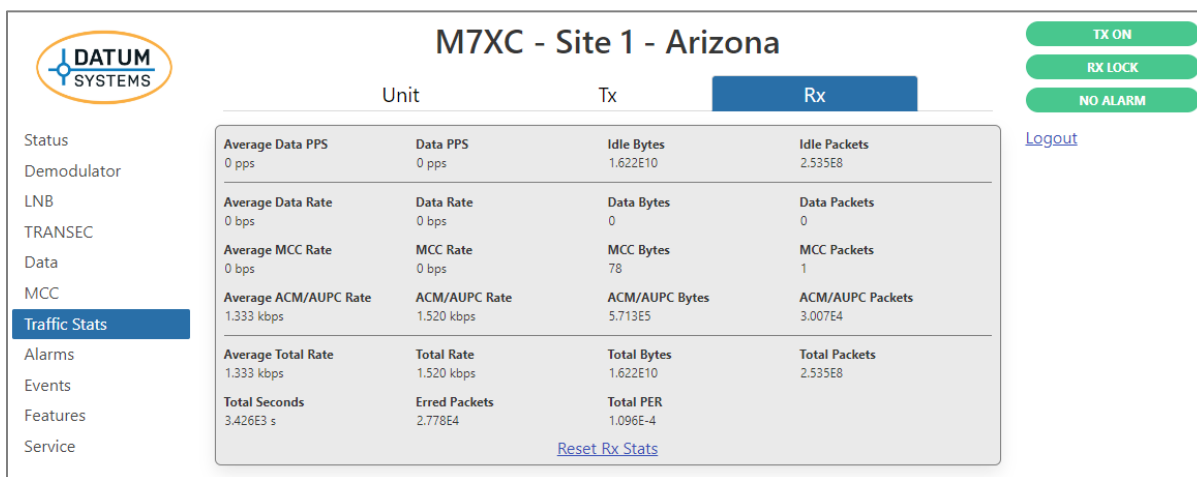


Figure 3-59 RX Traffic Stats Page

- **Average Data PPS:** Displays current statistics of the receive Average Data PPS
- **Data PPS:** Displays current statistics of the receive Data PPS
- **Idle Bytes:** Displays current statistics of the receive Idle Bytes
- **Idle Packets:** Displays current statistics of the receive Idle Packets
- **Average Data Rate:** Displays current statistics of the receive Average Data Rate
- **Data Rate:** Displays current statistics of the receive Data Rate
- **Data Bytes:** Displays current statistics of the receive Data Bytes
- **Data Packets:** Displays current statistics of the receive Data Packets
- **Average MCC Rate:** Displays current statistics of the receive Average MCC Rate
- **MCC Rate:** Displays current statistics of the receive MCC Rate
- **MCC Bytes:** Displays current statistics of the receive MCC Bytes
- **MCC Packets:** Displays current statistics of the receive MCC Packets
- **Average ACM/AUPC Rate:** Displays current statistics of the receive Average ACM/AUPC Rate
- **ACM/AUPC Rate:** Displays current statistics of the receive ACM/AUPC Rate
- **ACM/AUPC Bytes:** Displays current statistics of the receive ACM/AUPC Bytes
- **ACM/AUPC Packets:** Displays current statistics of the receive ACM/AUPC
- **Average Total Rate:** Displays current statistics of the receive Average of all traffic Rate
- **Total Rate:** Displays current statistics of the receive total Rate of all traffic
- **Total Bytes:** Displays current statistics of the receive Total Bytes
- **Total Packets:** Displays current statistics of the receive Total Packets
- **Total Seconds:** Time since last “Reset Rx Stats”
- **Erred Packets:** Total packets received with failed FCS or parity check

- Total PER: (Erred Packets / Total Packets)
- *Reset TX Stats*: By clicking on the “Reset TX Stats” button, all “Averaged” data will be reset

3.6.7 >RX >Alarms

The RX “Alarms” tab (Figure 3-60 RX Alarms Page) provides configurable action to be taken by the M7 modem when a RX alarm condition occurs.

For a list of the Alarm Settings Descriptions or Alarm Comparisons see (Table 3-1 Alarm Settings Description, Table 3-2 Alarm Settings Comparison)

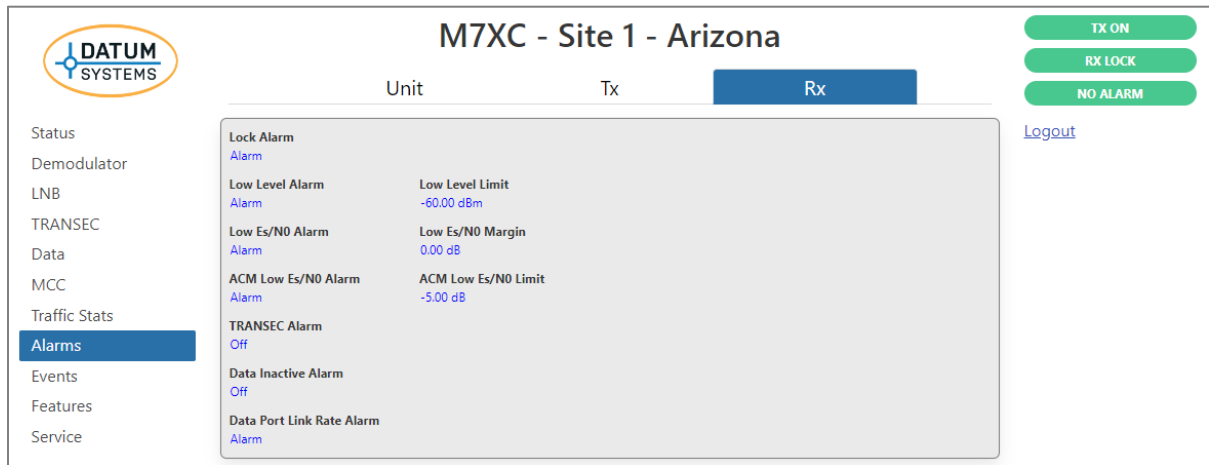


Figure 3-60 RX Alarms Page

- *Lock Alarm*: Triggers if the carrier is unlocked
- *Low Level Alarm*: Triggers if the locked carrier has an RSL (Receive Signal Level) reading below the >Low Level Limit setting.
- *Low Level Limit (dB)*: User setting of the RSL level that should trigger an Alarm or Event
- *Low Es/No Alarm*: Triggers if the locked carrier has an Es/No reading below the >Low Es/No Margin setting.
- *Low Es/No Margin (dB)*: User setting of the Es/No margin that should trigger an Alarm or Event
- *ACM Low Es/No Alarm*: Selection of alarm notification taken by the modem when the Es/No level of the receive channel is below the ACM Low Es/No Limit setting
- *ACM Low Es/No Limit (dBm)*: Configuration of the receive channel's Es/No level margin threshold that will trigger the ACM Low Es/No Alarm
- *TRANSEC Alarm*: Triggers when TRANSEC is not properly configured, mismatched or disabled when installed
- *Data Inactive Alarm*: Triggers when data is not egressing through the LAN Ethernet port
- *Data Port Link Rate Alarm*: Triggers when the modem WAN RX capacity exceeds the LAN port rate configuration of the modem.

3.6.8 >RX >Events

The RX Events tab (Figure 3-60 RX Alarms Page) provides configurable action to be taken by the M7 modem when a RX Event condition occurs.

For a list of the Event Settings Descriptions see (Table 3-3 Event Settings Description)

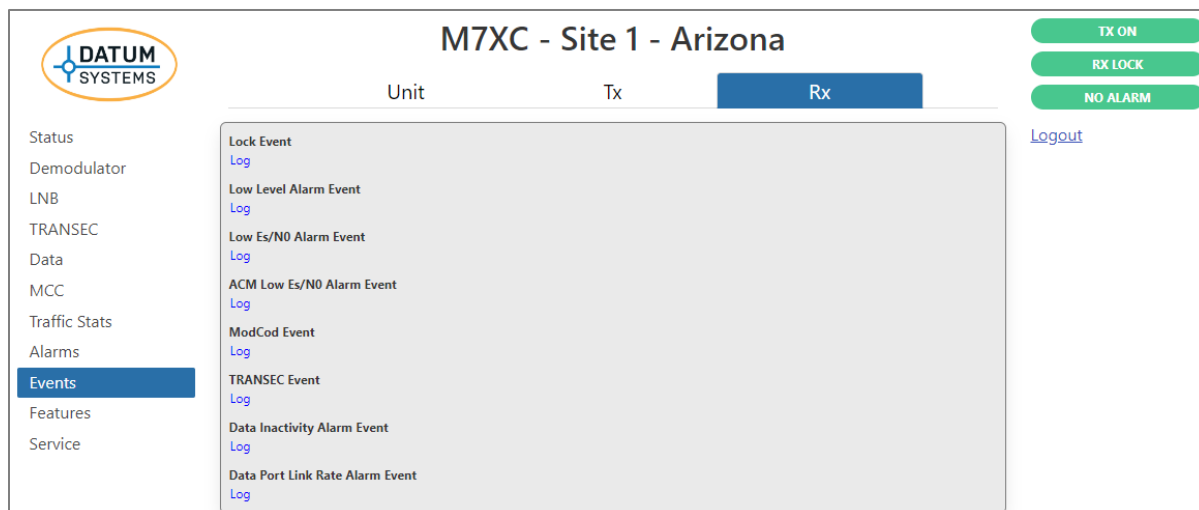


Figure 3-61 RX Alarms Page

- **Lock Event:** Triggers if the carrier is unlocked
- **Low Level Alarm Event:** Triggers if the locked carrier has an RSL (Receive Signal Level) reading below the >Low Level Limit setting.
- **Low Es/No Alarm Event :** Triggers if the locked carrier has an Es/No reading below the >Low Es/No Margin setting.
- **ModCod Event:** Triggers if an unknown or unexpected MODOCD is received
- **TRANSEC Event:** Triggers when TRANSEC is not properly configured, mismatched or disabled when installed
- **Data Inactive Alarm Event:** Triggers when data is not egressing through the LAN Ethernet port
- **Data Port Link Rate Alarm Event:** Triggers when the modem WAN RX capacity exceeds the LAN port rate configuration of the modem.

3.6.9 >RX >Features

The RX “Features” tab (Figure 3-62 RX Features Page) displays the current feature capability of the demodulator(s).

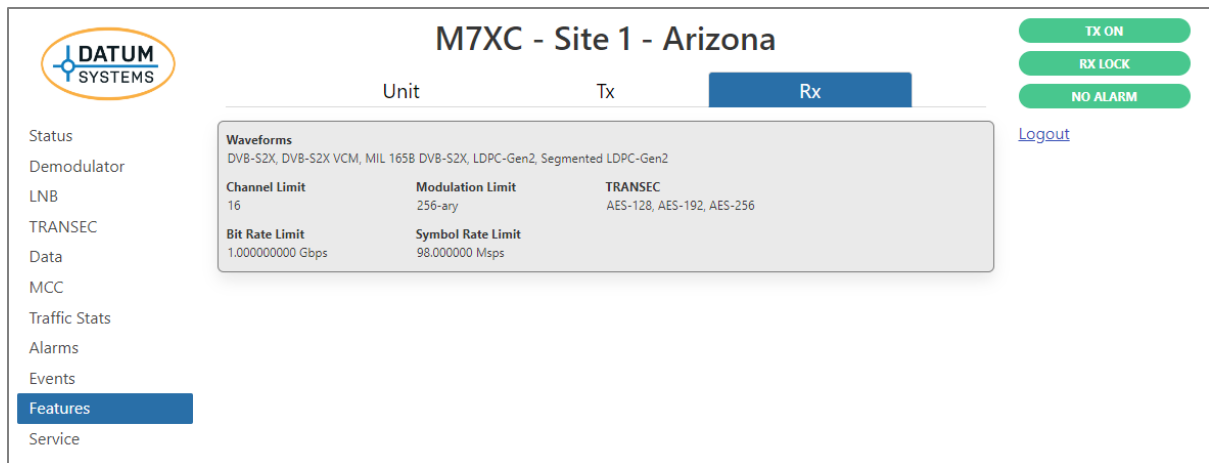


Figure 3-62 RX Features Page

- **Waveforms:** Displays the selections of receive waveforms available to the demodulator(s)
- **Channel Limit:** Displays the number of demodulators or channels (Segmented) that the unit is capable of enabling
- **Modulation Limit:** Displays all, or the maximum demodulation capability of the modulator
- **TRANSEC:** displays the levels of decryption that the unit is capable of when TRANSEC is optioned
- **Bit Rate Limit:** Display of the bit rate maximum capability of the demodulator
- **Symbol Rate Limit:** Displays of the symbol rate maximum of the demodulator

3.6.10 >RX >Service

The RX “Service” tab (Figure 3-63 RX Service) displays the temperature of the major demodulator system components.

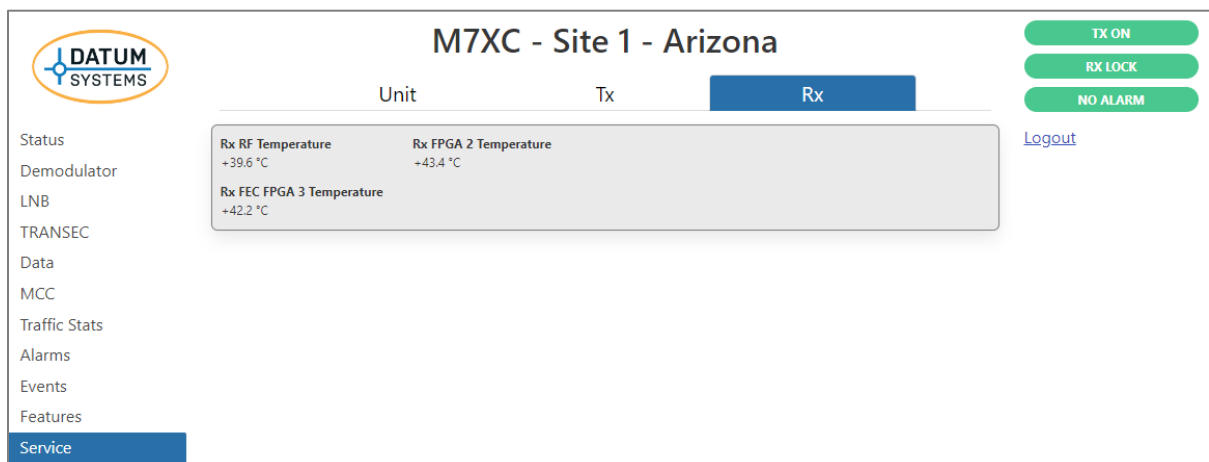


Figure 3-63 RX Service

- *RX RF Temperature*: Displays the current temperature of the RX RF section of the modem
- *RX FPGA 2 Temperature*: Displays current temperature of the RX FPGA 2
- *RX FPGA 3 Temperature*: Displays current temperature of the RX FPGA 3

3.7 Modem Configuration

Configuring the modem operating parameters is essential prior to placing the unit into service. The operating parameters can be set by using any of the control interfaces.

3.7.1 Configuring the Modem for Operation

The following description assumes that the modem configuration setup is to be performed manually at a depot location. The modem can also be automatically configured using a central controller SNMP interface. No control application software is provided by Datum Systems and is the responsibility of the user organization.

3.7.1.1 Setting the Modem Station ID Name

Each modem contains two unique identification entries available via any control interface. They are the unit serial number and the Unit Name or “Station ID”. The serial number is set at the factory and cannot be changed. The Station ID is entered and changed whenever necessary. This field allows identification of the modem with up to 28 characters.

The Station ID is entered directly as text in the >Unit >Status page, “Station ID” parameter and is shown in Figure 3-64 Unit Status Station ID Page below. Once entered, the Station ID will be displayed at the top of the web page with the Model Name.

The screenshot displays the web interface for the M7XC Compact Satellite Modem. At the top, the title 'M7XC - Site 1 - Arizona' is shown in a red box. Below it, the 'Unit' tab is selected, and the 'Status' page is active. The 'Station ID' field is highlighted with a red box, showing 'Site 1 - Arizona'. The left sidebar contains navigation links: Status, Event Log, Store/Recall, Data Port, Mgmt Port, HTTP, SNMP, Reference, Monitor, Alarms, Events, Management, Service, Satellite, and ACU. The top right has buttons for TX ON, RX LOCK, and NO ALARM. The main content area shows the following status information:

Summary Status		Reference Status
No Alarms Or Warnings		Internal, Ok
Model	Serial Number	Firmware Version
M7XC	52501	2.3.1-eng.1
Build	Station ID	
Tx: BR20001-3 Rev B3T, Rx: BR20002-1 Rev B2	Site 1 - Arizona	
Date Time UTC	Trial Activation Time	Power Input Voltage
2024-09-26 15:14:48	Not Set	+24.0 V
Mgmt Port Status	Data Port Status	
Connected, 1 Gb	Connected, 1 Gb	

Figure 3-64 Unit Status Station ID Page

3.7.2 Setting Essential Parameters

Setting of the parameters below is essential to achieving proper operation. An improper setting of any one of these parameters will result in failure to close the link with the far end.

There are many other parameters that might need to be set on the modem for operation within the system. These include, but not limited to, interface settings, alarm function settings and configuration displays. The discussion in the following sections is to be used as a starting point and is not expected to explain every possible configuration available in the modem. These basic parameters are here to serve as a minimum checklist for basic operation.

Unit Settings	TX Settings	RX Settings
>Data Port >Data Port Link	>Mod >Waveform	>Demod >Waveform
>Mgmt >Address/Mask	>Mod >Output	>Demod > Carrier Frequency
	>Mod > Carrier Frequency	>Demod >Roll Off
	>Mod >Roll Off	>Demod > Symbol Rate
	>Mod > Level	>Demod >Scrambler
	>Mod > Symbol Rate	
	>Mod >Frame Size	
	>Mod >Scrambler	

3.7.3 Using the Internal or External Reference

The modem is designed with an internal Oven Compensated Crystal Oscillator (OCXO) that has a nominal stability of 2×10^{-8} at normal operating temperatures, and ages less than 0.1 ppm per year. The internal reference determines the accuracy of all frequencies and symbol/data rates of the modem. If the accuracy is not sufficient, an external reference can also be used.

The external reference is applied at SMA connector J1 shown in Figure 3-65 Internal / External Reference.

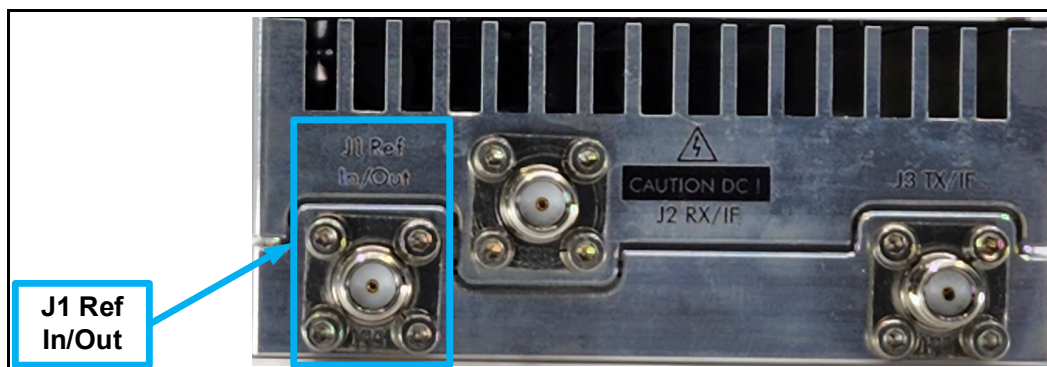


Figure 3-65 Internal / External Reference

The input and output reference frequency can be set to either 10 MHz or 50 MHz and can be selected using any of the control interfaces. An example of the web page configuration is shown in Figure 3-66 Unit Reference Page.

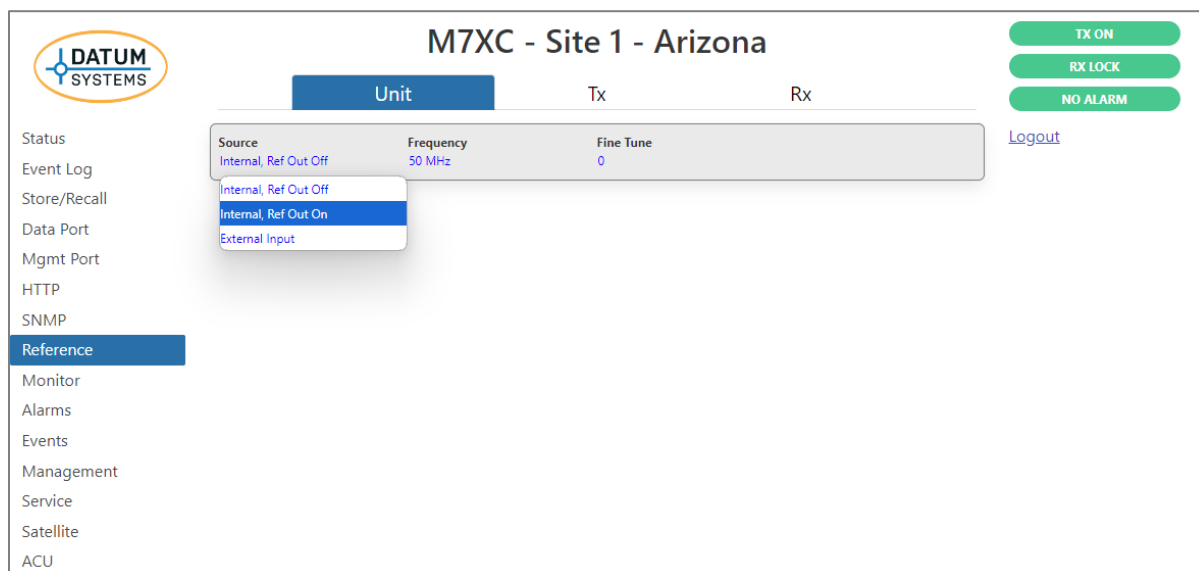


Figure 3-66 Unit Reference Page

The external reference input provides band-pass filtering from ~1 to 12 MHz. The external frequency reference source output must be low in phase noise.

3.8 Data Interface Technical Details

3.8.1 GigE Data Interface

The GigE Interface is a Layer 2 Bridge that provides high speed Internet capability to the M7XC. The GigE interface will not provide any IP routing functionality.

The purpose of the GigE is to act as an interface between the modem LAN ports and a point-to-point or point-to-multipoint WAN satellite network.

The physical interface connector is a RJ-45 10/100/1000BaseT copper interface. The Ethernet port can be activated or disabled by any of the control interfaces.

3.9 Built-in 1:1 Redundancy Mode Operation (Not Released)

Refer to Appendix J 1:1 redundancy mode operation (Not Currently Available).

4 Maintenance

4.1 Periodic Maintenance

The M7XC requires no mandatory periodic field maintenance or adjustments. The internal digital calibration data is held inside the modem EEPROM. There are no batteries or parts contained within the modem case requiring periodic service. The only moving part is the optional fan that is designed with a service life of 200,000+ hours. There is also no external fuse on the M7XC Modem.

4.1.1 Internal Reference Calibration

During factory testing and calibration, the modem unit is compared to a known in-house reference and calibrated. The default values are permanently stored representing this factory calibration. The unit may be offset from this factory value by using the manual tuning.

Manual tuning of the modem's reference is accomplished using the >Unit >Reference page, "Fine Tune" parameter and entering a value from +/- 2000 Hz.

The factory calibration may be restored by setting the >Unit >Reference page, "Fine Tune" parameter value to "0".

4.2 Modem Checkout

The following descriptions assume that the full system is in operation and the software is running properly on the central processor.

4.2.1 Initial Power-Up

⇒ **CAUTION!!!:** Before initial power-up of the modem, it is recommended to disconnect the transmit output cable from the satellite ground station equipment. This is especially true if the current modem configuration settings are unknown. Incorrect settings could disrupt existing communications traffic and potentially harmful results. New modems from the factory are normally shipped with a default configuration of setting the transmit carrier OFF.

The modem does not have a power switch so turning the unit "ON" is done by connecting the DC power to the DC input pins on the multi-function connector J4. At every power-up, the modem processor tests itself and several of its components before beginning its main program. The power-up diagnostics take approximately 5 seconds and show no results if successful. If a failure is detected, the indication may vary depending on the type of fault detected. A serious failure will result in the side panel Alarm LEDs flashing at a rate of approximately 4 times a second.

Most potential failures will result in the modem giving a description of the problem in the >Unit >Status web page.

The initial field checkout of the modem can be accomplished from the web pages.

4.2.2 Factory Default Parameters

New modems from the factory have default values placed into the non-volatile memory for operating parameters. If a Monitor/Control System does not configure the modem automatically via remote control, the modem can be easily configured from the front panel or through the HTTP interface using a standard internet browser. A modem can be returned to the factory default settings by using the command **<Unit: Store/Recall – Storage Location = Factory Default>**, then editing and choosing the “Factory Default” option.

The most common factory default parameters placed into the non-volatile memory are shown in Table 4-1 Factory Default Parameters.

Table 4-1 Factory Default Parameters

TX: >Waveform “Off”	Demodulator: >Waveform “Off”
Unit: >Data Port >Data Port Link “Auto” >Data Port >Offline Mode “Mute” >Mgmt Port >(no changes to params) >SNMP >Alert 1,2,3,4 Mode “Off” >SNMP >(no changes to other params) >Ref >Source “Internal, Ref Out Off” >Ref > Frequency “10MHz” >Ref > Fine Tune “0”	

4.3 Updating Modem Firmware

The M7XC modem series system firmware is updated via the web browser or through SNMP. If the unit requires a firmware update, contact Datum Systems customer service to request the firmware or goto the below download link to download the most current FW release for the M7XC modem.

4.3.1 Firmware Update using the Web Browser

The following section will provide a step-by-step process on how to update the M7XC Firmware Version using the Web browser interface.

Required Equipment:

1. A PC with a Connection to the Internet.

Required Software:

1. M7XC_Current_Software.zip

Downloadable from:

http://modem.datum-direct.com/M7XC/M7XC_Current_Software.7z

Extract the zipped folder, find and save the firmware file to a known location on your computer. The file should look something similar to: “FW-900014.M7XCv2.1.2-C.fw”

Before starting the update process, access the >Unit >Status page and verify / record the current installed Firmware Version see (Figure 4-1 Firmware Update Current FW Version).

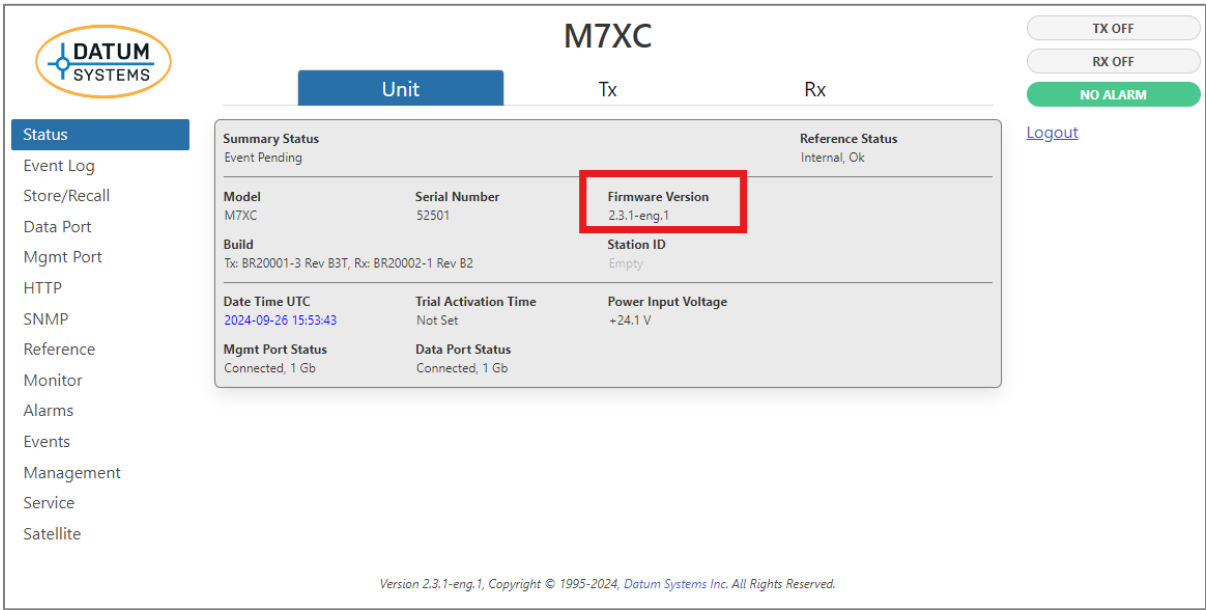


Figure 4-1 Firmware Update Current FW Version

Next move to the >Unit >Management page and click on the >Firmware Update “Choose File” button as show in (Figure 4-2 Firmware Update Choose File)



Figure 4-2 Firmware Update Choose File

After selecting the “Choose File” button your computers file system should open to locate the recently downloaded file. Locate and select the unzipped / extracted file with the name similar to “FW-900014.M7XCv2.1.2-C.fw”

Once the file has been selected the Web Interface should look like the below (Figure 4-3 Firmware Update: Upload New Firmware)



Figure 4-3 Firmware Update: Upload New Firmware

Once the desired Firmware is located and the “Upload Firmware” button is selected the modem will begin to load the new firmware into the unit. This process will take a few seconds and the progress of the FW update can be seen on the right side of the

Web interface as “FW-XXXXXX.M7XCXXX.fw – XX%” see (Figure 4-4 Firmware Update: Update Progress %)

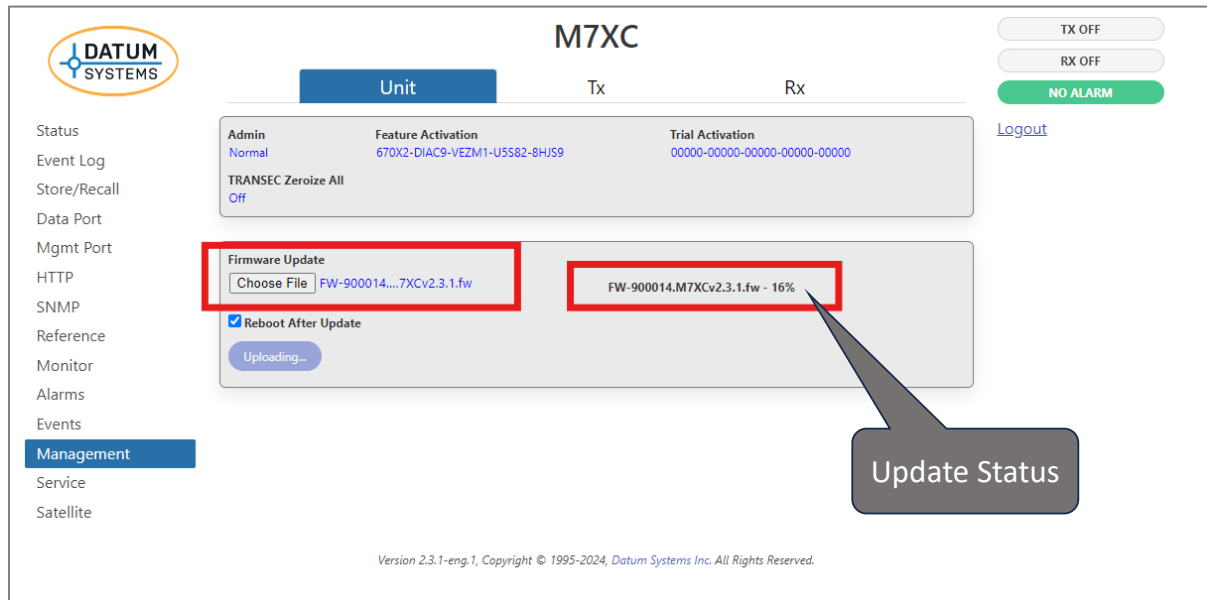


Figure 4-4 Firmware Update: Update Progress %

After the new Firmware has been loaded on the modem the modem will verify the Firmware authenticity before writing to the modem's internal FLASH memory. See Figure 4-5 Firmware Update: Verification and Figure 4-6 Firmware Update: Updating Modem Flash).



Figure 4-5 Firmware Update: Verification



Figure 4-6 Firmware Update: Updating Modem Flash

Once the modem has updated the Firmware you will be prompted to log back into the unit. See below Figure 4-7 Firmware Update: Log Back In

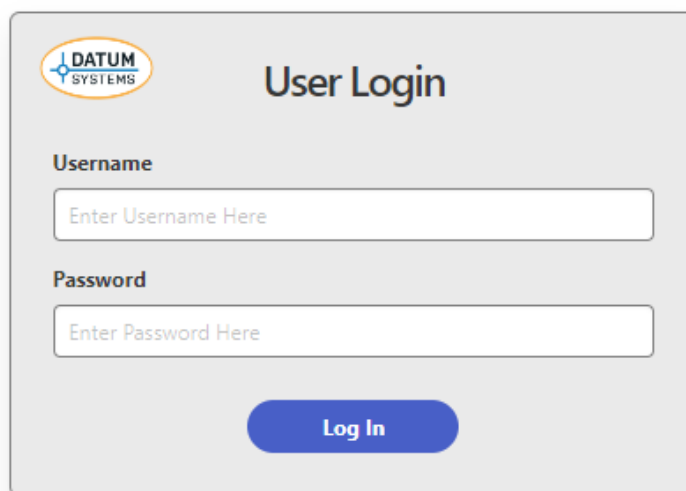


Figure 4-7 Firmware Update: Log Back In

Verify the new Firmware has been loaded onto the modem and is now running by comparing the FW Version now installed vs. the previously recorded Firmware version (Figure 4-1 Firmware Update Current FW Version).

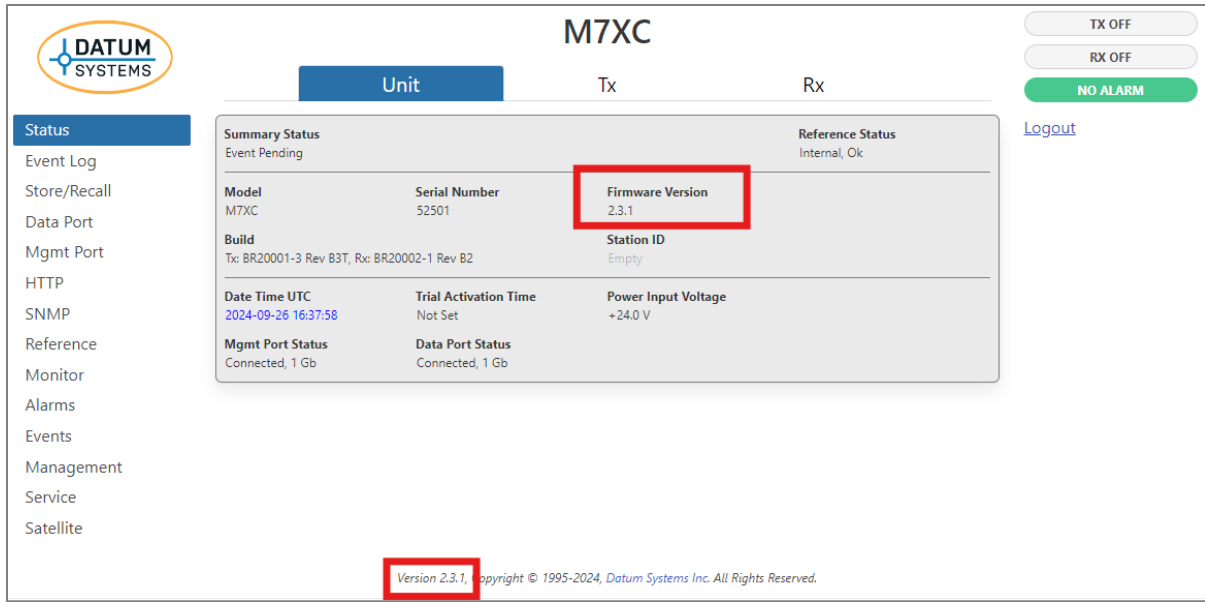


Figure 4-8 Firmware Update: Check New Firmware Version Installed

Upgrade is now complete!

4.4 Upgrading the Modem Features

Each of the modulator and demodulator cards in the M7XC series are available with a different set of features that enable a cost-effective approach for matching the modem capability to the system requirement. Upgrading from one set of features to another is a simple process that can be accomplished from any of the control interfaces.

The first step is to contact your local reseller or Datum Systems directly requesting a feature upgrade. When requesting an upgrade, the information required is:

- Modulator Serial Number
- Demodulator Serial Number
- Features desired to be upgraded

Upon confirmation of the upgrade purchase, a specific 20-digit code will be sent to enable the upgraded features. The code is unique to the serial number of the card to be updated and will not work in any other card.

The most common way to insert the feature code into the M7XC is via the web browser.

4.4.1 Using the web browser

To insert the code, go to the “Management” page in the Unit (Figure 4-9 Unit Features Page) tab and enter the 20-digit code directly in the “Feature Activation” box, then press the “Execute & Save” button. If correctly entered the modem will now display the new features in the various “Feature” (Figure 4-10 and Figure 4-11) tab parameter box.

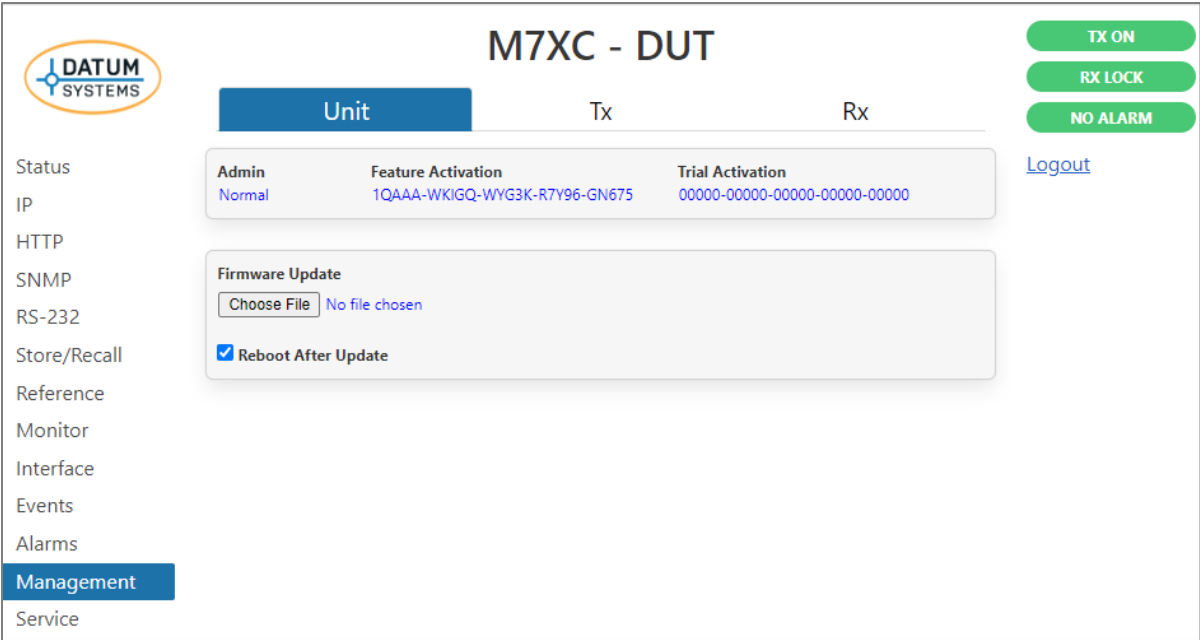


Figure 4-9 Unit Features Page

Feature sets can only be upgraded. There is no code to downgrade a feature set to a lower one.

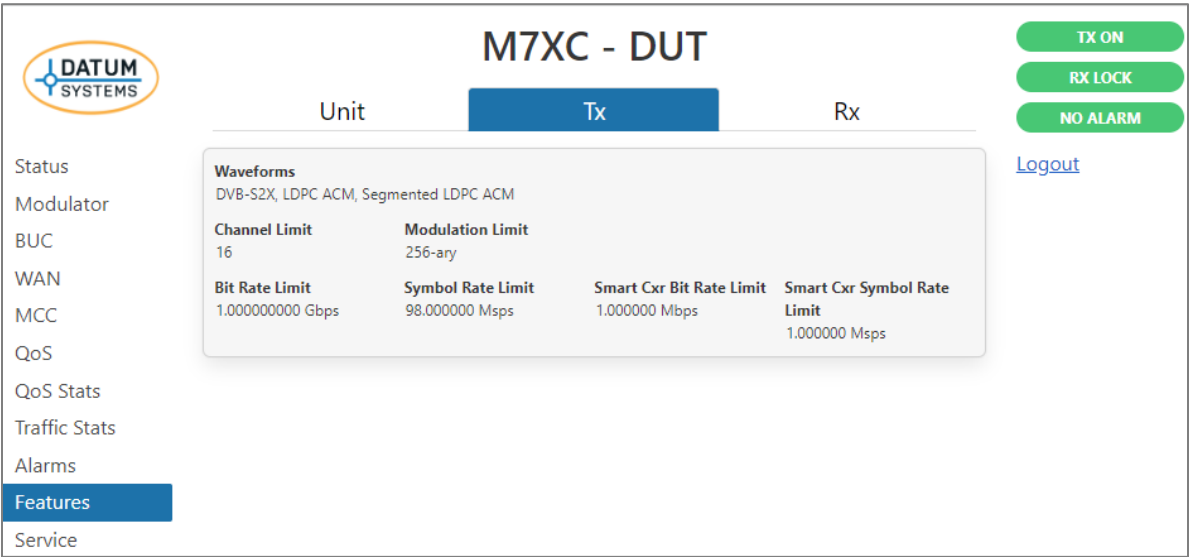


Figure 4-10 TX Features Installed Page

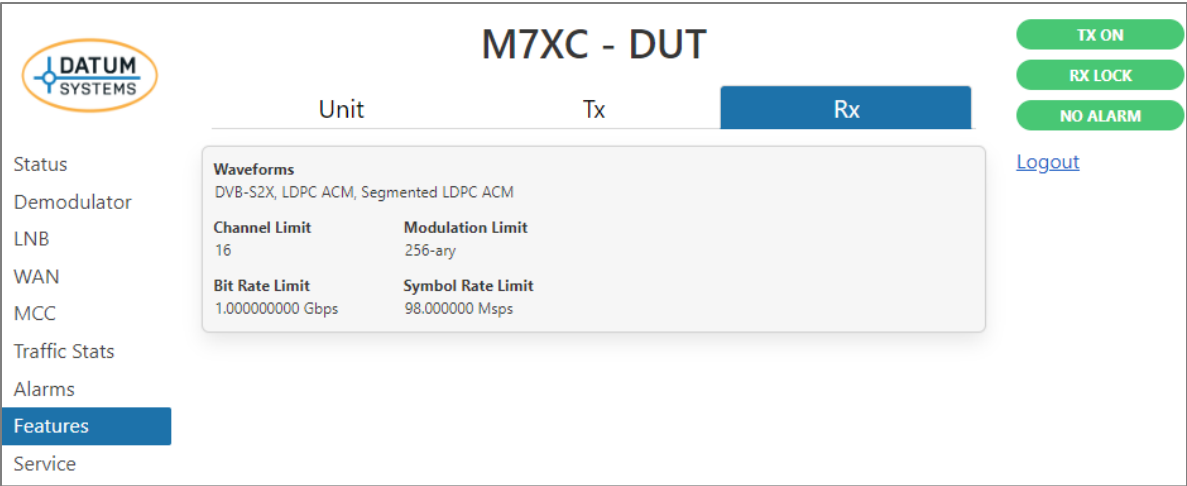


Figure 4-11 RX Features Installed Page

4.5 What do the abbreviations in the manual mean?

A good example is the display and manual representation "Redundcy SW Rqst". Unfortunately, the display does not hold enough characters to display the full text of "Redundancy Switch Request". Following is a list of abbreviations used.

Table 4-2 Abbreviations in Manual

Abbreviation	Full Text
1:1, 1:N, M:N	One for One, One for N and M for N. All redundancy switch types.
ACR	Automatic Configuration Recovery
AFC	Automatic Frequency Control
AGC	Automatic Gain Control
ALC	Automatic Level Control
Alm	Alarm
Alt	Alternate
AUPC	Automatic Uplink Power Control
AUFC	Automatic Uplink Frequency Control
BER, BERT	Bit Error Rate, Bit Error Rate Test
BUC	Block Up-Converter
Cal	Calibrate
Clk	Clock
Config	Configuration
Cntst	Contrast
CXR	Carrier
Dem	Demodulator
Dif	Differential
ESC	Engineering Service Channel
Erred	Errored
FEC	Forward Error Correction
Freq, Frq	Frequency
Frmt	Format
HSSI	High Speed Serial Interface (used with routers)
IDcOff, QDcOff	I and Q channel DC Offset
I/O	Input/Output

Intf	Interface
Keybrd	Keyboard
LCD	Liquid Crystal Display
LDPC & FLDPC	(Flexible) Low Density Parity Check FEC
LNB	Low Noise Block downconverter
LO	Local Oscillator
Loopbck	Loop-back
Lvl	Level
Max	Maximum
MCC	Modem Control Channel
Min	Minimum
Mod	Modulator
Mux	Multiplexer
Opt	Option
OverHd	Overhead
RCV, Rcv	Receive, into the Demodulator
Redundcy	Redundancy
Ref	Reference
Rqst	Request
RS, R-S	Reed-Solomon – Type of FEC
Sat	Satellite
SCPC	Single Channel Per Carrier
SER	Symbol Error Rate
SnIP	Satellite network Interface Processor, our name for an Ethernet Interface running Linux.
SW, Sw	Switch
Sync	Synchronous or Synchronization
SysClk	System Clock
Ter, Terr	Terrestrial – Line side of modem
TPC	Turbo Product Codes – Type of FEC
Tst	Test
USB	Universal Serial Bus
VSAT	Very Small Aperture Terminal
XMT, Xmt	Transmit, from the Modulator

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5 Appendix List

- 5.1 Appendix A: M7XC Datasheet / Specification**
- 5.2 Appendix B: M7XC SNMP Operation**
- 5.3 Appendix C: M7XC Smart Carrier Cancelling**
- 5.4 Appendix D: M7XC MCC Operation**
- 5.5 Appendix E: M7XC ACM Operation**
- 5.6 Appendix F: M7XC Pt-Pt, Pt-Multi-Pt VCM and ACM Operation**
- 5.7 Appendix G: M7XC Open AMIP Operation**
- 5.8 Appendix H: M7XC Segmented Mode Operation**
- 5.9 Appendix I: M7XC Demod Unlock Daisy Chain Operation**
- 5.10 Appendix J: M7XC Direct Sequence Spread Spectrum (DSSS) Operation**
- 5.11 Appendix K: M7XC 1:1 Redundancy Operation**
- 5.12 Appendix L: TRANSEC Operation**
- 5.13 Appendix M: SSH Operation**